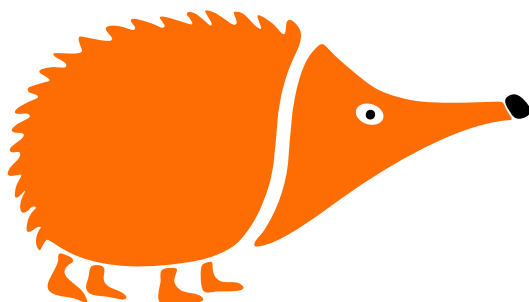




EchidnaShield

Documentación archivada de echidna.es

Echidna Educación · echidna.es · octubre de 2026

Contenidos con licencia CC BY-SA 4.0. Esta placa ya no se fabrica; su sucesora es la EchidnaBlack2.

Índice

La placa

Presentación

Alimentación

Modo sensores / Modo MkMk

Componentes

LEDs

Pulsadores

Joystick

Sensor de luz (LDR)

LED RGB

Audio

Acelerómetro

Conexiones MkMk

Complementos

Complementos

Servomotor de posición

Servomotor continuo

Sensor de temperatura LM35

Bluetooth

Documentación

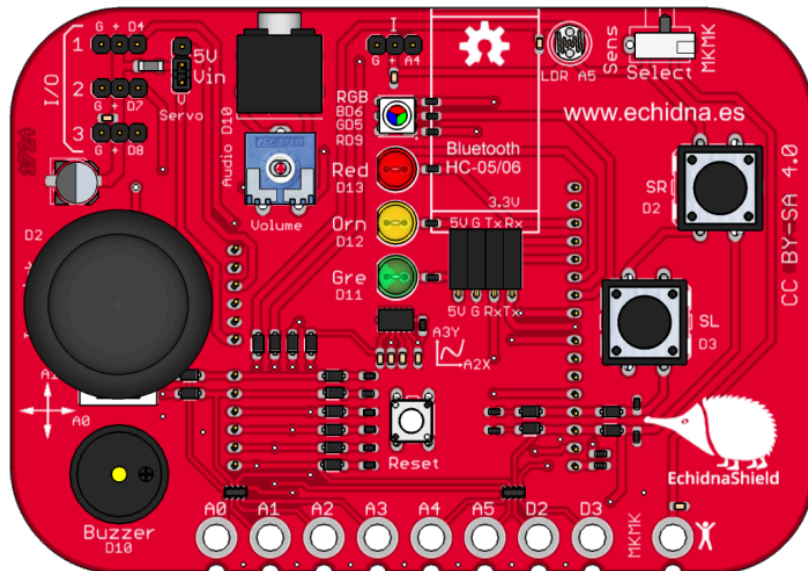
Documentación técnica

Anexos: hojas de características

La placa

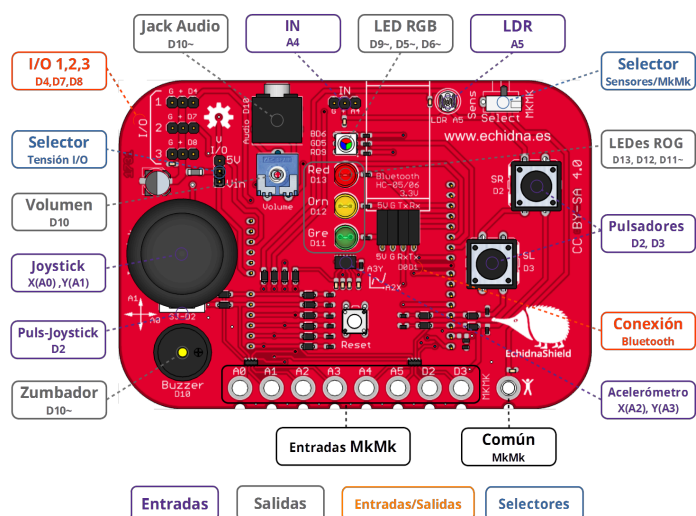
Presentación

Características



- Escudo para Arduino
- Sensores integrados: pulsadores, joystick, acelerómetro, luz
- Actuadores integrados: LEDs, LED RGB, audio
- Flexible: entradas y salidas disponibles para complementos
- 8 Entradas tipo Makey Makey
- Conexión BlueTooth
- Open Source Hardware Certification OSHWA UID ES000007

Esquema de componentes



Componentes

Pulsadores

Es un componente electromecánico que permite abrir o cerrar un circuito con un solo estado estable

Joystick

Es un mando que consta de dos potenciómetros uno para el eje X y otro para el eje Y.

Sensor Luz LDR

“Light Dependent Resistor”, es una resistencia cuyo valor depende de la cantidad de luz que incide sobre ella.

LED RGB

Light Emitting Diode y Red Green Blue, Son 3 LEDes Rojo, Verde y Azul en la misma cápsula.

Audio

Dispone de dos salidas para reproducir audio: el zumbador y el jack, ambos con un regulador de volumen.

MkMk

Echidna dispone de 8 Conectores MkMk que le permiten detectar gran variedad de objetos.

Acelerómetro

Sensor de aceleraciones basado en condensadores dentro de una estructura

LEDs ROG

Light Emitting Diode se usan como testigos (indicadores) y como fuente de iluminación.

Modo Sensores/Modo MkMk

Alimentación

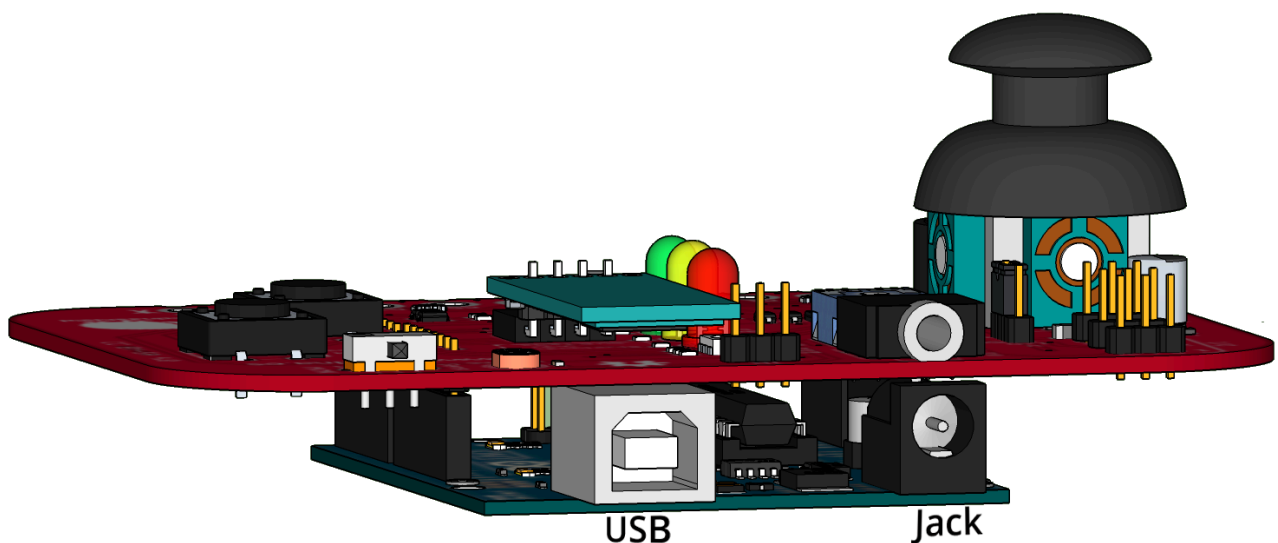
Complementos

Documentación

Alimentación

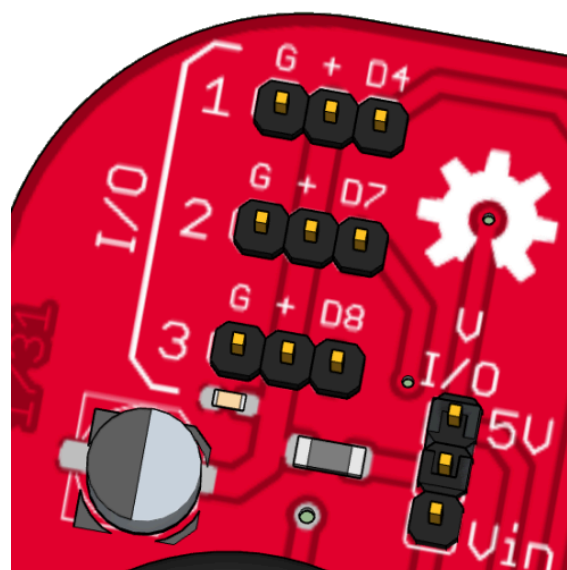
Formas de alimentar la placa

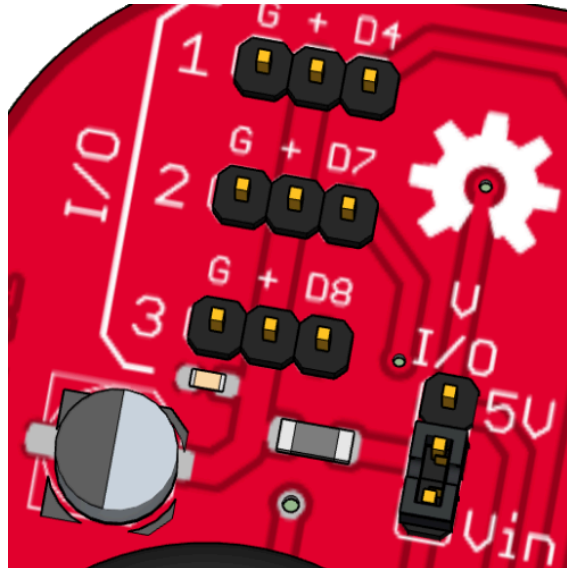
1. Mediante la **conexión USB**: esta forma es válida para la mayoría de las funciones, toda la placa recibe una tensión regulada de 5V proporcionada por el cable USB, generalmente con un límite de 500mA.
2. Mediante el **Jack de alimentación**: todas las partes reciben una tensión regulada internamente de 5V y 1A máx. Nos da la posibilidad de seleccionar V_{in} para alimentar los pines I/O con la potencia que proporciona el alimentador externo (no superar 1A de corriente constante)



Selector de alimentación

El jumper de alimentación permite seleccionar la alimentación de las I/O.





Selector 5v

En caso de querer alimentar las I/O desde los 5 Volts procedentes del Arduino el jumper debe estar colocado en esta posición. Aconsejado para sensores externos que necesiten una tensión estabilizada.

¡No utilizar la alimentación 5V cuando los servos a controlar consuman más de 300 mA!* De lo contrario se sobrepasaría el regulador de Arduino.

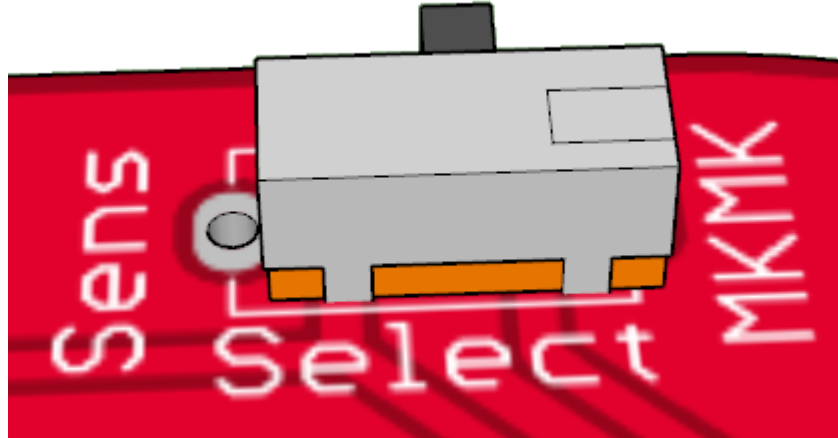
Selector alimentación Vin (Alimentación externa)

Aconsejado para alimentar servos u otros dispositivos conectados a I/O que consuman más de 300 mA con alimentación externa por el Jack de alimentación.

La alimentación de los servos cuenta con un filtro L-C para evitar que lleguen los parásitos de los motores al procesador.

Modo sensores / Modo MkMk

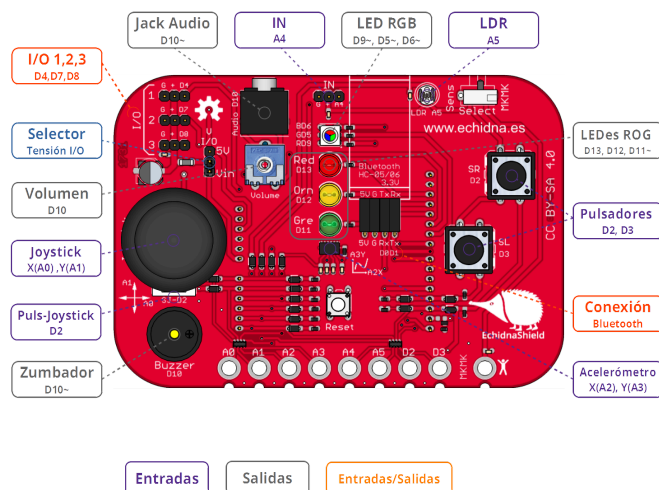
Selector Modo Sensores/Modo MkMk



Es un conmutador que selecciona entre el Modo Sensores y el Modo Makey Makey.

Modo sensores

En el modo sensores tenemos activo todos los componentes de la placa exceptuando las conexiones MkMk.



Modo MkMk

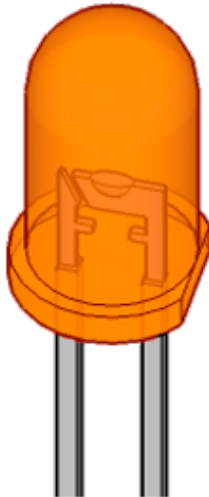
En este modo tenemos activas todas las salidas, las I/O y las conexiones MkMk.

Componentes

LEDs

Descripción

Acrónimo de **Light Emitting Diode**, basado en el fenómeno de electroluminiscencia. Se usan como testigos (indicadores) y como fuente de iluminación.



Funcionamiento

Al aplicar tensión emite luz, por lo tanto podemos controlar su encendido apagado en digital “1” enciende, “0” apaga y su brillo en las salidas PWM

Todos los LEDes se conectan con una resistencia serie “ R_s ” para controlar la tensión/corriente aplicada.



Pines

- D11~ «Gre»: salida digital y PWM
- D12 «Orn»: salida digital

- D13 «Red»: salida digital

Digital: se enciende con un «1» y se apaga con un «0».

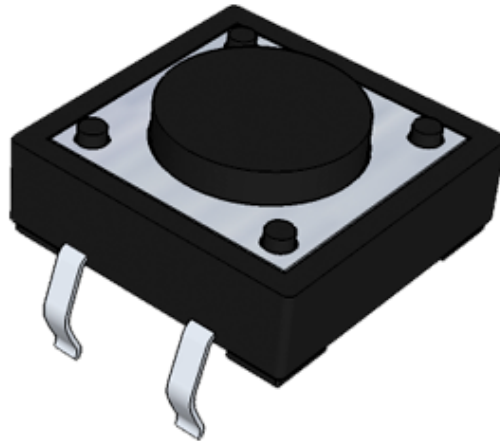
PWM: en la salida D11~ «Gre» se puede regular la intensidad luminosa con un valor de 0 a 255.

Hoja de características (anexo 1)

Pulsadores

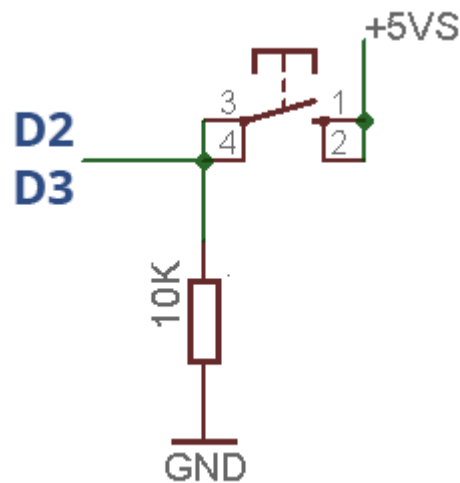
Descripción

Es un componente electromecánico que permite abrir o cerrar un circuito con un solo estado estable.



Funcionamiento

Está conectado con una resistencia a masa formando un circuito denominado pull-down, que proporciona un “0” (0V) sin pulsar y un “1” (5V) cuando pulsamos.



Pines

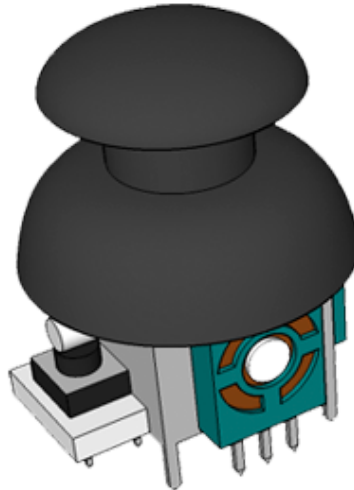
Entradas digitales: **D2** «SR» y **D3** «SL». Dan 0 si el pulsador no está pulsado y 1 si está pulsado.

Hoja de características (anexo 2)

Joystick

Descripción

Es un mando que consta de dos potenciómetros uno para el eje X y otro para el eje Y. Este modelo cuenta además con un pulsador.

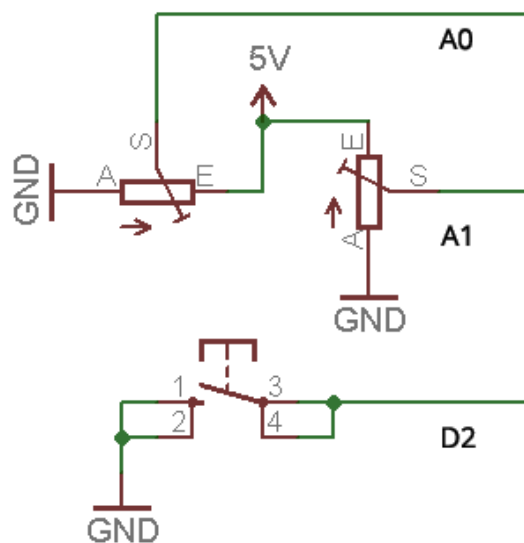


Funcionamiento

Permite transferir el movimiento del mando en una tensión proporcional entre 0-4,88V (0-999 de lectura analógica) * en la salida de cada eje (X, Y).

En la posición de reposo los valores rondan 2,5V (520 de lectura analógica) *, el pulsador del joystick está conectado con el pulsador «SR» (D2).

* Debido a las tolerancias de los componentes estos valores pueden ser distintos en cada placa.



Pines

- A0: eje X del joystick, entrada analógica
- A1: eje Y del joystick, entrada analógica
- D2: pulsador del joystick, entrada digital

Valores que da el joystick:

- En reposo, valores en torno a 512 en los dos ejes.
- En el eje X, 0 a la izquierda y 1023 a la derecha.
- En el eje Y, 0 abajo y 1023 arriba.

Hoja de características (anexo 3)

Sensor de luz (LDR)

Descripción

LDR es el acrónimo de “Light Dependent Resistor”, es una resistencia cuyo valor depende de la cantidad de luz que incide sobre ella. Generalmente se fabrica con sulfuro de cadmio.



Funcionamiento

La LDR proporciona valores altos de resistencia (en torno al $M\Omega$) con poca luz y valores bajos con mucha luz. Gracias a conectarla con una resistencia en serie forma un divisor de tensión que consigue invertir la lógica del funcionamiento, de forma que con mucha luz proporciona valores altos de tensión $4,9V = 999$ (de lectura analógica) * y valores bajos con poca luz ($0,0V = 000$).

* Debido a las tolerancias de los componentes estos valores pueden ser distintos en cada placa.



Pines

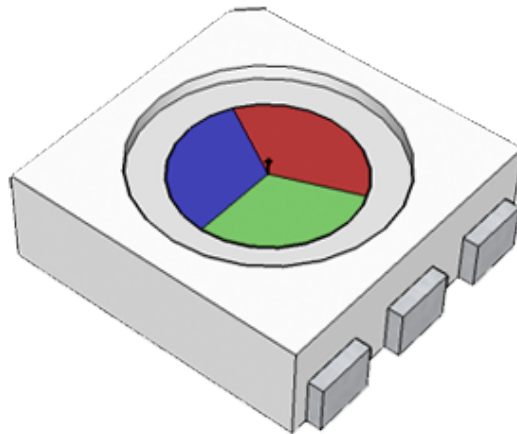
A5 «LDR»: entrada analógica.

Hoja de características

LED RGB

Descripción

LED RGB acrónimos de Light Emitting Diode y Red Green Blue, es decir son tres LEDes Rojo, Verde y Azul en la misma cápsula.



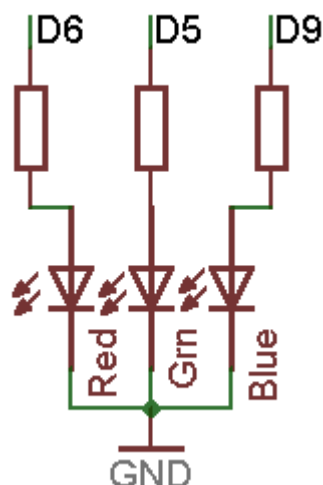
Funcionamiento

El LED RGB podemos controlarlo digitalmente, encendiendo o apagando cada LED y ajustar la luminosidad mediante PWM “analógico”.

Control digital: podemos encender y apagar cada uno de los LED RGB, lo que nos permite formar $2^3=8$ colores diferentes.

Control “analógico” PWM: podemos controlar el brillo de cada LED, variando su intensidad luminosa desde 0 a 255, lo que permite formar $256 \times 256 \times 256$, más de 16 millones de colores.

Todos los LEDes se conectan con una resistencia serie “Rs” para controlar la tensión/corriente aplicada.



Pines

- D9~ «RGB_R»: salida digital y PWM
- D5~ «RGB_G»: salida digital y PWM
- D6~ «RGB_B»: salida digital y PWM

Como salidas PWM, con un valor entre 0 y 255 se regula la intensidad de cada LED y, con ella, el color que emite.

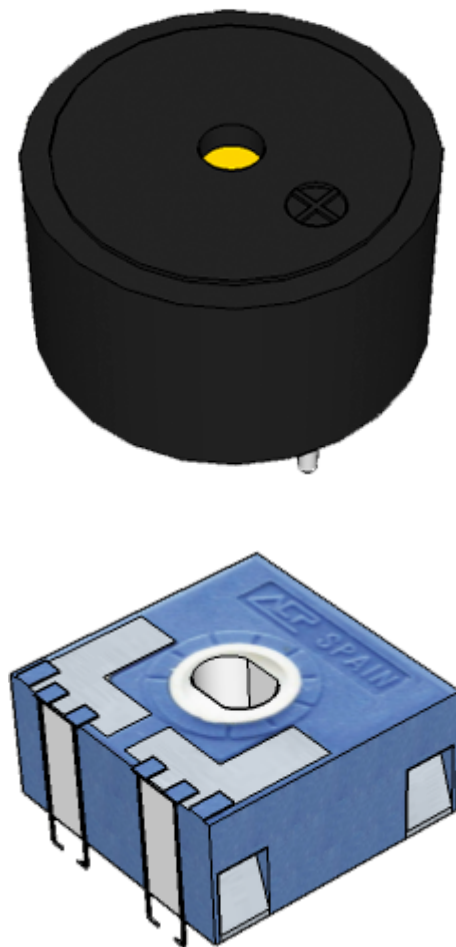
Hoja de características (anexo 4)

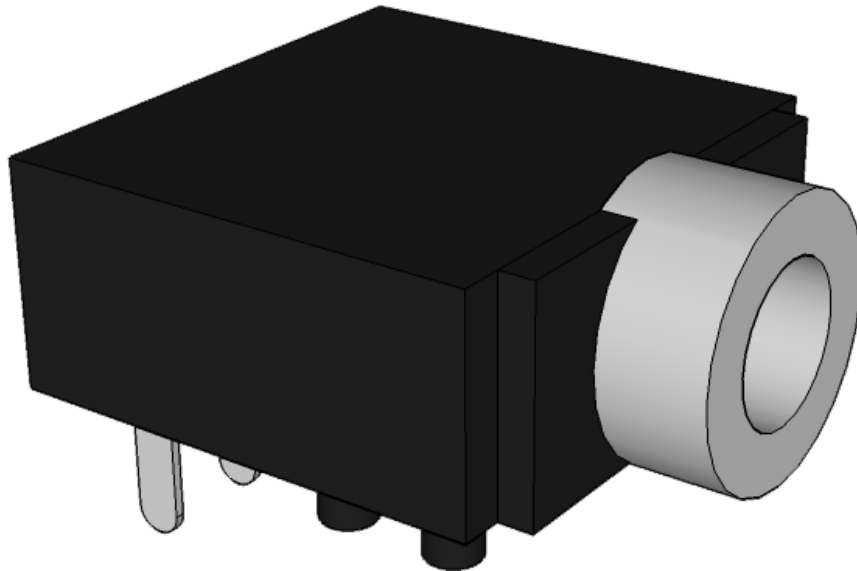
Audio

Descripción

Disponemos de dos salidas para reproducir audio, el zumbador y el jack al que podemos conectar auriculares o altavoces autoamplificados. Al conectar una clavija de audio en el jack se desconecta el zumbador.

Además contamos con un potenciómetro que permite ajustar el volumen del sonido.



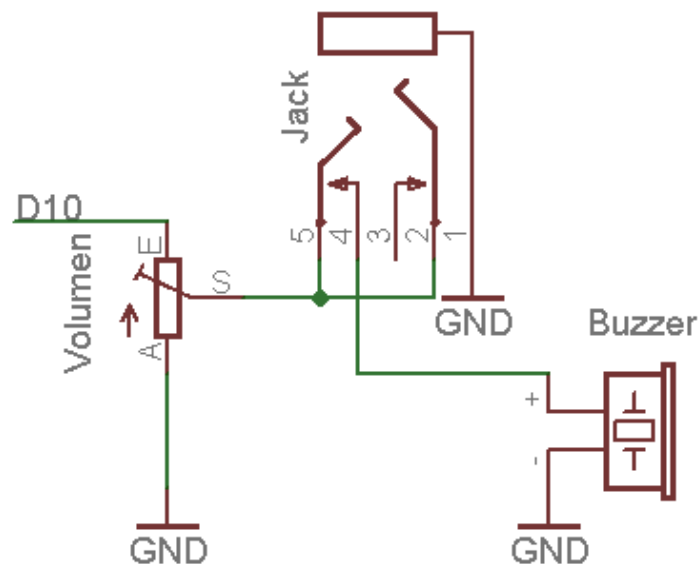


Funcionamiento

En D10~ podemos reproducir señales entre 31 Hz y 20 KHz.

El zumbador al ser excitado por una señal con una frecuencia determinada vibra reproduciendo un sonido. Hay que tener en cuenta que su frecuencia central es de 2,3 KHz, con lo que si nos alejamos mucho de esa frecuencia no sonará con calidad. (Do = C4 = 262 Hz) (cuidado con el volumen, a bajo nivel el zumbador puede no resultar audible).

En la salida del jack conectamos un equipo externo de audio que nos permite reproducir las frecuencias de entrada con más calidad (cuidado con el volumen, puede dañar los auriculares e incluso llegar al umbral doloroso en los oídos).



Pines

D10~ «Buzz»: salida PWM. Con un valor entre 0 y 255 se modula la frecuencia del sonido.

Hoja de características (anexo 5)

Acelerómetro

Descripción

Sensor de aceleraciones basado en condensadores diferenciales dentro de una estructura micro-mecanizada.

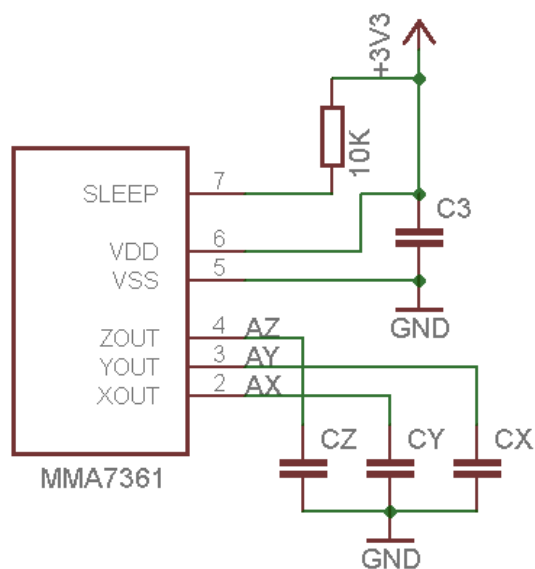
Proporciona una tensión que depende de la aceleración.



Funcionamiento

Mide la aceleración en los ejes (X, Y) de Echidna, en la posición de reposo proporciona un valor de $1,56V = 320$ (de lectura analógica)* y en sus extremos los valores $0,8V = 159$ - $2,40V = 490$.

* Debido a las tolerancias de los componentes estos valores pueden ser distintos en cada placa.



Pines

- A2 «Ace_X»: entrada analógica

- A3 «Ace_Y»: entrada analógica

Hoja de características (anexo 6)

Conexiones MkMk

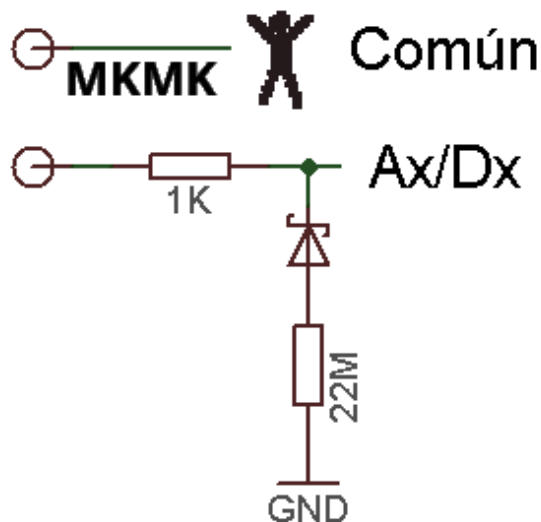
Descripción

Una conexión MkMk es un conector que permite detectar gran variedad de objetos al conectarlos entre una entrada y el común. Echidna dispone de 8 conexiones MkMk.



Funcionamiento

Cada conexión MkMk es parte de un divisor de tensión con una resistencia muy elevada, por lo que al conectar un elemento que conduce electricidad, aunque sea tenuemente, al cerrar el circuito con el punto común, la tensión de la entrada sube y es detectada. La tensión umbral de detección depende de cada material/persona.



Pines

- A0 MkMk0: entrada analógica
- A1 MkMk1: entrada analógica
- A2 MkMk2: entrada analógica
- A3 MkMk3: entrada analógica
- A4 MkMk4: entrada analógica
- A5 MkMk5: entrada analógica
- D2 MkMk6: entrada digital
- D3 MkMk7: entrada digital



23

Complementos

Complementos

Estos son algunos de los **componentes** que se pueden usar para complementar la funcionalidades de **EchidnaShield**, para conectarlos se disponen de las **I/O y la entrada analógica IN**.

Servomotor Posición

Son motores de corriente continua que permiten posicionarlo en un ángulo entre 0 y 180°.

Servomotor Continuo

Son motores de corriente continua con una reductora y electrónica de control que permiten controlar el sentido de giro.

Sensor Temperatura

Es un sensor de temperatura calibrado cuya salida es lineal, y donde cada 10 mV equivale a 1°C

Infrarrojos distancia

Es un sensor de distancia que proporciona una tensión según la cantidad de infrarrojo que rebota en una superficie.

Complemento Conexiones MxMx

Pinzas cocodrilo y cinta conductiva

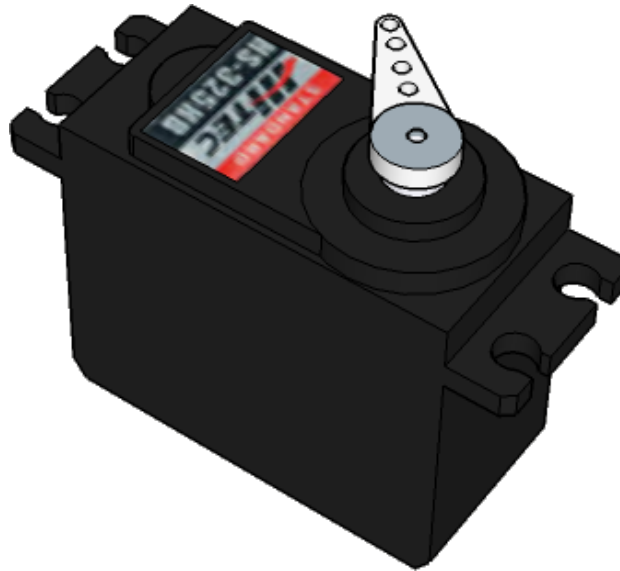
Bluetooth

Es un transceptor que permite conectar dispositivos Bluetooth a la placa Arduino.

Servomotor de posición

Descripción

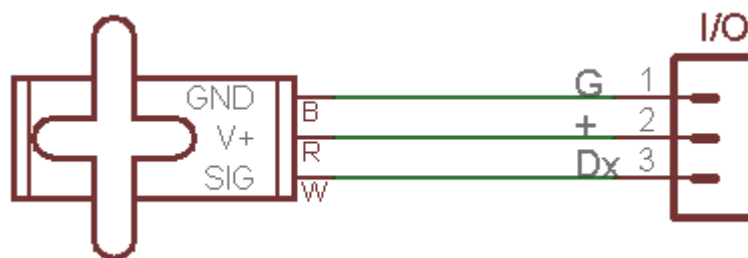
Son motores de corriente continua con una reductora y electrónica de control que permiten posicionarlo en un ángulo entre 0 y 180°.



Funcionamiento

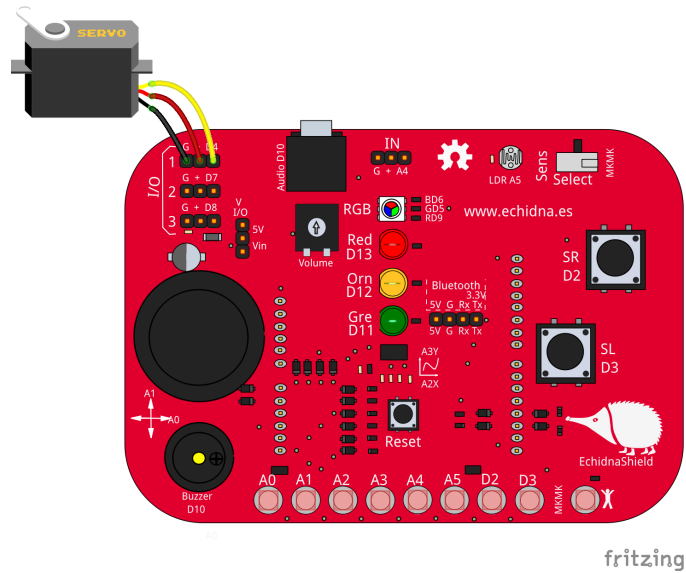
Se controla mediante pulsos de 1-2 ms en periodos de 20 ms.

En la práctica se usa una librería que nos permite indicar directamente el ángulo.



Elementos de conexión

Se conecta directamente con sus cables. Es aconsejable colocar el jumper de alimentación en Vin y alimentar Arduino mediante el jack de alimentación.



Pines

- I/O 1: pin D4
- I/O 2: pin D7
- I/O 3: pin D8

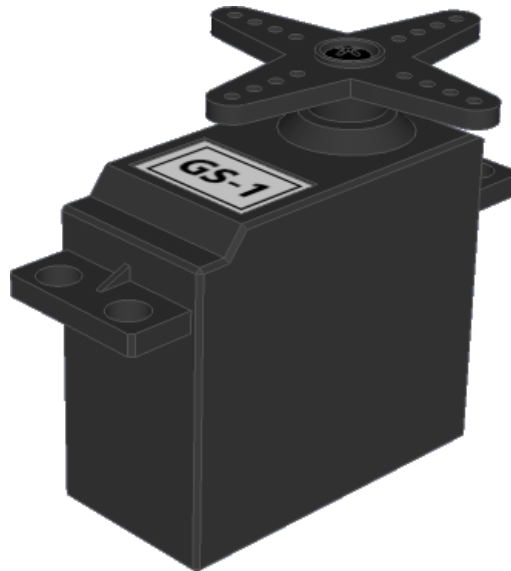
El servomotor de posición se controla indicando el ángulo, de 0 a 180 grados.

Hoja de características (anexo 7)

Servomotor continuo

Descripción

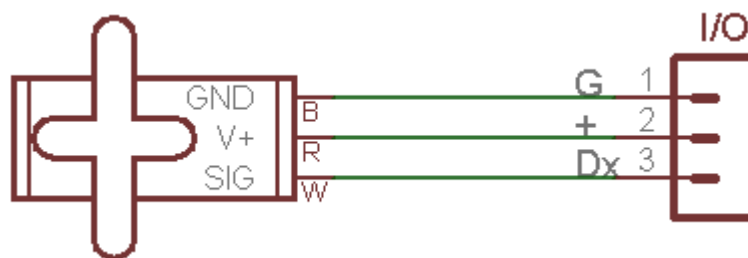
Son motores de corriente continua con una reductora y electrónica de control que permiten controlar el sentido de giro.



Funcionamiento

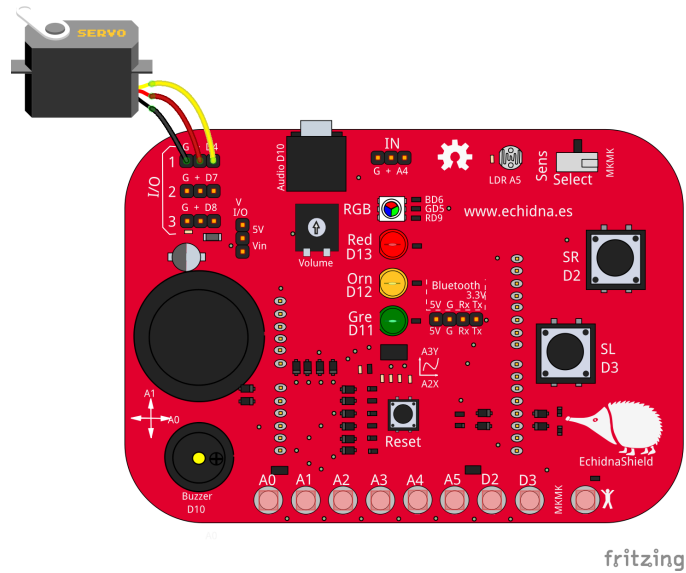
Se controla mediante pulsos de 1-2 ms en periodos de 20 ms.

En la práctica se usa una librería que nos permite indicar directamente el el sentido de giro.



Elementos de conexión

Se conecta directamente con sus cables. Es aconsejable colocar el jumper de alimentación en Vin y alimentar Arduino mediante el jack de alimentación.



fritzing

Pines

- I/O 1: pin D4
- I/O 2: pin D7
- I/O 3: pin D8

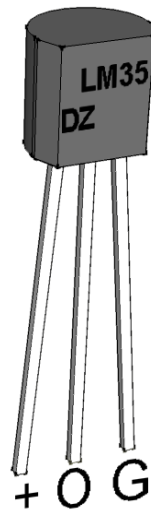
El servomotor continuo se controla indicando el sentido de giro.

Hoja de características (anexo 8)

Sensor de temperatura LM35

Descripción

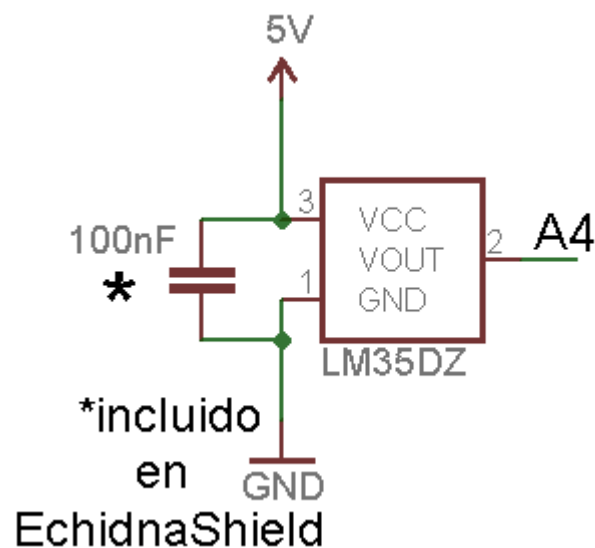
Es un sensor de temperatura calibrado cuya salida es lineal y donde cada 10 mv equivale a 1 °C



Funcionamiento

Se conecta directamente al conector IN (5V, GND, A4). Proporciona 10 mv por °C. Teniendo en cuenta que 5V son 1024 “pasos” en la medida analógica, podemos obtener la temperatura mediante la siguiente operación:

$$\text{temperatura} = (\text{lectura Analógica} * 5.0 * 100.0) / 1024.0$$



Elementos de conexión

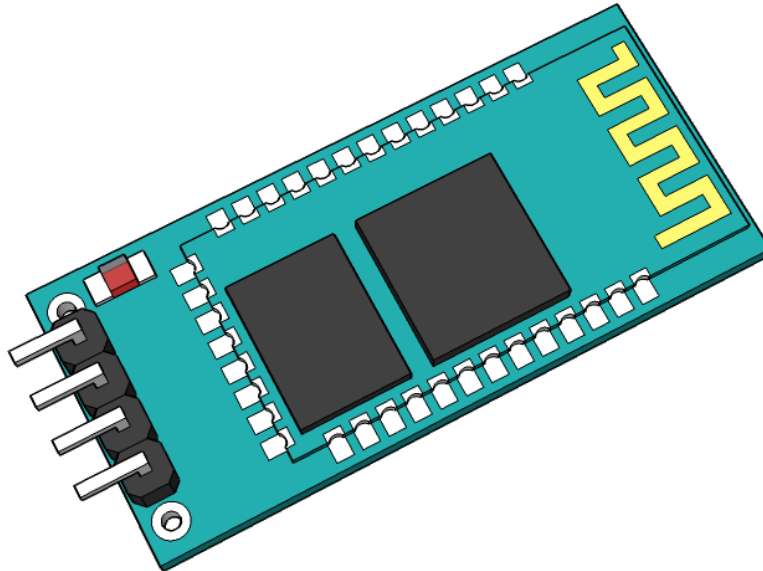
Para conectar el sensor necesitamos tres cables dupont-dupont hembra-hembra. También se puede usar una protoboard y tres cables dupont-dupont macho hembra.



Bluetooth

Descripción

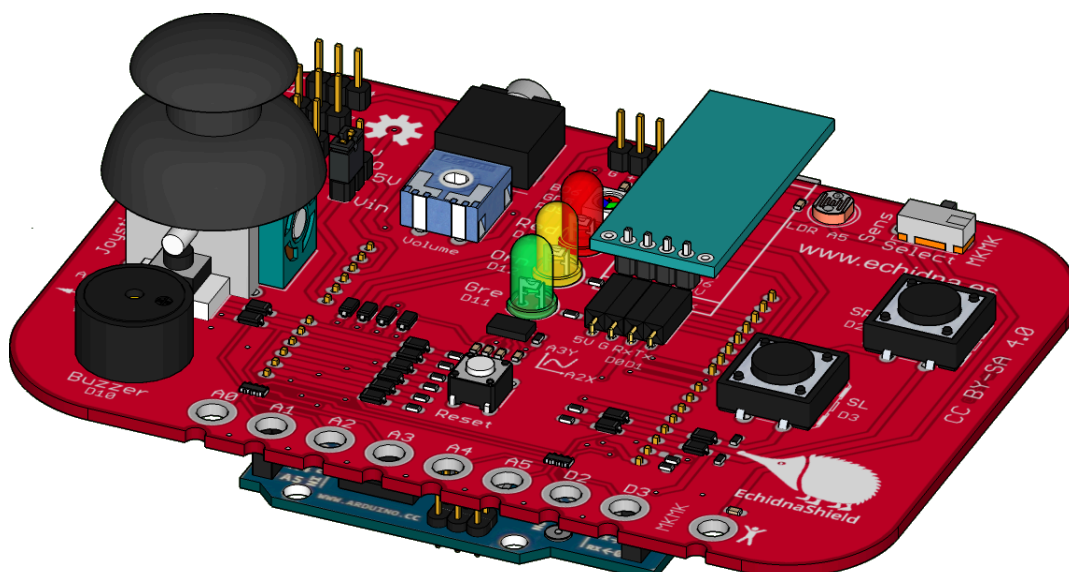
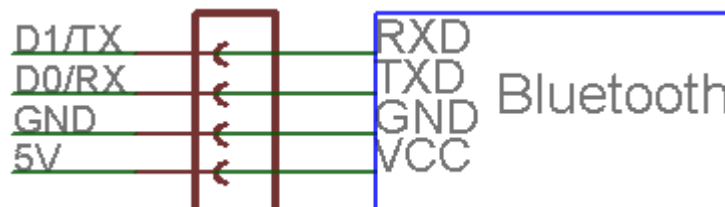
Es un transceptor que permite conectar dispositivos Bluetooth a la placa Arduino.



Funcionamiento

Envía y recibe datos a través del puerto serie vía Bluetooth

Esquema y conexión



Pines

- D0 (Rx): recepción de datos
- D1 (Tx): transmisión de datos

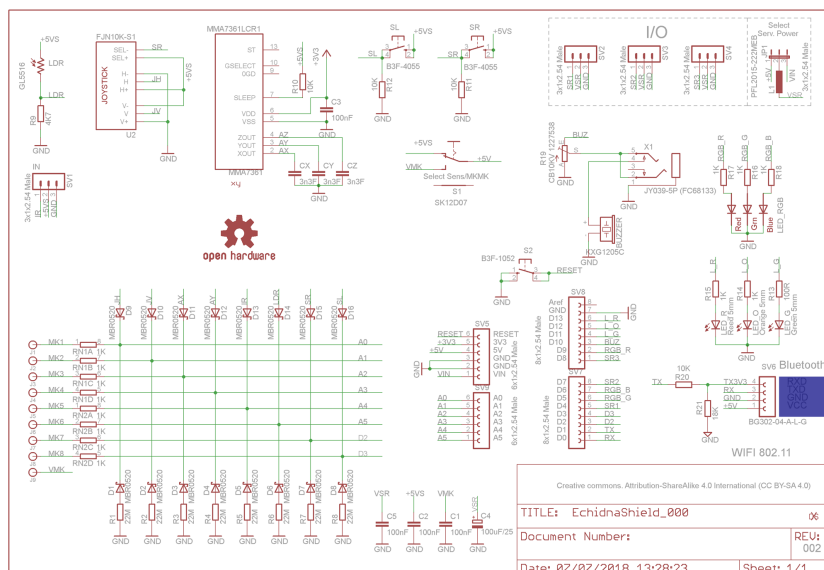
Se comunica a través del puerto serie Tx/Rx, es necesario programar Arduino previamente a la conexión del Bluetooth, ya que si no este bloquea la comunicación por cable.

Hoja de características (anexo 10)

Documentación

Documentación técnica

Esquemático



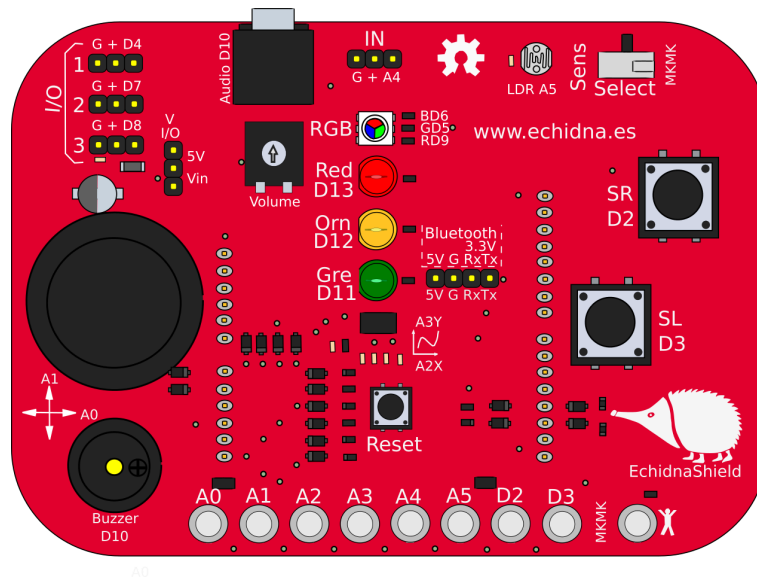
Open Source Hardware Certification

EchidnaShield está certificada como Open Source Hardware con registro OSHWA UID ES000007



Licencia

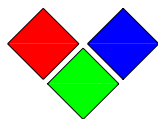
La placa Echidna Shield se publica con la licencia CERN-OHL-W 2.0. No se permite su reproducción bajo la marca Echidna.



Anexos

Hojas de características de los componentes, en las páginas siguientes y en este orden:

1. LEDs: Hoja de características (DataSheet-LED.pdf)
2. Pulsadores: Hoja de características (DataSheet-button.pdf)
3. Joystick: Hoja de características (DataSheet-Joystick.pdf)
4. LED RGB: Hoja de características (DataSheet-LEDRGB.pdf)
5. Audio: Hoja de características (F-CM12P-LF.pdf)
6. Acelerómetro: Hoja de características (DataSheet-MMA7361L.pdf)
7. Servomotor de posición: Hoja de características (DatasheetServoPosicion.pdf)
8. Servomotor continuo: Hoja de características (DataSheeetServoContinuo.pdf)
9. Sensor de temperatura LM35: Hoja de características (lm35.pdf)
10. Bluetooth: Hoja de características (hc06.pdf)



深圳市昱申科技有限公司

CHINA YOUNG SUN LED TECHNOLOGY CO., LTD.

TEL: (86) 755-28079401 28079402 28079403 28079404 28079405
FAX: (86) 755-28079407 E-mail: info@100LED.com Web: www.100LED.com

Model No.: YSL-R531R3D-D2

Applications:

- Decorations
- Bill Insperctor
- Inceticial Lights
- Medical Appliance

Absolute Maximum Ratings: (Ta=25°C) .

ITEMS	Symbol	Absolute Maximum Rating	Unit
Forward Current	I _F	20	mA
Peak Forward Current	I _{FP}	30	mA
Suggestion Using Current	I _{SU}	16-18	mA
Reverse Voltage (V _R =5V)	I _R	10	uA
Power Dissipation	P _D	105	mW
Operation Temperature	T _{OPR}	-40 ~ 85	°C
Storage Temperature	T _{STG}	-40 ~ 100	°C
Lead Soldering Temperature	T _{SOL}	Max. 260°C for 3 Sec. Max. (3mm from the base of the expoxy bulb)	

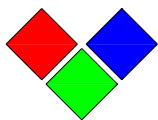
Absolute Maximum Ratings: (Ta=25°C)

ITEMS	Symbol	Test condition	Min.	Typ.	Max.	Unit
Forward Voltage	V _F	I _F =20mA	1.8	---	2.2	V
Wavelenength (nm) or TC(k)	Δ λ	I _F =20mA	620	---	625	nm
*Luminous intensity	I _v	I _F =20mA	150	---	200	mcd
50% Viewing Angle	2 θ 1/2	I _F =20mA	40	---	60	deg

Address: 5/F, Building B, Anzhilong Indl., Qinghua East Road., Longhua Town, Shenzhen CHINA. 518109

www.100LED.com

ONE HUNDRED LED
PERFECT LED



深圳市昱申科技有限公司

CHINA YOUNG SUN LED TECHNOLOGY CO., LTD.

TEL: (86) 755-28079401 28079402 28079403 28079404 28079405

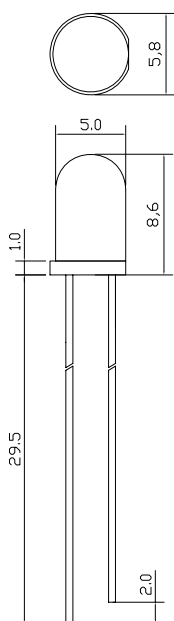
FAX: (86) 755-28079407 E-mail: info@100LED.com Web: www.100LED.com

Light Degradation in mcd: ($I_F=20mA$)

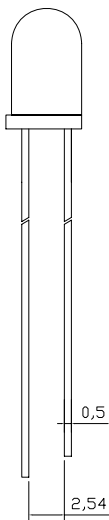
Colors	Light Degradation in mcd after Different Hours					
	216 Hrs	360 Hrs	792 Hrs	1104 Hrs	1992 Hrs	2328 Hrs
Red	1.52%	-1.22%	-3.10%	-4.68%	-5.72%	-8.27%
Yellow	-1.71%	-2.97%	-5.93%	-8.13%	-8.90%	-11.10%
Blue	3.13%	-0.33%	-3.84%	-8.23%	-21.32%	-24.92%
Green	-8.02%	-9.78%	-14.25%	-17.37%	-20.79%	-22.30%
Hours	48 Hrs	168 Hrs	336 Hrs	360Hrs	720 Hrs	1008 Hrs
Cool White	10.56%	6.72%	-2.29%	-7.68%	-17.32%	-22.48%
Pure White	13.66%	8.22%	-1.45%	-8.50%	-19.52%	-25.26%
Warm White	3.02%	-4.38%	-15.18%	-21.15%	-27.19%	-29.97%

Mechanical Dimensions:

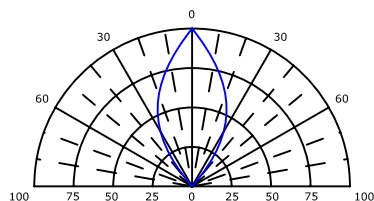
- All dimension are in mm, tolerance is $\pm 0.2mm$ unless otherwise noted
- An epoxy meniscus may extend about 1.5mm down the leads.
- Burr around bottom of epoxy may be 0.5mm Maximum



Unit: mm

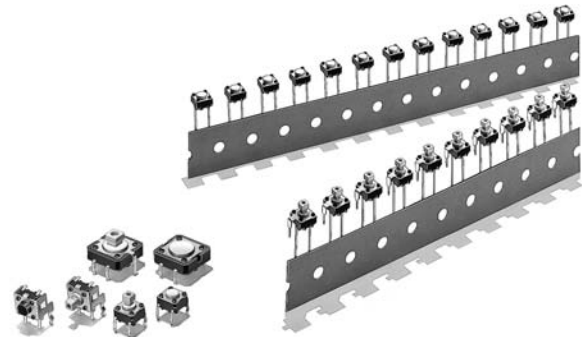


Viewing Angle Drawing



**Through-hole-mounting Switches
in a Wide Range of Models:
6 × 6 mm, 12 × 12 mm, Side-operated
Models, Gold-plated Contacts, and
Radial Tape**

- Extended mechanical/electrical durability: 10 x 10⁶ operations for 12 x 12 mm type and 1 x 10⁶ operations for the 6 x 6 mm type
- Taped radial type, vertical type and high force types are available.
- Gold plated models available for increased contact reliability, resistance to corrosive gas and insulation failure prevention for ion migration in harsh environments
- B32-series Key Tops mount to models with projected plungers.

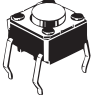
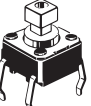
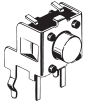


RoHS Compliant

■ List of Models

6 × 6 mm Models

Type	Contact material	Plunger	Height	Operating force (OF)	Plunger color	Bags			
						Without ground terminal	Minimum packing unit	With ground terminal	Minimum packing unit
Standard: B3F-1000 Series	Silver plated	Flat type 	4.3 mm	0.98 N {100 gf}	Ivory	B3F-1000	100 pcs	B3F-1100	100 pcs
				1.47 N {150 gf}	Yellow	B3F-1002		B3F-1102	
				2.55 N {260 gf}	Orange	B3F-1005		B3F-1105	
				4.9 N {500 gf}	Red	B3F-1006		---	
			5.0 mm	0.98 N {100 gf}	Black	B3F-1020		B3F-1120	
				1.47 N {150 gf}	Gray	B3F-1022		B3F-1122	
				2.55 N {260 gf}	Pink	B3F-1025		B3F-1125	
				4.9 N {500 gf}	Blue	B3F-1026		---	
			5.0 mm (7.5-mm pitch)	0.98 N {100 gf}	Black	---		B3F-1110	
			7.0 mm	0.98 N {100 gf}	Black	B3F-1060		---	
				1.47 N {150 gf}	Yellow	B3F-1062		---	
				0.98 N {100 gf}	Black	B3F-1070		---	
				1.47 N {150 gf}	Yellow	B3F-1072		---	
			9.5 mm	2.55 N {260 gf}	Orange	B3F-1075		---	
				0.98 N {100 gf}	Ivory	B3F-1050		B3F-1150	
				1.47 N {150 gf}	Yellow	B3F-1052		B3F-1152	
				2.55 N {260 gf}	Orange	B3F-1055		B3F-1155	
				4.9 N {500 gf}	Red	B3F-1056		---	
		Projected type 	7.3 mm	0.98 N {100 gf}	Ivory	B3F-1050		B3F-1150	
				1.47 N {150 gf}	Yellow	B3F-1052		B3F-1152	
				2.55 N {260 gf}	Orange	B3F-1055		B3F-1155	
				4.9 N {500 gf}	Red	B3F-1056		---	

Type	Contact material	Plunger	Height	Operating force (OF)	Plunger color	Bags			
						Without ground terminal	Minimum packing unit	With ground terminal	Minimum packing unit
Side-operated: B3F-3000 Series	Silver plated	 Flat type	3.15 mm	0.98 N {100 gf}	Ivory	---	100 pcs	B3F-3100	100 pcs
				1.47 N {150 gf}	Yellow	---		B3F-3102	
				2.55 N {260 gf}	Orange	---		B3F-3105	
			3.85 mm	0.98 N {100 gf}	Black	---		B3F-3120	
				1.47 N {150 gf}	Gray	---		B3F-3122	
				2.55 N {260 gf}	Pink	---		B3F-3125	
		 Projected type	6.15 mm	0.98 N {100 gf}	Ivory	---		B3F-3150	
				1.47 N {150 gf}	Yellow	---		B3F-3152	
				2.55 N {260 gf}	Orange	---		B3F-3155	
High-reliability gold-plated: B3F-1000-G Series	Gold plated	 Flat type	4.3 mm	1.76 N {180 gf}	Yellow	B3F-1002-G		B3F-1102-G	
			5.0 mm		Gray	B3F-1022-G		B3F-1122-G	
			7.0 mm		Yellow	B3F-1062-G (see note)		---	
			9.5 mm		Yellow	B3F-1072-G (see note)		---	
Side-operated with highly reliable gold-plated: B3F-3000-G Series	Gold plated	 Projected type	7.3 mm		Yellow	B3F-1052-G		---	
		 Flat type	3.85 mm		Gray	---		B3F-3122-G (see note)	

Note: Bulk Packaged, 100 Switches per bag. Order in multiples of the package quantity.

12 × 12 mm Models

Type	Contact material	Plunger (or LED color)	Height	Operating force (OF)	Plunger color	Bags			
						Without ground terminal	Minimum packing unit	With ground terminal	Minimum packing unit
Standard: B3F-4000 Series	Silver plated	Flat type	4.3 mm	1.27 N {130 gf}	Ivory	B3F-4000	100 pcs	B3F-4100	100 pcs
				2.55 N {260 gf}	Yellow	B3F-4005		B3F-4105	
		Projected type	7.3 mm	1.27 N {130 gf}	Ivory	B3F-4050		B3F-4150	
				2.55 N {260 gf}	Yellow	B3F-4055		B3F-4155	
Long durability: B3F-5000 Series	Silver plated	Flat type	4.3 mm	1.27 N {130 gf}	Blue	B3F-5000		B3F-5100	
		Projected type	7.3 mm		Blue	B3F-5050		B3F-5150	
High reliability gold-plated: B3F-5001 Series	Gold plated	Flat type	4.3 mm	1.27 N {130 gf}	Blue	B3F-5001		B3F-5101	
		Projected type	7.3 mm		Blue	B3F-5051		B3F-5151	

Note: Bulk Packaged, 100 switches per bag. Order in multiples of the package quantity.

6 × 6 mm Radial Models (Taping Specifications)

Type	Contact material	Plunger	Height	Operating force (OF)	Plunger color	Taped Radial			
						Without ground terminal	Minimum packing unit	With ground terminal	Minimum packing unit
Taped Radial: B3F-6000 Series	Silver plated	Flat type	4.3 mm	0.98 N {100 gf}	Ivory	B3F-6000	1,000 pcs	B3F-6100	1,000 pcs
				1.47 N {150 gf}	Yellow	B3F-6002		B3F-6102	
			5.0 mm	0.98 N {100 gf}	Black	B3F-6020		B3F-6120	
				1.47 N {150 gf}	Gray	B3F-6022		B3F-6122	
		Projected type	7.3 mm	0.98 N {100 gf}	Ivory	B3F-6050		B3F-6150	
				1.47 N {150 gf}	Yellow	B3F-6052		B3F-6152	

Note: The switches are tape packaged in units of 1,000 per package. Order in multiples of the package size. Switches are not sold individually.

■ Ratings/Characteristics

Rating (resistive load)	1 to 50 mA, 3 to 24 VDC (B3F-G: 100 μ A to 50 mA, 3 to 24 VDC)
Minimum applicable load (reference value)	10 μ A at 1 VDC (resistive load)
Ambient operating temperature	-25°C to +70°C at 60%RH max. (with no icing or condensation)
Ambient operating humidity	35% to 85% (at +5 to +35°C)
Contact form	SPST-NO
Contact resistance (initial value)	100 m Ω max.
Insulation resistance	100 M Ω min. (at 250 VDC with insulation tester)
Dielectric strength	500 VAC, 50/60 Hz for 1 min
Bounce time	5 ms max.
Vibration resistance	Malfunction: 10 to 55 Hz, 1.5 mm double amplitude
Shock resistance	Destruction: 1,000 m/s ² {approx. 100G} max. Malfunction: 100 m/s ² {approx. 10G} max.
Durability	B3F-1000, B3F-3000, B3F-6000: 1,000,000 operations min (OF: 0.98 N {100 gf}) (B3F-1070: 500,000 operations min) 300,000 operations min (OF: 1.47 N {150 gf}) 100,000 operations min (OF: 2.55 N {260 gf}) 50,000 operations min (OF: 4.9 N {500 gf}) B3F-4000: 3,000,000 operations min (OF: 1.27 N {130 gf}) 1,000,000 operations min (OF: 2.55 N {260 gf}) B3F-5000/5001: 10,000,000 operations min. B3F-G: 300,000 operations min.
Weight	6 $\times$ 6 mm models: approx. 0.25 g 12 $\times$ 12 mm models (standard types): approx. 0.85 g Radial models: approx. 0.25 g
Degree of protection	IEC IP00
Washing	Not possible

■ Operating Characteristics

6 $\times$ 6 mm Models

Item	B3F-1000, B3F-3000, B3F-6000				B3F-G
	0.98 N	1.47 N	2.55 N	4.9 N	1.76 N
Operating force (OF)	B3F-1□□0 B3F-3□□0 B3F-6□□0	B3F-1□□2 B3F-3□□2 B3F-6□□2	B3F-1□□5 B3F-3□□5	B3F-10□6	B3F-1□□2-G B3F-3□□2-G
Operating force (OF)	0.98 $\pm$ 0.29 N {100 $\pm$ 30 gf}	1.47 $\pm$ 0.49 N {150 $\pm$ 50 gf}	2.55 $\pm$ 0.69 N {260 $\pm$ 70 gf}	4.9 $\pm$ 1.47 N {500 $\pm$ 150 gf}	1.76 $\pm$ 0.49 N {180 $\pm$ 50 gf}
Releasing force (RF)	0.2 N {20 gf} min.	0.49 N {50 gf} min.	0.49 N {50 gf} min.	0.7 N {70 gf} min.	0.49 N {50 gf} min.
Pretravel (PT)	0.25 ^{+0.2} / _{-0.1} mm				0.25 ^{+0.2} / _{-0.1} mm

12 $\times$ 12 mm Models

Item	B3F-4000, B3F-5000, B3F-5001	
	1.27 N	2.55 N
Operating force (OF)	B3F-4□□0 B3F-5□□0 B3F-5□□1	B3F-4□□5
Operating force (OF)	1.27 $\pm$ 0.49 N {130 $\pm$ 50 gf}	2.55 $\pm$ 0.69 N {260 $\pm$ 70 gf}
Releasing force (RF)	0.29 N {30 gf} min.	0.49 N {50 gf} min.
Pretravel (PT)	0.3 ^{+0.2} / _{-0.1} mm	

■ Dimensions (Unit: mm)

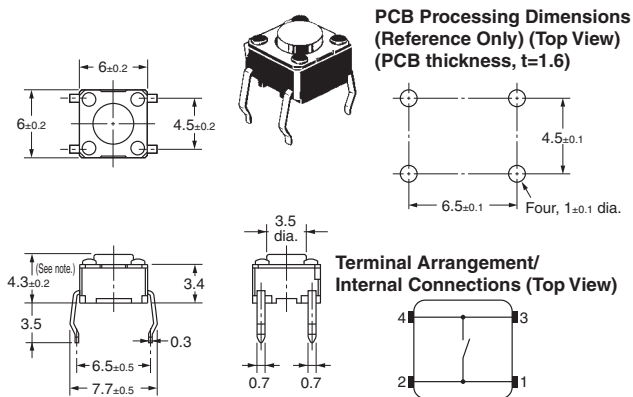
Note: The numbers used for terminals in the following graphics are indicated in the "Bottom View" diagram below. In this diagram, the Switch is rotated so that the terminals are on the right and left-hand sides, and the OMRON logo appears the right way up. (Except Side-operated and Radial Models)



6 × 6 mm Models

Standard, Flat Plunger Type (without Ground Terminal)

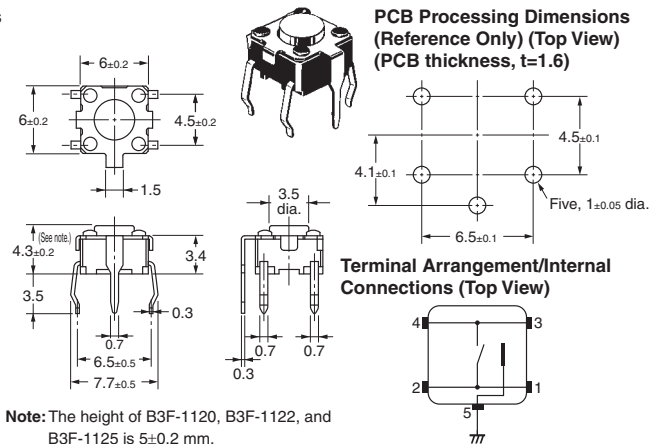
B3F-1000, B3F-1002, B3F-1005, B3F-1006
B3F-1020 (See note.), B3F-1022 (See note.),
B3F-1025 (See note.), B3F-1026 (See note.)
B3F-1002-G, B3F-1022-G (See note.)



Note: The height of B3F-1020, B3F-1022, B3F-1025, and B3F-1026 is 5 ± 0.2 mm.

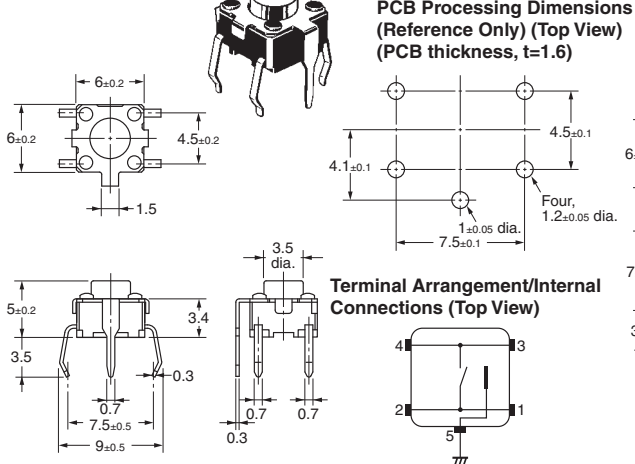
Standard, Flat Plunger Type (with Ground Terminal)

B3F-1100, B3F-1102, B3F-1105
B3F-1120 (See note.), B3F-1122 (See note.)
B3F-1125 (See note.)
B3F-1102-G, B3F-1122-G (See note.)



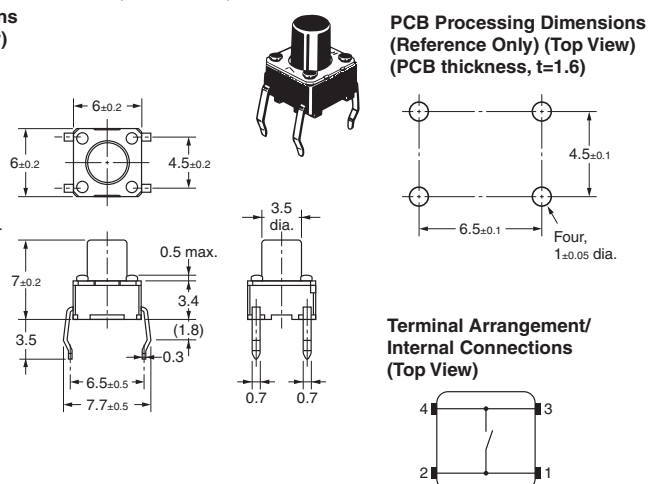
Standard, Flat Plunger Type (with Ground Terminal, Pitch: 7.5 mm)

B3F-1110



Standard, Flat Plunger Type (without Ground Terminal)

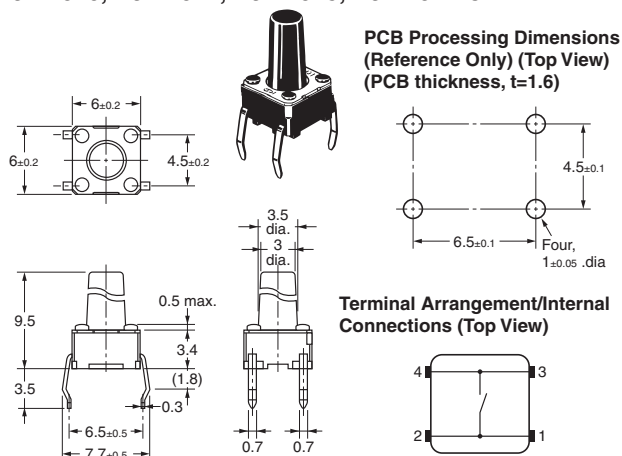
B3F-1060, B3F-1062, B3F-1062-G



Note: Unless otherwise specified, a tolerance of ±0.4 mm applies to all dimensions. No terminal numbers are indicated on the Switches.

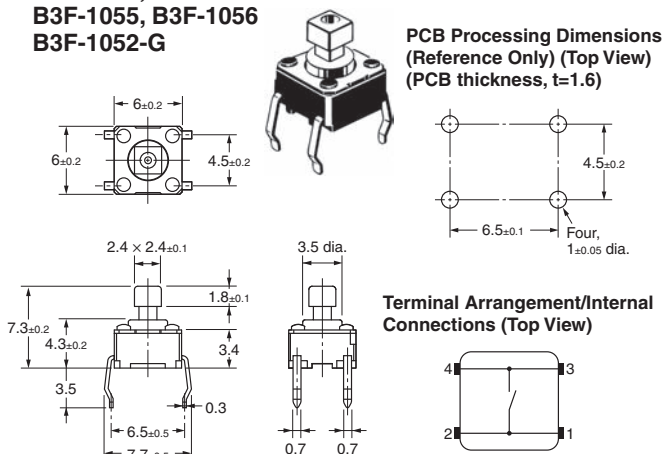
Standard, Flat Plunger Type (without Ground Terminal)

B3F-1070, B3F-1072, B3F-1075, B3F-1072-G



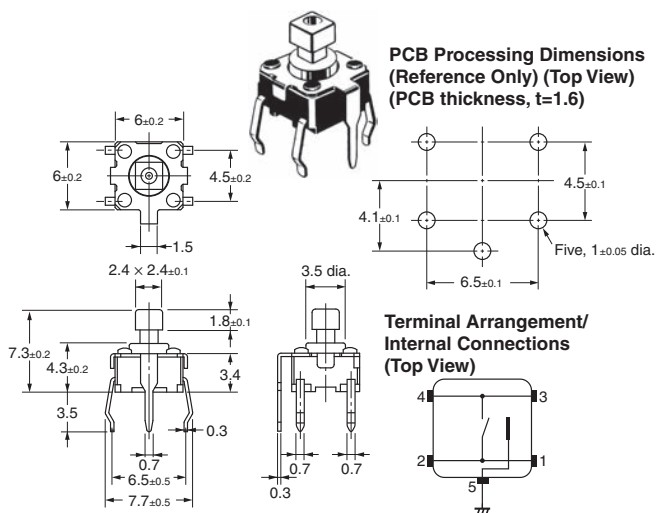
Standard, Projected Plunger Type (without Ground Terminal)

B3F-1050, B3F-1052
B3F-1055, B3F-1056
B3F-1052-G



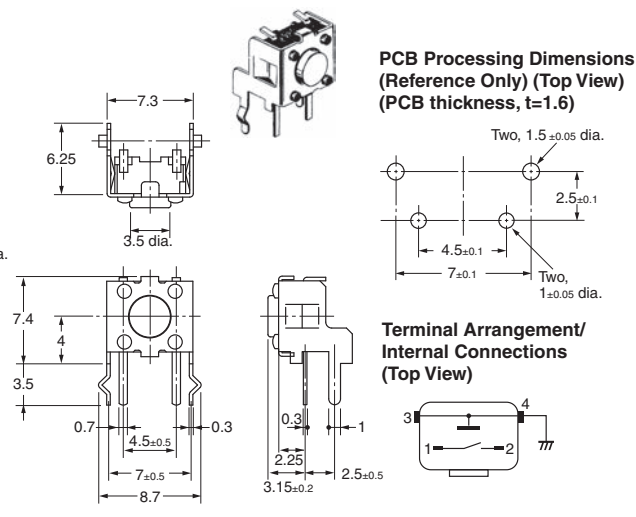
Standard, Projected Plunger Type (with Ground Terminal)

B3F-1150, B3F-1152, B3F-1155



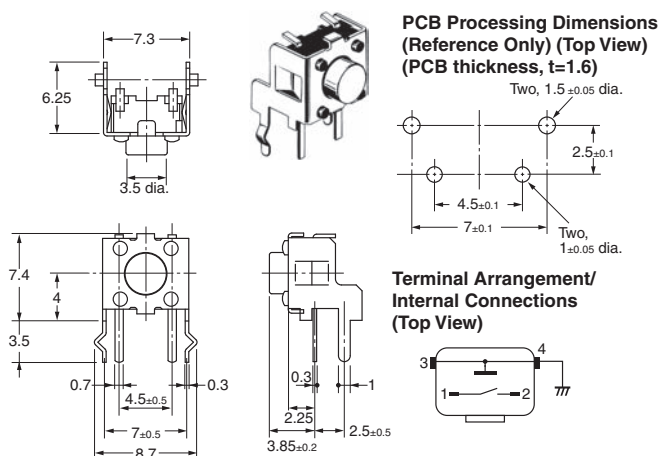
Side-operated, Flat Plunger Type

B3F-3100, B3F-3102, B3F-3105



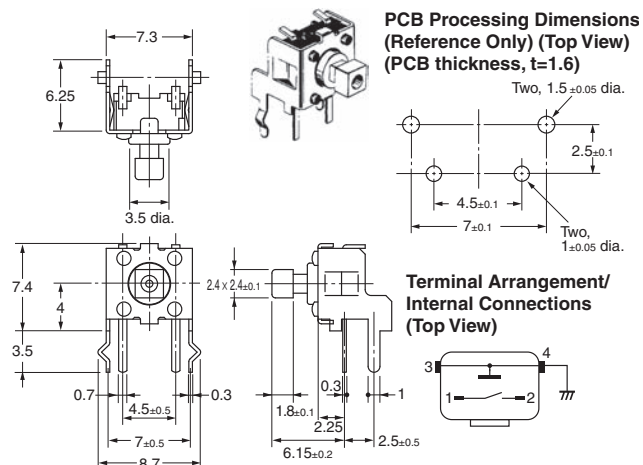
Side-operated, Flat Plunger Type (Height: 3.85 mm)

B3F-3120, B3F-3122, B3F-3125, B3F-3122-G



Side-operated, Projected Plunger Type

B3F-3150, B3F-3152, B3F-3155

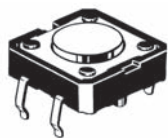


Note: Unless otherwise specified, a tolerance of ± 0.4 mm applies to all dimensions. No terminal numbers are indicated on the Switches.

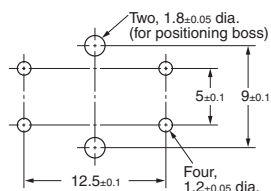
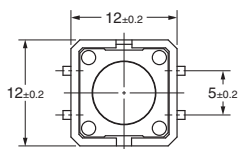
12 × 12 mm Models

Standard, Long-durability,
and High-reliability Models
Flat Plunger Type
(without Ground Terminal)

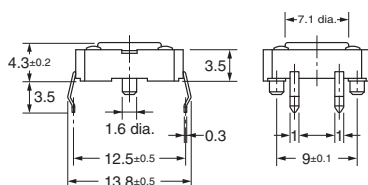
B3F-4000, B3F-4005,
B3F-5000, B3F-5001



PCB Processing Dimensions
(Reference Only) (Top View)
(PCB thickness, $t=1.6$)

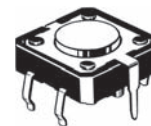


Terminal Arrangement/
Internal Connections
(Top View)

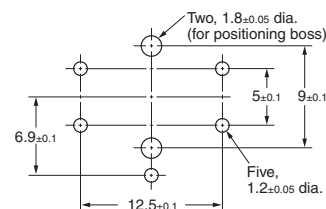
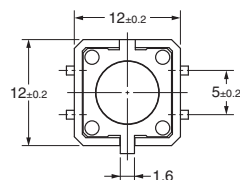


Standard, Long-durability,
and High-reliability Models
Flat Plunger Type
(with Ground Terminal)

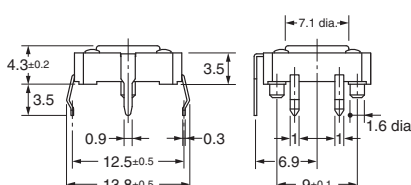
B3F-4100, B3F-4105,
B3F-5100, B3F-5101



PCB Processing Dimensions
(Reference Only) (Top View)
(PCB thickness, $t=1.6$)

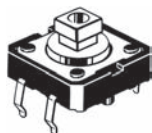


Terminal Arrangement/
Internal Connections
(Top View)

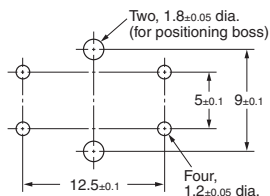
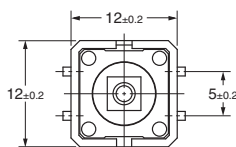


Standard, Long-durability,
and High-reliability Models
Projected Plunger Type
(without Ground Terminal)

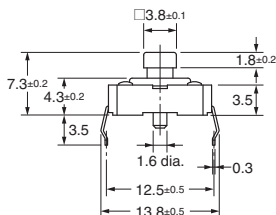
B3F-4050, B3F-4055,
B3F-5050, B3F-5051



PCB Processing Dimensions
(Reference Only) (Top View)
(PCB thickness, $t=1.6$)

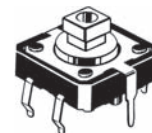


Terminal Arrangement/
Internal Connections
(Top View)

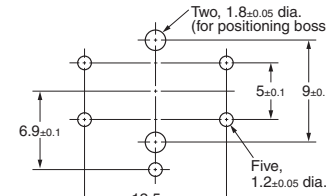
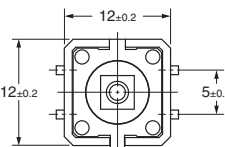


Standard, Long-durability,
and High-reliability Models
Projected Plunger Type
(with Ground Terminal)

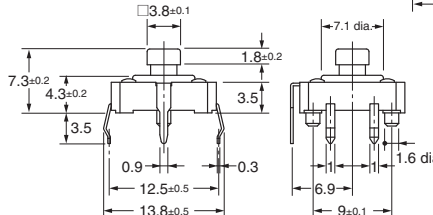
B3F-4150, B3F-4155,
B3F-5150, B3F-5151



PCB Processing Dimensions
(Reference Only) (Top View)
(PCB thickness, $t=1.6$)

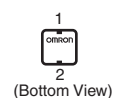


Terminal Arrangement/
Internal Connections
(Top View)



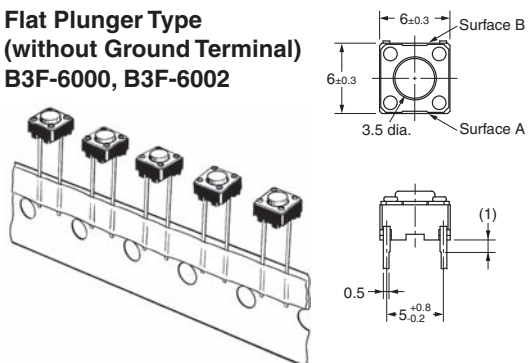
Note: Unless otherwise specified, a tolerance of ± 0.4 mm applies to all dimensions.

Note: The numbers used for terminals in the following graphics are indicated in the "Bottom View" diagram below. In this diagram, the Switch is rotated so that the terminals are on the right and left-hand sides, and the OMRON logo appears the right way up.



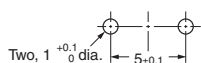
6 mm × 6 mm Radial Types (Taping Specifications): Sold in Units of 1,000 Switches

Flat Plunger Type (without Ground Terminal) B3F-6000, B3F-6002

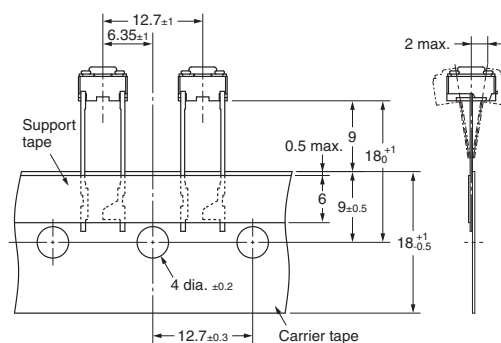
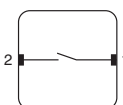


Note: The tape is random between surface A and surface B.

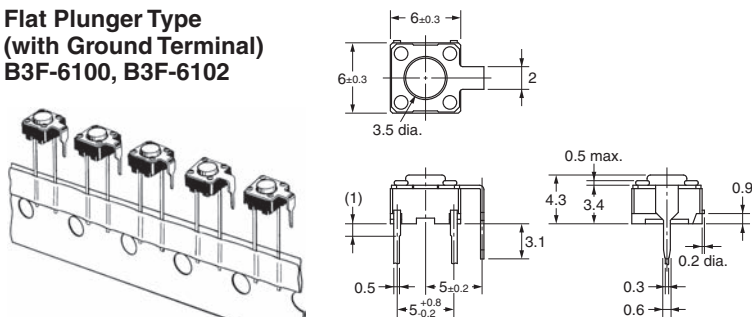
**PCB Processing Dimensions
(Reference Only) (Top View)
(PCB thickness, t=1.6)**



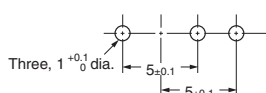
**Terminal Arrangement
/Internal Connections
(Top View)**



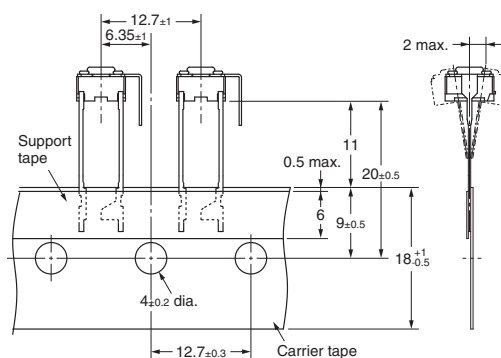
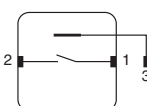
Flat Plunger Type (with Ground Terminal) B3F-6100, B3F-6102



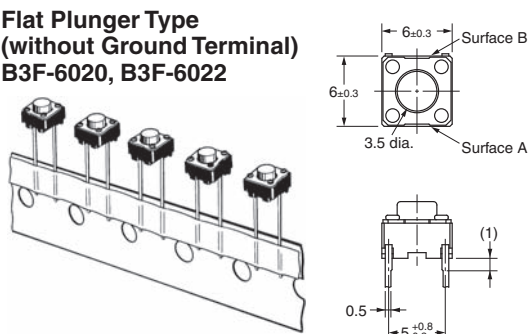
**PCB Processing Dimensions
(Reference Only) (Top View)
(PCB thickness, t=1.6)**



**Terminal Arrangement
/Internal Connections
(Top View)**



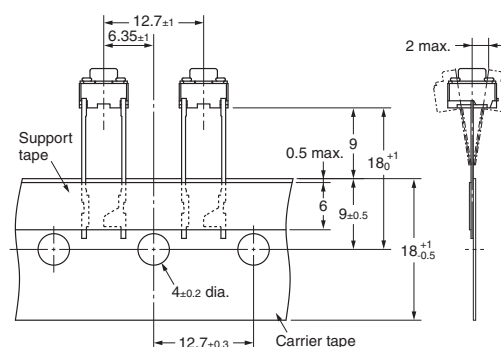
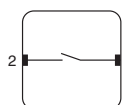
Flat Plunger Type (without Ground Terminal) B3F-6020, B3F-6022



**PCB Processing Dimensions
(Reference Only) (Top View)
(PCB thickness, t=1.6)**

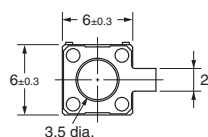
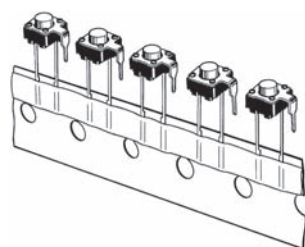


**Terminal Arrangement
/Internal Connections
(Top View)**

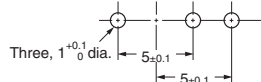


Note: Unless otherwise specified, a tolerance of ± 0.4 mm applies to all dimensions. No terminal numbers are indicated on the Switches.

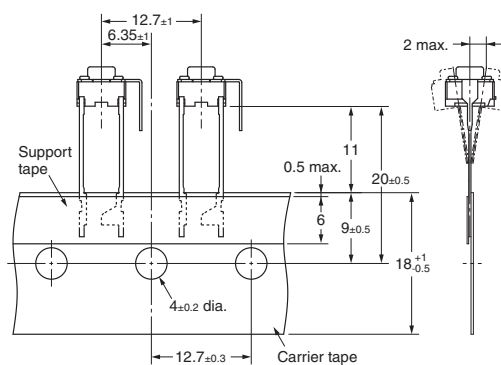
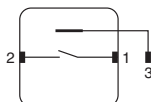
Flat Plunger Type (with Ground Terminal) B3F-6120, B3F-6122



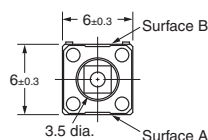
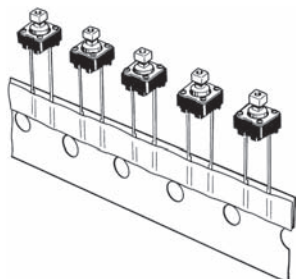
**PCB Processing Dimensions
(Reference Only) (Top View)**
(PCB thickness, $t=1.6$)



**Terminal Arrangement
/Internal Connections
(Top View)**



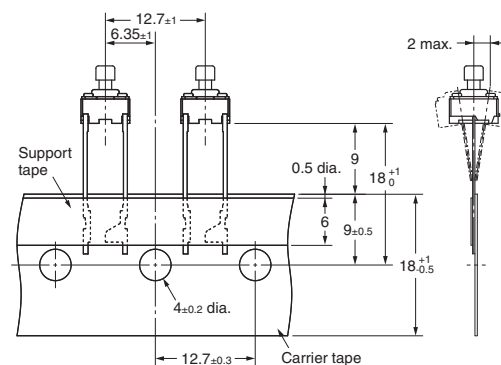
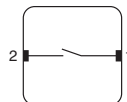
Projected Plunger Type (without Ground Terminal) B3F-6050, B3F-6052



**PCB Processing Dimensions
(Reference Only) (Top View)**
(PCB thickness, $t=1.6$)

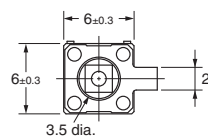
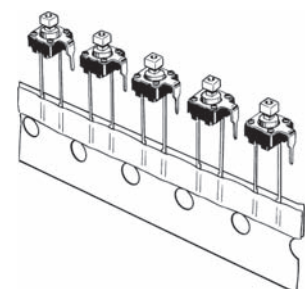


**Terminal Arrangement
/Internal Connections
(Top View)**

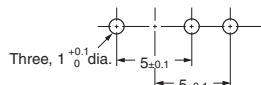


Note: The tape is random between surface A and surface B.

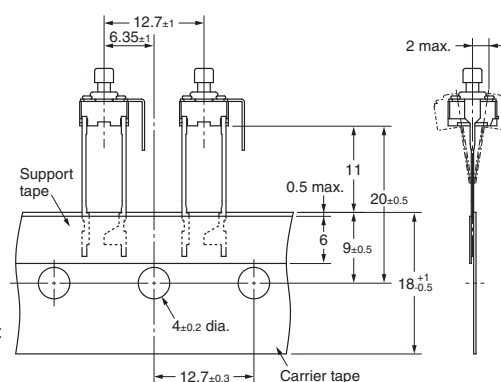
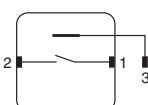
Projected Plunger Type (with Ground Terminal) B3F-6150, B3F-6152



**PCB Processing Dimensions
(Reference Only) (Top View)**
(PCB thickness, $t=1.6$)



**Terminal Arrangement
/Internal Connections
(Top View)**



Note: Unless otherwise specified, a tolerance of ± 0.4 mm applies to all dimensions. No terminal numbers are indicated on the Switches.

■ Key Tops

B32-series Key Tops are available for projected plungers. Refer to the Datasheet of B32 for details.

■ Precautions

Be sure to read the safety precautions common to all Tactile Switches for correct use.

- Application examples provided in this document are for reference only. In actual applications, confirm equipment functions and safety before using the product.
- Consult your OMRON representative before using the product under conditions which are not described in the manual or applying the product to nuclear control systems, railroad systems, aviation systems, vehicles, combustion systems, medical equipment, amusement machines, safety equipment, and other systems or equipment that may have a serious influence on lives and property if used improperly. Make sure that the ratings and performance characteristics of the product provide a margin of safety for the system or equipment, and be sure to provide the system or equipment with double safety mechanisms.

Note: Do not use this document to operate the Unit.

OMRON Corporation

Electronic and Mechanical Components Company

Contact: www.omron.com/ecb

Cat. No. A070-E1-08
1014(0207)(O)



Feature

1. Superior operational feeling free from gaps in operating directions
Ideal for finger-tip operation.
2. Integrated tact switch for selection contributes to the improvement of user-friendliness of the set.
3. Long Life Type Available.

FJN10K Series Joystick Potentiometer

Electrical characteristics

- Total Resistance.....5K Ω ~100K Ω
- Total Resistance Tolerance..... $\pm 20\%$
- Resistance Taper.....B taper
- Rated Power.....Taper B:0.0125W
- Withstand Voltage.....1 minute at AC 250V
- Insulation Resistance.....More than 100M Ω at DC 250V for 1 minute
- Max. Operating Voltage.....Taper B:AC 50V or DC 5.0V

Mechanical characteristics

- Figure of Lever Operation.....Circular Operating
- Stopper Strength3.1Kgf.cm Min 3Sec.
- Operating Angle of Lever..... $60^\circ \pm 6^\circ$
- Push-Pull Strength.....Push/Pull:5.0Kgf Min 3Sec.
- Operating Force.....160 $\pm$ 40gf.cm
- Accuracy of Rest Position..... $\pm 5^\circ$

Durability

- Operating Life.....Standard Life: 1Million Cycles min
Long Life: 2 Million Cycles min
- Operating Temperature Range..... $-10^\circ\text{C} \sim +70^\circ\text{C}$

Tact Switch specification

- Ratings.....12 V DC, 50 mA
- Operating Force.....820 $\pm$ 260gf
- Travel.....0.5+0.5/-0.4mm
- Operating Life.....100,000Cycles min

HOW TO ORDER

FJN10K-	N	1	B	100K	29	8
---------	---	---	---	------	----	---

Model Series	
Code	Switch Option
N	No Switch
S	With Tact Switch

Code	Shaft Length
1	L7.2

Total Resistance	
5K	
10K	
20K	
50K	
100K	

Electric Effective Angle	
See Note1	

Code	Operation Life
(Blank)	Std. Life
8	Long Life

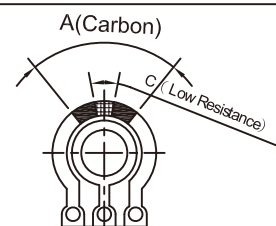
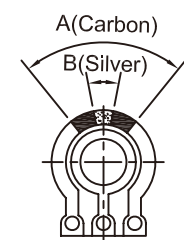
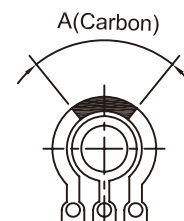
Code	Resistance Taper
B	Linear (0B)

Note 1: Electronic Effective Angle

ANGLE A	Code	ANGLE A	Code	ANGLE A	Code	ANGLE A	Code
15°	e5	45°	B5				
20°	d0	50°	C0				
25°	d5	53°	C3				
28°	d8	55°	C5				
30°	A0	60°	D0				
32°	A2	65°	D5				
35°	A5						
38°	A8						
40°	B0						

ANGLE A	Code	ANGLE A	Code	ANGLE A	Code	ANGLE A	Code
60°	01	30°	11	40°	20	45°	48
38°	03	50°	12	25°	22	32°	51
38°	04	85°	13	35°	26	20°	52
50°	05	100°	14	40°	30	45°	55
53°	06	80°	15	147°	32	48° ⁺⁰ ₋₃	56
45°	07	60°	16	70°	33	45°	73
32°	08	25°	17	28°	34	46°	74
53°	09	50°	18	35°	35		
45°	10	45°	19	42°	44		

ANGLE A	Code	ANGLE A	Code	ANGLE A	Code	ANGLE A	Code
45°	25	35°	38	45°	54	53°	64
38°	27	50°	42	35°	60		
45°	28	40°	43	54°	59		
60°	29	53°	47	60°	66		
40°	31	30°	53	32°	71		



碳膜层
(Carbon)



硫化层
(Anti-Sulfurated)

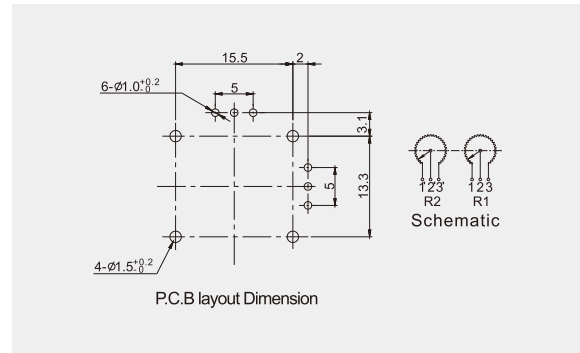
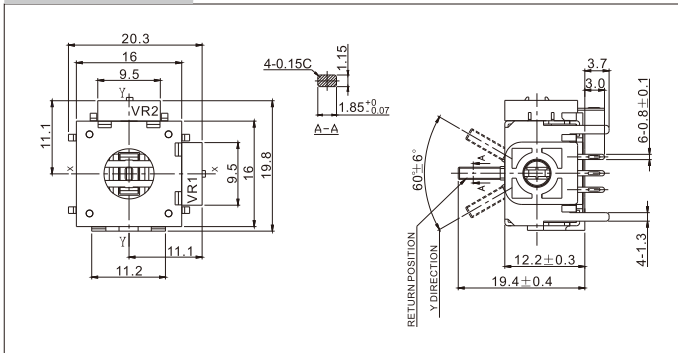


银层
(Silver)

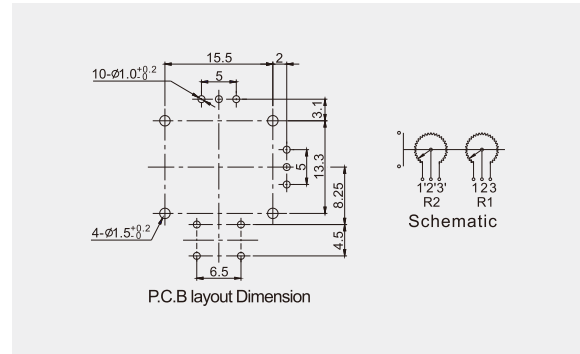
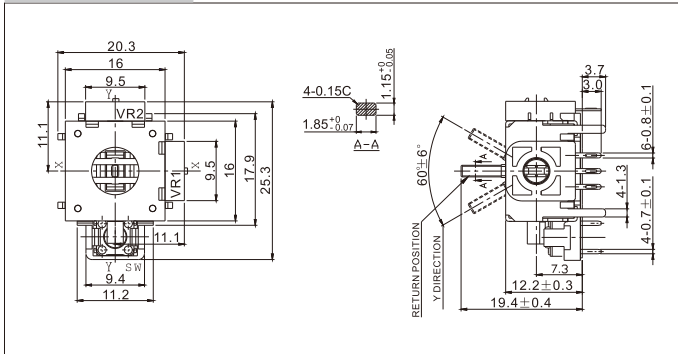


低碳层
(Low Resistance)

FJN10K-N1 Metal Shaft, No Switch, PC Pins



FJN10K-S1 Metal Shaft with Tact Switch, PC Pins



Cree® PLCC6 3-in-1 SMD LED CLP6C-FKB



PRODUCT DESCRIPTION

Cree PLCC full-color LEDs offer high-intensity light output and a wide viewing angle in an industry-standard package. Designed to work in a wide array of environmental conditions, Cree PLCC full-color LEDs are suited for indoor video screen, decorative lighting and amusement applications.

FEATURES

- Size (mm):6.0 x 5.0
- Dominant Wavelength:
Red (619 - 624nm)
Green (520 - 540nm)
Blue (460 - 480nm)
- Luminous Intensity (mcd)
Red (560 - 1120)
Green (1120 - 2240)
Blue (280 - 560)
- Lead-Free
- RoHS Compliant

APPLICATIONS

- Full-Color Video Screen
- Decorative lighting
- Amusement

ABSOLUTE MAXIMUM RATINGS ($T_A = 25^\circ\text{C}$)

Items	Symbol	Absolute Maximum Rating			Unit
		R	G	B	
Forward Current ^{Note 1}	I_F	50	50	50	mA
Peak Forward Current ^{Note 2}	I_{FP}	200	100	100	mA
Reverse Voltage	V_R	5	5	5	V
Power Dissipation	P_D	130	200	200	mW
Operation Temperature	T_{opr}	-40 ~ +100			$^\circ\text{C}$
Storage Temperature	T_{stg}	-40 ~ +100			$^\circ\text{C}$
Junction Temperature	T_J	110	110	110	$^\circ\text{C}$
Junction/ambient 1 chip on	R_{THJA}	450	400	450	$^\circ\text{C/W}$
Junction/ambient 3 chips on	R_{THJA}	650	580	680	$^\circ\text{C/W}$
Junction/solder point 1 chip on	R_{THJS}	300	280	300	$^\circ\text{C/W}$
Junction/solder point 3 chips on	R_{THJS}	450	430	480	$^\circ\text{C/W}$
Electrostatic Discharge Classification(MIL-STD-883E)	ESD	1000 V			

Note: 1.Single-color light.
2.Pulse width ≤ 0.1 msec, duty $\leq 1/10$.

TYPICAL ELECTRICAL & OPTICAL CHARACTERISTICS ($T_A = 25^\circ\text{C}$)

Characteristics	Condition	Symbol	Values			Unit
			R	G	B	
Dominant Wavelength	$I_F = 20$ mA	λ_{DOM}	619~624	520~540	460~480	nm
Spectral bandwidth at 50% I_{REL} max	$I_F = 20$ mA	$\Delta \lambda$	24	38	28	nm
Forward Voltage	$I_F = 20$ mA	$V_{F(avg)}$	2.0	3.2	3.2	V
		$V_{F(max)}$	2.6	4.0	4.0	V
Luminous Intensity	$I_F = 20$ mA	$I_{V(min)}$	560	1120	280	mcd
		$I_{V(avg)}$	700	1600	400	mcd
Reverse Current (max)	$V_R = 5$ V	I_R	10	10	10	μA

INTENSITY BIN LIMIT ($I_F = 20 \text{ mA}$)

Red			Green			Blue		
Bin Code	Min.(mcd)	Max.(mcd)	Bin Code	Min.(mcd)	Max.(mcd)	Bin Code	Min.(mcd)	Max.(mcd)
K	560	710	P	1120	1400	G	280	355
M	710	900	Q	1400	1800	H	355	450
N	900	1120	R	1800	2240	J	450	560

Tolerance of measurement of luminous intensity is $\pm 10\%$.

COLOR BIN LIMIT ($I_F = 20 \text{ mA}$)

Red			Green			Blue		
Bin Code	Min.(nm)	Max.(nm)	Bin Code	Min.(nm)	Max.(nm)	Bin Code	Min.(nm)	Max.(nm)
RB	619	624	G7	520	525	B3	460	465
			G8	525	530	B4	465	470
			G9	530	535	B5	470	475
			Ga	535	540	B6	475	480

Tolerance of measurement of dominant wavelength is $\pm 1 \text{ nm}$.

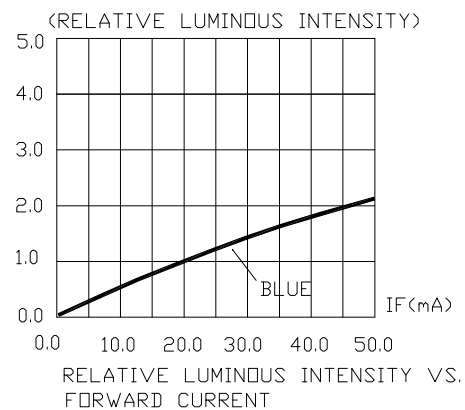
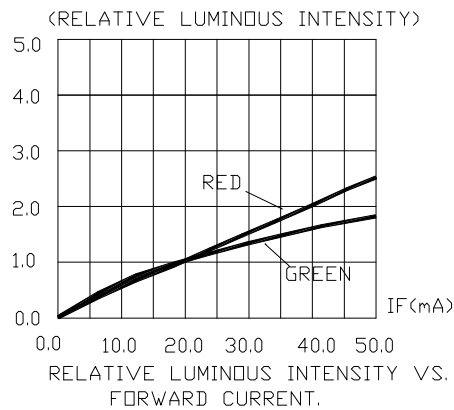
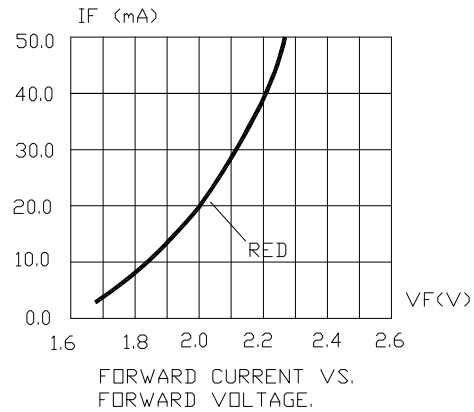
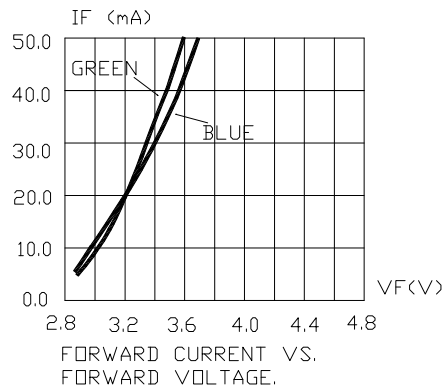
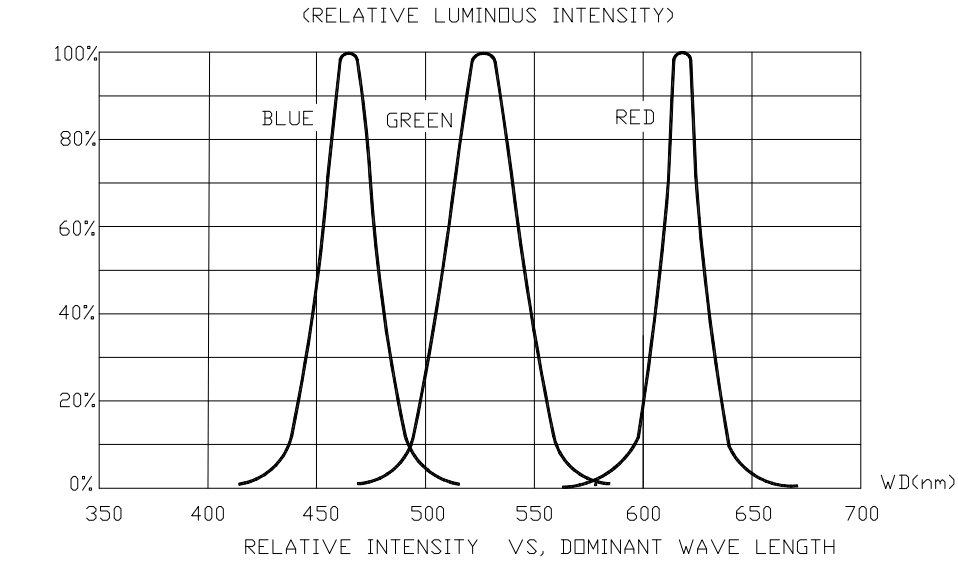
ORDER CODE TABLE*

Kit Number	Color	Luminous Intensity (mcd)		Dominant Wavelength (nm)				Package
		Min.	Max.	Color Bin	Min. (nm)	Color Bin	Max. (nm)	
CLP6C-FKB-CKNPRGJBB7a363	Red	560	1120	RB	619	RB	624	Reel
	Green	1120	2240	G7	520	Ga	540	Reel
	Blue	280	560	B3	460	B6	480	Reel
CLP6C-FKB-CK1P1G1BB7R3R3	Red	Any 1 Intensity bin from K(560) - N(1120)		RB	619	RB	624	Reel
	Green	Any 1 Intensity bin from P(1120) - R(2240)		Any 1 hue bin from G7(520) - Ga(540)				Reel
	Blue	Any 1 Intensity bin from G(280) - J(560)		Any 1 hue bin from B3(460) - B6(480)				Reel
CLP6C-FKB-CM1Q1H1BB7R3R3	Red	Any 1 Intensity bin from M(710) - N(1120)		RB	619	RB	624	Reel
	Green	Any 1 Intensity bin from Q(1400) - R(2240)		Any 1 hue bin from G7(520) - Ga(540)				Reel
	Blue	Any 1 Intensity bin from H(355) - J(560)		Any 1 hue bin from B3(460) - B6(480)				Reel

Notes:

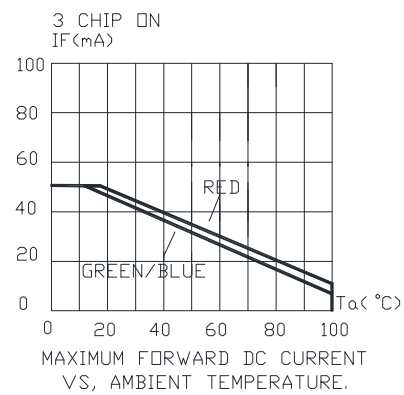
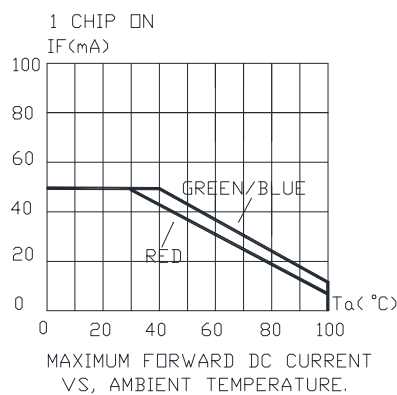
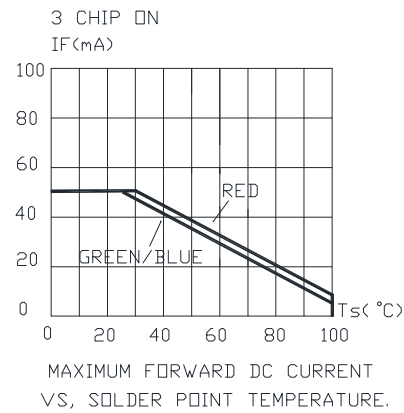
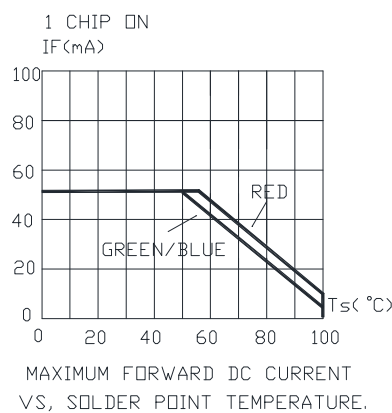
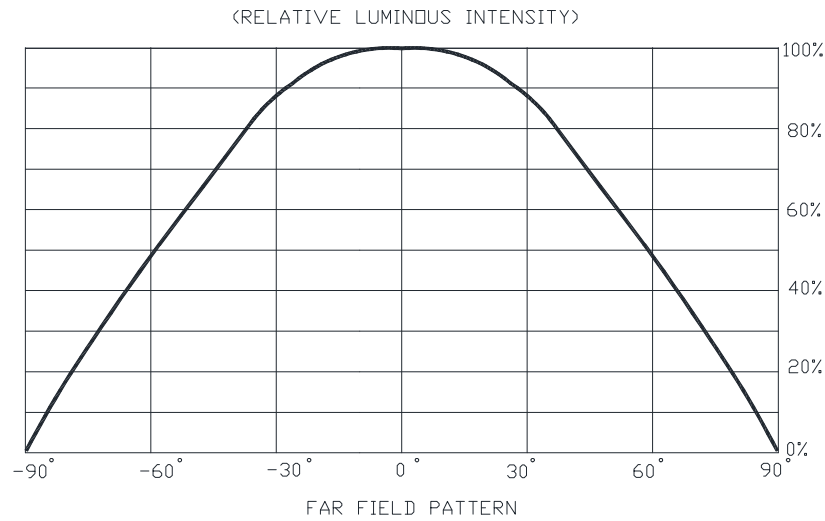
1. The above kit numbers represent the order codes which include multiple intensity-bin and color-bin codes. Only one intensity-bin code and one color-bin code will be shipped on each reel. Single intensity-bin code and single color bin code will be orderable in certain quantities. For example, any 1 intensity bin from K - N means only 1 intensity bin (K or M or N) will be shipped by Cree. For example, any 1 color bin from G7 - Ga means only 1 color bin (G7 or G8 or G9 or Ga) will be shipped by Cree.
2. Please refer to the "Cree LED Lamp Reliability Test Standards" document for reliability test conditions.
3. Please refer to the "Cree LED Lamp Soldering & Handling" document for information about how to use this LED product safely.

GRAPHS



The above data are collected from statistical figures that do not necessarily correspond to the actual parameters of each single LED. Hence, these data will be changed without further notice.

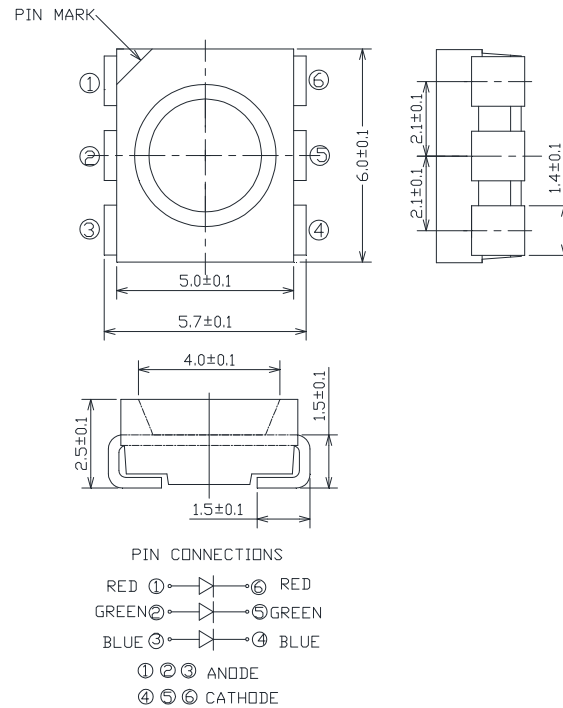
GRAPHS



The above data are collected from statistical figures that do not necessarily correspond to the actual parameters of each single LED. Hence, these data will be changed without further notice.

MECHANICAL DIMENSIONS

All dimensions are in mm.



NOTES

RoHS Compliance

The levels of environmentally sensitive, persistent biologically toxic (PBT), persistent organic pollutants (POP), or otherwise restricted materials in this product are below the maximum concentration values (also referred to as the threshold limits) permitted for such substances, or are used in an exempted application, in accordance with EU Directive 2002/95/EC on the restriction of the use of certain hazardous substances in electrical and electronic equipment (RoHS), as amended through April 21, 2006.

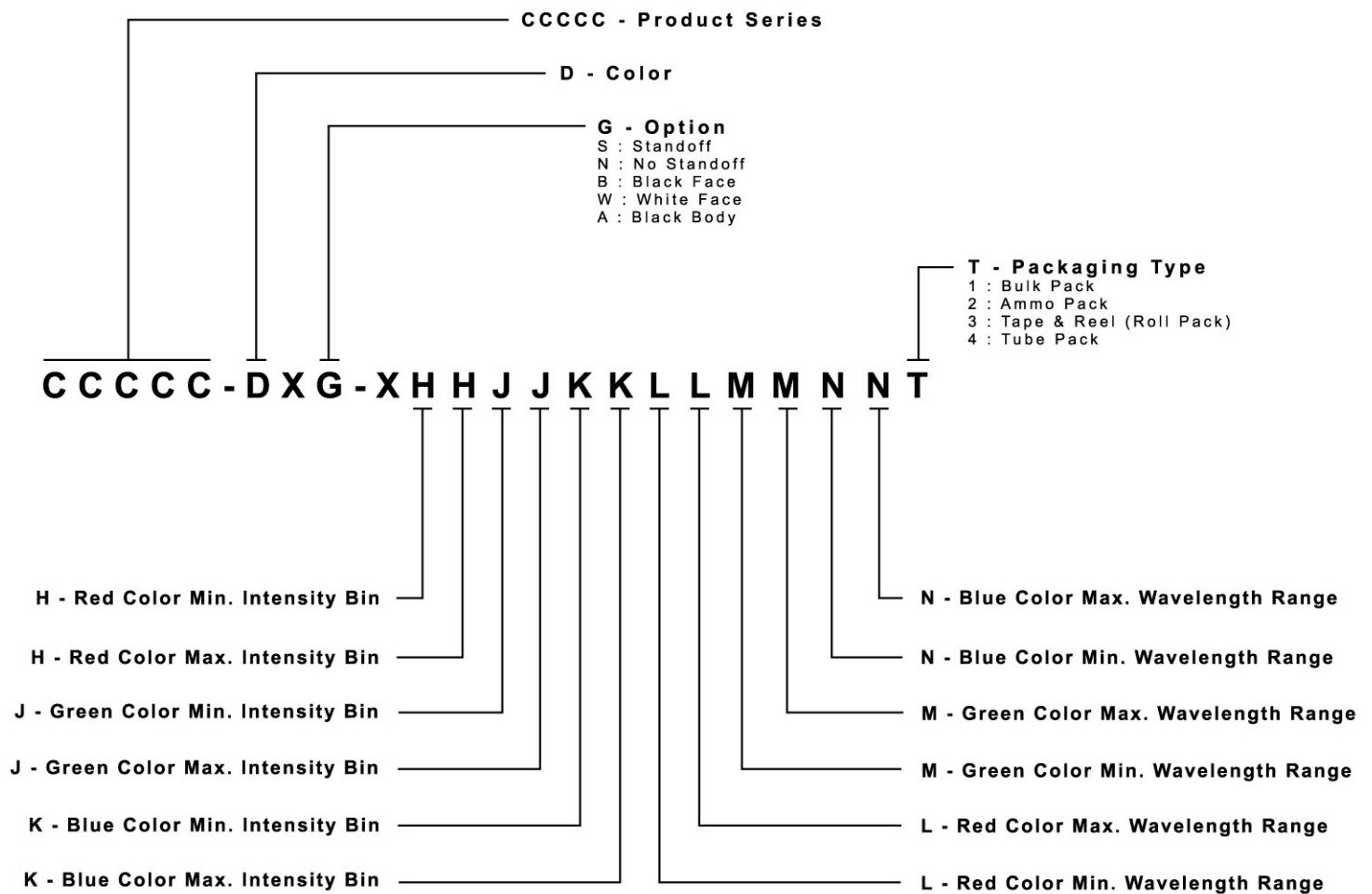
Vision Advisory Claim

Users should be cautioned not to stare at the light of this LED product. The bright light can damage the eye.

KIT NUMBER SYSTEM

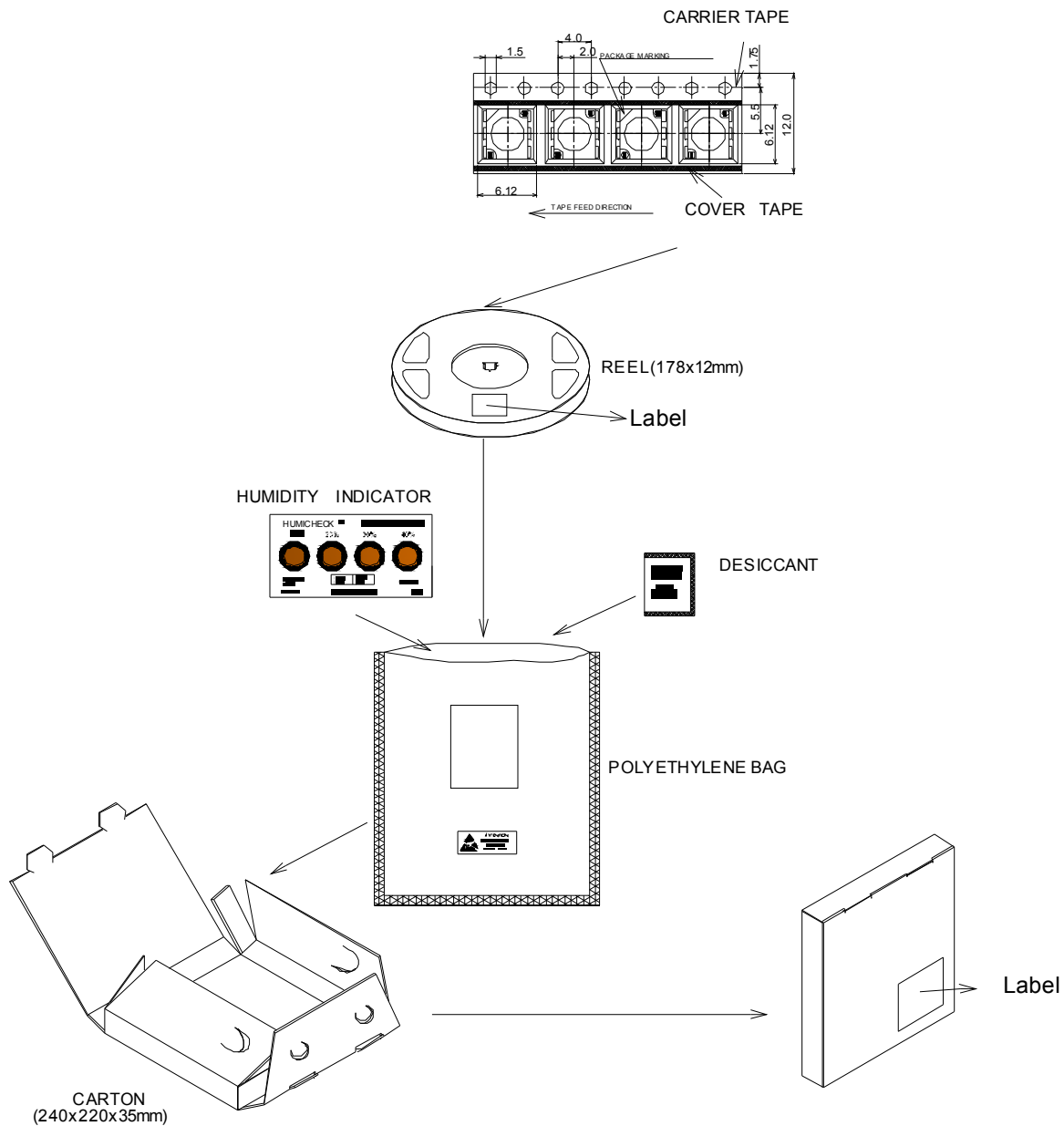
Cree LED lamps are tested and sorted into performance bins. A bin is specified by ranges of color, forward voltage, and brightness. Sorted LEDs are packaged for shipping in various convenient options. Please refer to the "Cree LED Lamp Packaging Standard" document for more information about shipping and packaging options.

Cree LEDs are sold by order codes in combinations of bins called kits. Order codes are configured in the following manner:



PACKAGING

- The boxes are not water resistant and they must be kept away from water and moisture.
- The LEDs are packed in cardboard boxes after packaging in normal or anti-electrostatic bags.
- Cardboard boxes will be used to protect the LEDs from mechanical shocks during transportation.
- The reel pack is applied in SMD LED.
- Max 900 pcs per reel.



F/CM12P

Characteristics *Kennwerte*

Condition/Remark *Bedingungen/Kommentar*

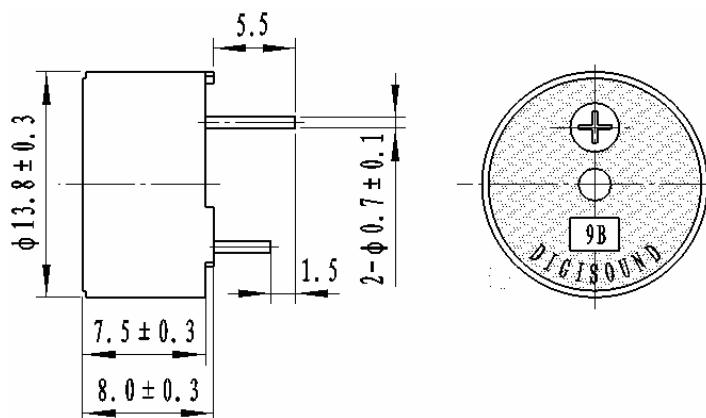
Value *Wert*

Dimension <i>Abmessungen</i>	ø x H	13,8 x 7,5 mm
Rated voltage <i>Nennspannung</i>	DC	12 V
Operating voltage <i>Betriebsspannung</i>	DC	3...16 V
Rated current <i>Nennstrom</i>	Note 1	≤ 12 mA
Sound pressure level <i>Lautstärke</i>	Note 1	85 dB
Resonant frequency <i>Resonanzfrequenz</i>	+ 10%	4k±500 Hz
Operating temperature <i>Betriebstemperatur</i>		-20...+70°C
Storage temperature <i>Lagertemperatur</i>		-30...+80°C
Strength of Pin <i>Pin-Festigkeit</i>		1 kg
Weight <i>Gewicht</i>		0,8g
ROHS compliant		Yes
Housing material <i>Gehäusematerial</i>		PBT
Polarity marking	“+”-sign at the top of housing	
<i>Polaritätskennzeichnung</i>	„+“-Zeichen auf dem Gehäuse	

Note 1: Value applying rated voltage (DC)

DIMENSIONS (UNIT: mm)

Tolerance: ±0.5mm Except Specified



Halbzeug, Werkstoff

Untolerierte Maße

Zeichn.-Nr.

090924.1H

Maßstab

Ersatz f.
Zeichn.

	Tag	Name	And. zust.	And.-Mittlg. Nr.	Tag	Name
gezeichnet						
bearbeitet	24.09.09	KH				
geprüft						
normgepr.						

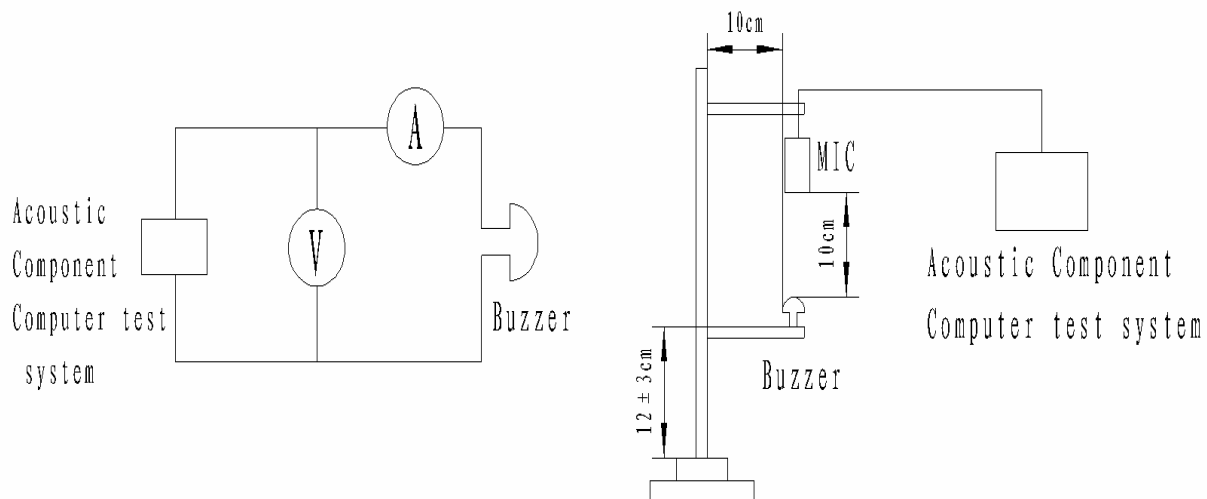
F/CM12P

Diese Zeichnung ist unser Eigentum. Vervielfältigung
unbefugte Verwertung, Mitteilung an andere ist
strafbar und Schadensersatzpflichtig

Vervielfält.-Pause
Nr.

Arbeitspause Nr.

TEST METHOD:

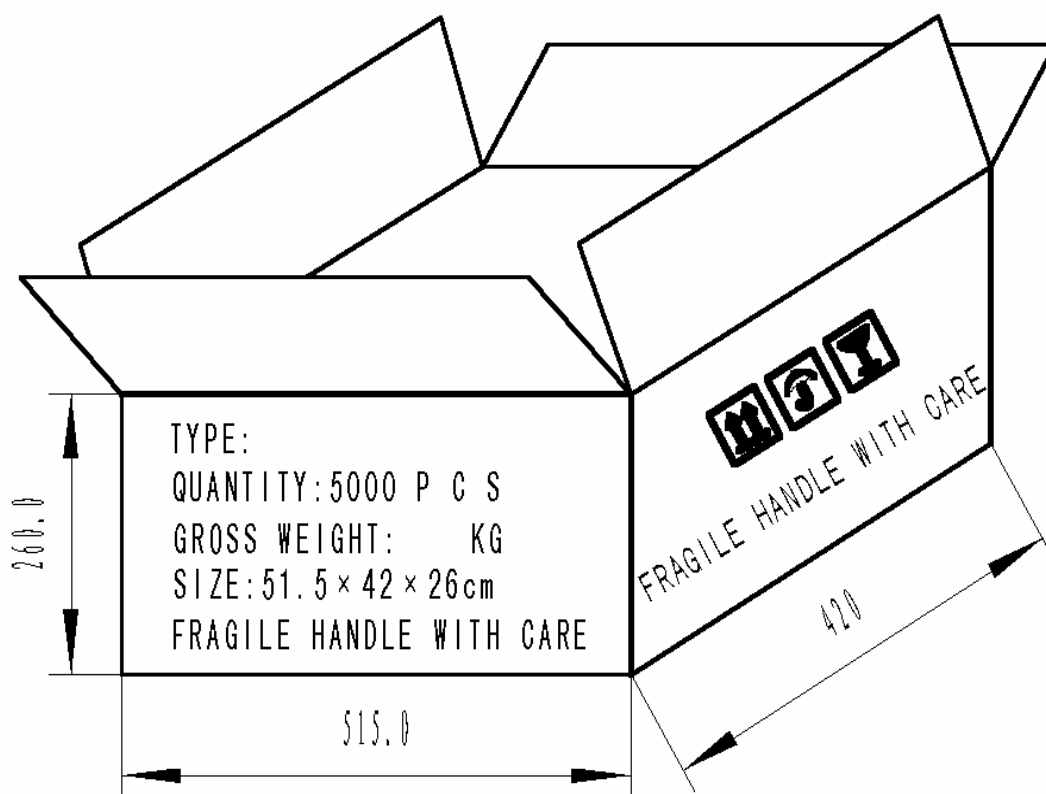


RELIABLY TEST:

NO.	ITEM	TESTING CONDITION	VARIANCE AFTER TEST
1	Humidity	50±5°C, 93(+2/-3)%RH, 96HRS	All specifications must be satisfied after the test.
2	<u>High temp.</u>	+80±2°C, 96HRS	
3	Low temp.	-30±2°C, 96HRS	
4	Temperature cycling	-30±2°C, 30minutes room temp. 15minutes +80±2°C, 30minutes room temp. 15minutes 5 cycles	
5	Drop test	3 times from height of 70cm onto the surface of 10mm thick wooden board.	
6	Vibration test	Make the test for the directions of X Y and Z (total 0.5 hours). To-and-fro. sweep time (from 10 to 55 Hz and then 55 to 10) under single amplitude of 10mm is 3minute.	
7	Solder heat resistance	The part leads (pins) shall be immersed in molten solder maintained at 250±10°C for a period of 30 seconds.	After the test part shall meet specifications without any degradation in appearance and performance.

		Halbzeug, Werkstoff		Untolerierte Maße		Zeichn.-Nr. 090924.1H	
				Maßstab			
Vervielfält.-Pause Nr.		Ersatz f. Zeichn.					
	Tag	Name	And. zust.	And.-Mittlg. Nr.	Tag	Name	F/CM12P
gezeichnet							
bearbeitet	24.09.09	KH					
geprüft							
normgepr.							

PACKING:



NOTES:

1. 100 PCS per box
2. Total 50 boxes per carton
3. Total 5000 PCS carton
4. Volume: 51.5 x 42 x 26 cm

Diese Zeichnung ist unser Eigentum. Vervielfältigung unbefugte Verwertung, Mitteilung an andere ist strafbar und Schadensersatzpflichtig

		Halbzeug, Werkstoff				Untolerierte Maße		Zeichn.-Nr.	
						Maßstab		090924.1H	
Vervielfält.-Pause Nr.								Ersatz f. Zeichn.	
	Tag	Name	And. zust.	And.-Mittlg. Nr.	Tag	Name	F/CM12P		
gezeichnet									
bearbeitet	24.09.09	KH							
geprüft									
Arbeitspause Nr.									

±1.5g, ±6g Three Axis Low-g Micromachined Accelerometer

The MMA7361L is a low power, low profile capacitive micromachined accelerometer featuring signal conditioning, a 1-pole low pass filter, temperature compensation, self test, 0g-Detect which detects linear freefall, and g-Select which allows for the selection between 2 sensitivities. Zero-g offset and sensitivity are factory set and require no external devices. The MMA7361L includes a Sleep Mode that makes it ideal for handheld battery powered electronics.

Features

- 3mm x 5mm x 1.0mm LGA-14 Package
- Low Current Consumption: 400 μ A
- Sleep Mode: 3 μ A
- Low Voltage Operation: 2.2 V – 3.6 V
- High Sensitivity (800 mV/g @ 1.5g)
- Selectable Sensitivity (± 1.5 g, ± 6 g)
- Fast Turn On Time (0.5 ms Enable Response Time)
- Self Test for Freefall Detect Diagnosis
- 0g-Detect for Freefall Protection
- Signal Conditioning with Low Pass Filter
- Robust Design, High Shocks Survivability
- RoHS Compliant
- Environmentally Preferred Product
- Low Cost

Typical Applications

- 3D Gaming: Tilt and Motion Sensing, Event Recorder
- HDD MP3 Player: Freefall Detection
- Laptop PC: Freefall Detection, Anti-Theft
- Cell Phone: Image Stability, Text Scroll, Motion Dialing, E-Compass
- Pedometer: Motion Sensing
- PDA: Text Scroll
- Navigation and Dead Reckoning: E-Compass Tilt Compensation
- Robotics: Motion Sensing

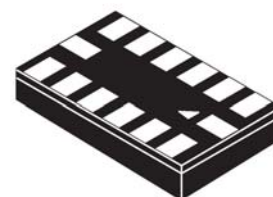
ORDERING INFORMATION

Part Number	Temperature Range	Package Drawing	Package	Shipping
MMA7361LT	-40 to +85°C	1977-01	LGA-14	Tray
MMA7361LR1	-40 to +85°C	1977-01	LGA-14	7" Tape & Reel
MMA7361LR2	-40 to +85°C	1977-01	LGA-14	13" Tape & Reel

MMA7361L

**MMA7361L: XYZ AXIS
ACCELEROMETER**
 ± 1.5 g, ± 6 g

Bottom View



**14 LEAD
LGA
CASE 1977-01**

Top View

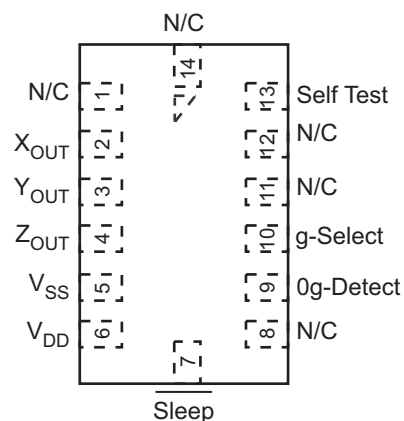


Figure 1. Pin Connections

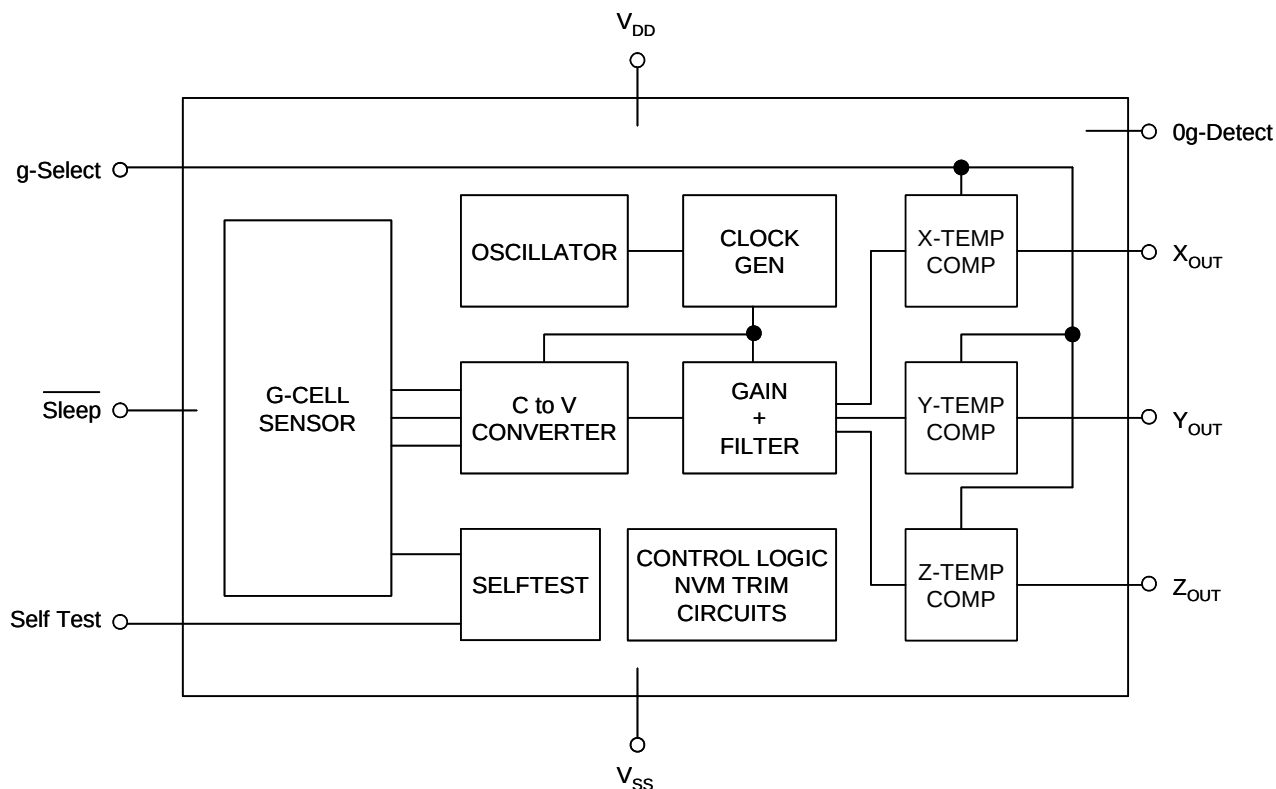


Figure 2. Simplified Accelerometer Functional Block Diagram

Table 1. Maximum Ratings

(Maximum ratings are the limits to which the device can be exposed without causing permanent damage.)

Rating	Symbol	Value	Unit
Maximum Acceleration (all axis)	g_{max}	± 5000	g
Supply Voltage	V_{DD}	-0.3 to +3.6	V
Drop Test ⁽¹⁾	D_{drop}	1.8	m
Storage Temperature Range	T_{stg}	-40 to +125	°C

1. Dropped onto concrete surface from any axis.

ELECTRO STATIC DISCHARGE (ESD)

WARNING: This device is sensitive to electrostatic discharge.

Although the Freescale accelerometer contains internal 2000 V ESD protection circuitry, extra precaution must be taken by the user to protect the chip from ESD. A charge of over 2000 volts can accumulate on the human body or associated test equipment. A charge of this magnitude can

alter the performance or cause failure of the chip. When handling the accelerometer, proper ESD precautions should be followed to avoid exposing the device to discharges which may be detrimental to its performance.

Table 2. Operating Characteristics

Unless otherwise noted: $-40^{\circ}\text{C} \leq T_A \leq 85^{\circ}\text{C}$, $2.2\text{ V} \leq V_{DD} \leq 3.6\text{ V}$, Acceleration = 0g, Loaded output⁽¹⁾

Characteristic	Symbol	Min	Typ	Max	Unit
Operating Range ⁽²⁾					
Supply Voltage ⁽³⁾	V_{DD}	2.2	3.3	3.6	V
Supply Current ⁽⁴⁾	I_{DD}	—	400	600	μA
Supply Current at Sleep Mode ⁽⁴⁾	I_{DD}	—	3	10	μA
Operating Temperature Range	T_A	-40	—	+85	$^{\circ}\text{C}$
Acceleration Range, X-Axis, Y-Axis, Z-Axis					
g-Select: 0	g_{FS}	—	± 1.5	—	g
g-Select: 1	g_{FS}	—	± 6.0	—	g
Output Signal					
Zero-g ($T_A = 25^{\circ}\text{C}$, $V_{DD} = 3.3\text{ V}$) ^{(5), (6)}	V_{OFF}	1.485	1.65	1.815	V
Zero-g ⁽⁴⁾	V_{OFF}, T_A	-2.0	± 0.5	+2.0	$\text{mg}/^{\circ}\text{C}$
Sensitivity ($T_A = 25^{\circ}\text{C}$, $V_{DD} = 3.3\text{ V}$)					
1.5g	$S_{1.5g}$	740	800	860	mV/g
6g	S_{6g}	190.6	206	221.5	mV/g
Sensitivity ⁽⁴⁾	S, T_A	-0.0075	± 0.002	+0.0075	$\%/^{\circ}\text{C}$
Bandwidth Response					
XY	f_{-3dBXY}	—	400	—	Hz
Z	f_{-3dBZ}	—	300	—	Hz
Output Impedance	Z_O	—	32	—	$\text{k}\Omega$
0g-Detect	$0g_{detect}$	-0.4	0	+0.4	g
Self Test					
Output Response					
X_{OUT}, Y_{OUT}	Δg_{STXY}	+0.05	-0.1	—	g
Z_{OUT}	Δg_{STZ}	+0.8	+1.0	+1.2	g
Input Low	V_{IL}	V_{SS}	—	$0.3 V_{DD}$	V
Input High	V_{IH}	$0.7 V_{DD}$	—	V_{DD}	V
Noise					
Power Spectral Density RMS (0.1 Hz – 1 kHz) ⁽⁴⁾	n_{PSD}	—	350	—	$\mu\text{g}/\sqrt{\text{Hz}}$
Control Timing					
Power-Up Response Time ⁽⁷⁾	$t_{RESPONSE}$	—	1.0	2.0	ms
Enable Response Time ⁽⁸⁾	t_{ENABLE}	—	0.5	2.0	ms
Self Test Response Time ⁽⁹⁾	t_{ST}	—	2.0	5.0	ms
Sensing Element Resonant Frequency					
XY	$f_{GCELLXY}$	—	6.0	—	kHz
Z	f_{GCELLZ}	—	3.4	—	kHz
Internal Sampling Frequency	f_{CLK}	—	11	—	kHz
Output Stage Performance					
Full-Scale Output Range ($I_{OUT} = 3\text{ }\mu\text{A}$)	V_{FSO}	$V_{SS}+0.1$	—	$V_{DD}-0.1$	V
Nonlinearity, $X_{OUT}, Y_{OUT}, Z_{OUT}$	NL_{OUT}	-1.0	—	+1.0	%FSO
Cross-Axis Sensitivity ⁽¹⁰⁾	$V_{XY, XZ, YZ}$	-5.0	—	+5.0	%

- For a loaded output, the measurements are observed after an RC filter consisting of an internal 32k Ω resistor and an external 3.3nF capacitor (recommended as a minimum to filter clock noise) on the analog output for each axis and a 0.1 μF capacitor on V_{DD} - GND. The output sensor bandwidth is determined by the Capacitor added on the output. $f = 1/2\pi * (32 \times 10^3) * C$. $C = 3.3\text{ nF}$ corresponds to $BW = 1507\text{ Hz}$, which is the minimum to filter out internal clock noise.
- These limits define the range of operation for which the part will meet specification.
- Within the supply range of 2.2 and 3.6 V, the device operates as a fully calibrated linear accelerometer. Beyond these supply limits the device may operate as a linear device but is not guaranteed to be in calibration.
- This value is measured with g-Select in 1.5g mode.
- The device can measure both + and – acceleration. With no input acceleration the output is at midsupply. For positive acceleration the output will increase above $V_{DD}/2$. For negative acceleration, the output will decrease below $V_{DD}/2$.
- For optimal 0g offset performance, adhere to AN3484 and AN3447
- The response time between 10% of full scale V_{DD} input voltage and 90% of the final operating output voltage.
- The response time between 10% of full scale Sleep Mode input voltage and 90% of the final operating output voltage.
- The response time between 10% of the full scale self test input voltage and 90% of the self test output voltage.
- A measure of the device's ability to reject an acceleration applied 90° from the true axis of sensitivity.

PRINCIPLE OF OPERATION

The Freescale accelerometer is a surface-micromachined integrated-circuit accelerometer.

The device consists of a surface micromachined capacitive sensing cell (g-cell) and a signal conditioning ASIC contained in a single package. The sensing element is sealed hermetically at the wafer level using a bulk micromachined cap wafer.

The g-cell is a mechanical structure formed from semiconductor materials (polysilicon) using semiconductor processes (masking and etching). It can be modeled as a set of beams attached to a movable central mass that move between fixed beams. The movable beams can be deflected from their rest position by subjecting the system to an acceleration (Figure 3).

As the beams attached to the central mass move, the distance from them to the fixed beams on one side will increase by the same amount that the distance to the fixed beams on the other side decreases. The change in distance is a measure of acceleration.

The g-cell beams form two back-to-back capacitors (Figure 3). As the center beam moves with acceleration, the distance between the beams changes and each capacitor's value will change, ($C = A\epsilon/D$). Where A is the area of the beam, ϵ is the dielectric constant, and D is the distance between the beams.

The ASIC uses switched capacitor techniques to measure the g-cell capacitors and extract the acceleration data from the difference between the two capacitors. The ASIC also signal conditions and filters (switched capacitor) the signal, providing a high level output voltage that is ratiometric and proportional to acceleration.

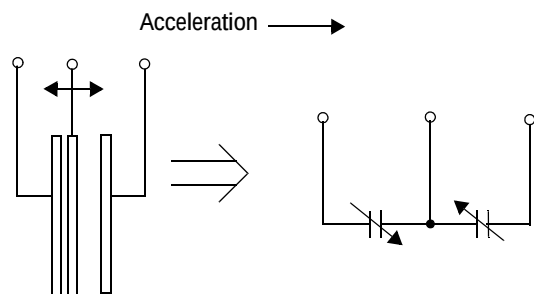


Figure 3. Simplified Transducer Physical Model

SPECIAL FEATURES

0g-Detect

The sensor offers a 0g-Detect feature that provides a logic high signal when all three axes are at 0g. This feature enables the application of Linear Freefall protection if the signal is connected to an interrupt pin or a poled I/O pin on a microcontroller.

Self Test

The sensor provides a self test feature that allows the verification of the mechanical and electrical integrity of the accelerometer at any time before or after installation. This feature is critical in applications such as hard disk drive

protection where system integrity must be ensured over the life of the product. Customers can use self test to verify the solderability to confirm that the part was mounted to the PCB correctly. To use this feature to verify the 0g-Detect function, the accelerometer should be held upside down so that the z-axis experiences -1g. When the self test function is initiated, an electrostatic force is applied to each axis to cause it to deflect. The x- and y-axis are deflected slightly while the z-axis is trimmed to deflect 1g. This procedure assures that both the mechanical (g-cell) and electronic sections of the accelerometer are functioning.

g-Select

The g-Select feature allows for the selection between two sensitivities. Depending on the logic input placed on pin 10, the device internal gain will be changed allowing it to function with a 1.5g or 6g sensitivity (Table 3). This feature is ideal when a product has applications requiring two different sensitivities for optimum performance. The sensitivity can be changed at anytime during the operation of the product. The g-Select pin can be left unconnected for applications requiring only a 1.5g sensitivity as the device has an internal pull-down to keep it at that sensitivity (800mV/g).

Table 3. g-Select Pin Description

g-Select	g-Range	Sensitivity
0	1.5g	800 mV/g
1	6g	206 mV/g

Sleep Mode

The 3 axis accelerometer provides a Sleep Mode that is ideal for battery operated products. When Sleep Mode is active, the device outputs are turned off, providing significant reduction of operating current. A low input signal on pin 7 (Sleep Mode) will place the device in this mode and reduce the current to 3 μ A typ. For lower power consumption, it is recommended to set g-Select to 1.5g mode. By placing a high input signal on pin 7, the device will resume to normal mode of operation.

Filtering

The 3 axis accelerometer contains an onboard single-pole switched capacitor filter. Because the filter is realized using switched capacitor techniques, there is no requirement for external passive components (resistors and capacitors) to set the cut-off frequency.

Ratiometricity

Ratiometricity simply means the output offset voltage and sensitivity will scale linearly with applied supply voltage. That is, as supply voltage is increased, the sensitivity and offset increase linearly; as supply voltage decreases, offset and sensitivity decrease linearly. This is a key feature when interfacing to a microcontroller or an A/D converter because it provides system level cancellation of supply induced errors in the analog to digital conversion process.

BASIC CONNECTIONS

Pin Descriptions

Top View

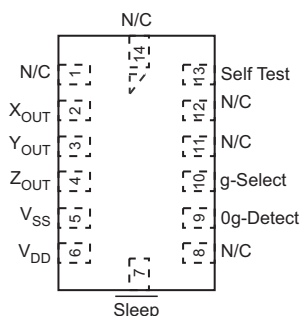


Figure 4. Pinout Description

Table 4. Pin Descriptions

Pin No.	Pin Name	Description
1	N/C	No internal connection Leave unconnected
2	X _{OUT}	X direction output voltage
3	Y _{OUT}	Y direction output voltage
4	Z _{OUT}	Z direction output voltage
5	V _{SS}	Power Supply Ground
6	V _{DD}	Power Supply Input
7	Sleep	Logic input pin to enable product or Sleep Mode
8	NC	No internal connection Leave unconnected
9	0g-Detect	Linear Freefall digital logic output signal
10	g-Select	Logic input pin to select g level
11	N/C	Unused for factory trim Leave unconnected
12	N/C	Unused for factory trim Leave unconnected
13	Self Test	Input pin to initiate Self Test
14	N/C	Unused for factory trim Leave unconnected

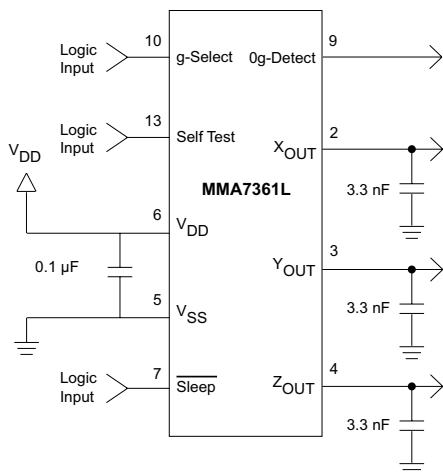


Figure 5. Accelerometer with Recommended Connection Diagram

PCB Layout

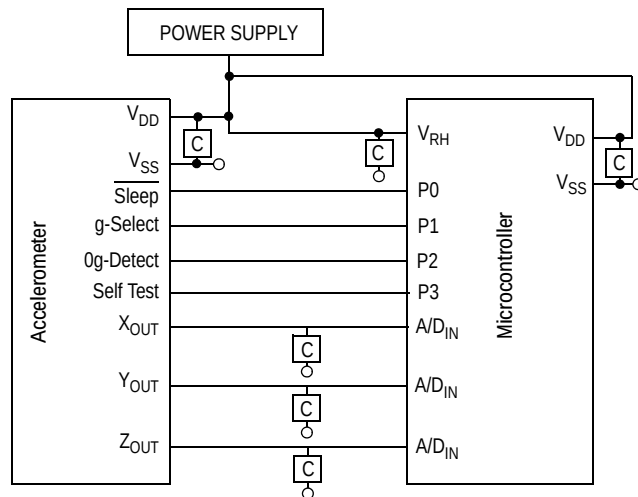
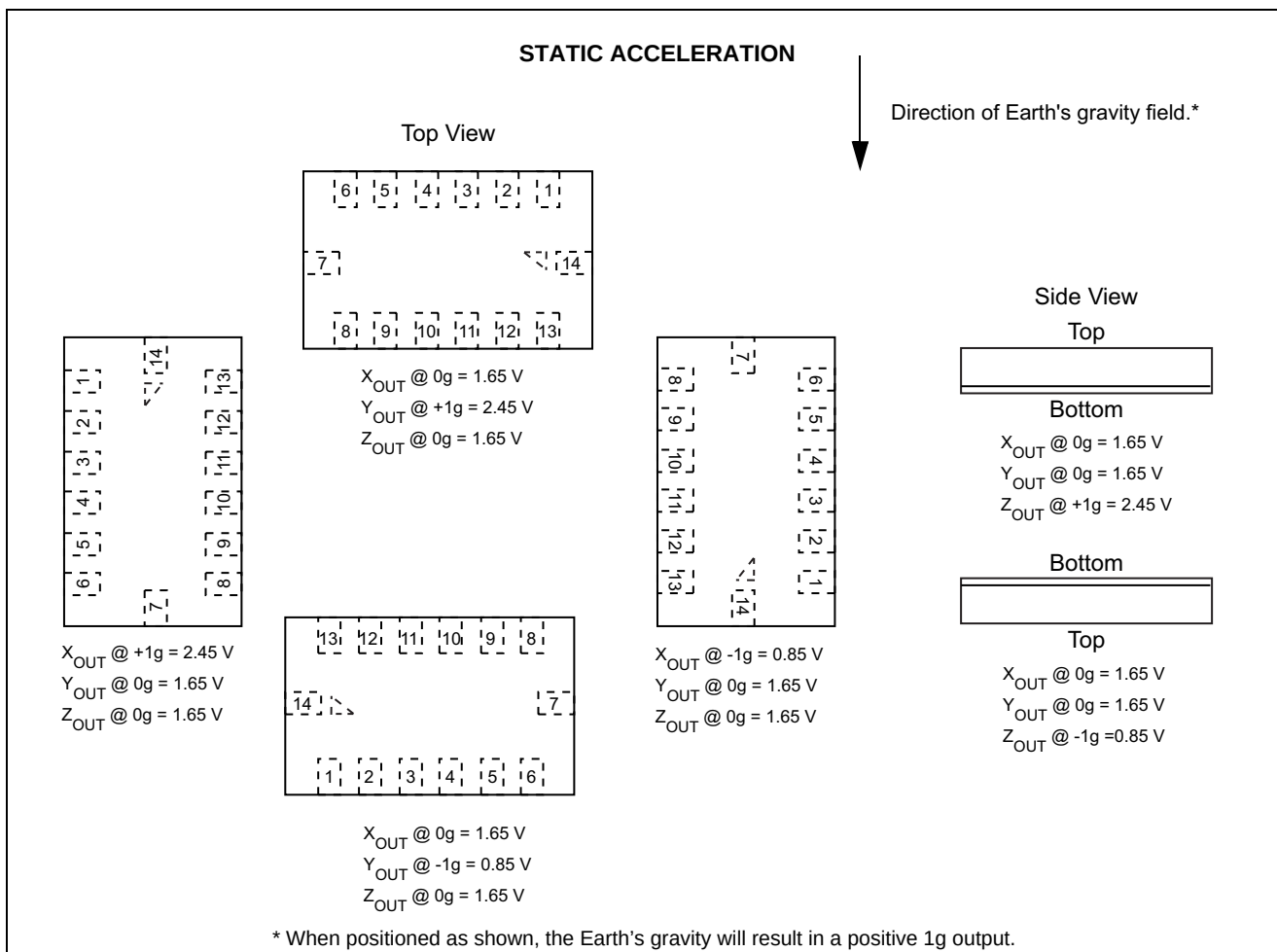
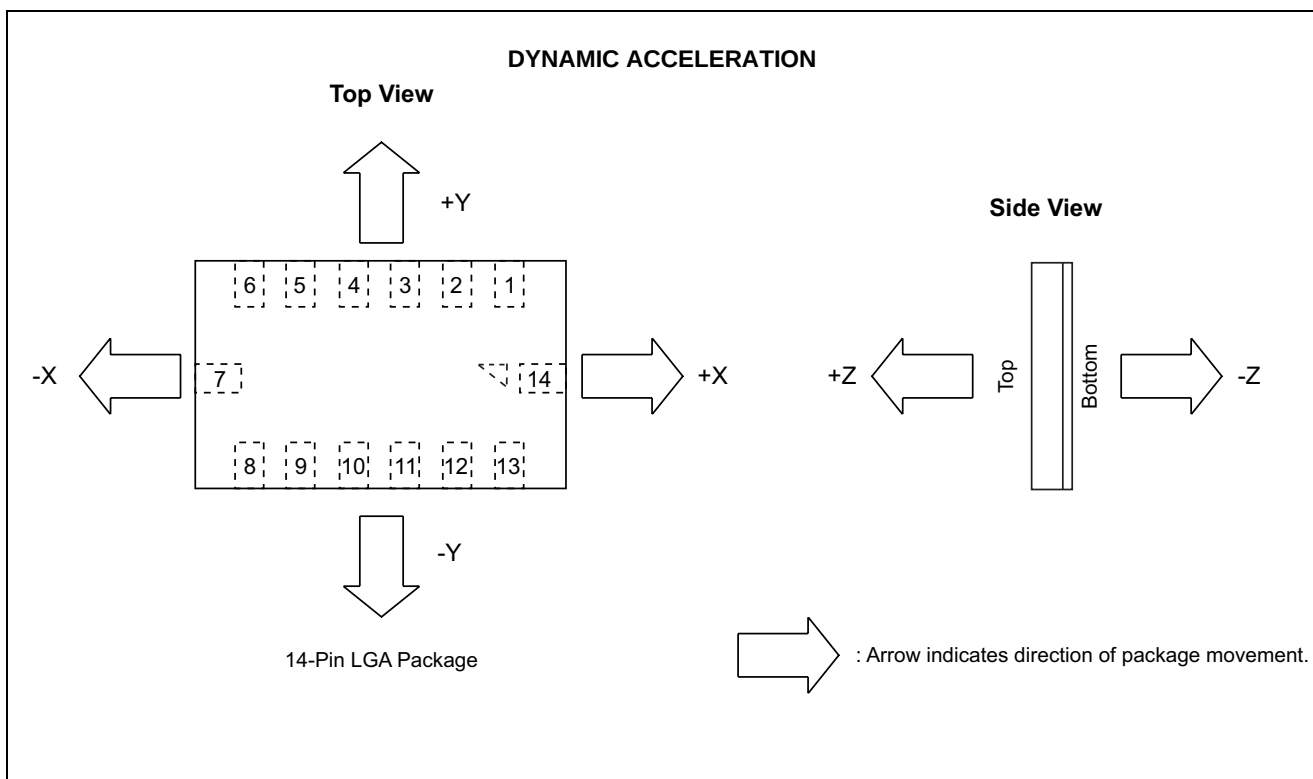


Figure 6. Recommended PCB Layout for Interfacing Accelerometer to Microcontroller

NOTES:

1. Use 0.1 µF capacitor on V_{DD} to decouple the power source.
2. Physical coupling distance of the accelerometer to the microcontroller should be minimal.
3. Place a ground plane beneath the accelerometer to reduce noise, the ground plane should be attached to all of the open ended terminals shown in Figure 6.
4. Use a 3.3nF capacitor on the outputs of the accelerometer to minimize clock noise (from the switched capacitor filter circuit).
5. PCB layout of power and ground should not couple power supply noise.
6. Accelerometer and microcontroller should not be a high current path.
7. A/D sampling rate and any external power supply switching frequency should be selected such that they do not interfere with the internal accelerometer sampling frequency (11 kHz for the sampling frequency). This will prevent aliasing errors.
8. 10MΩ or higher is recommended on X_{OUT}, Y_{OUT} and Z_{OUT} to prevent loss due to the voltage divider relationship between the internal 32 kΩ resistor and the measurement input impedance.



Top View

$X_{OUT} @ 0g = 1.65\text{ V}$
 $Y_{OUT} @ -1g = 0.85\text{ V}$
 $Z_{OUT} @ 0g = 1.65\text{ V}$

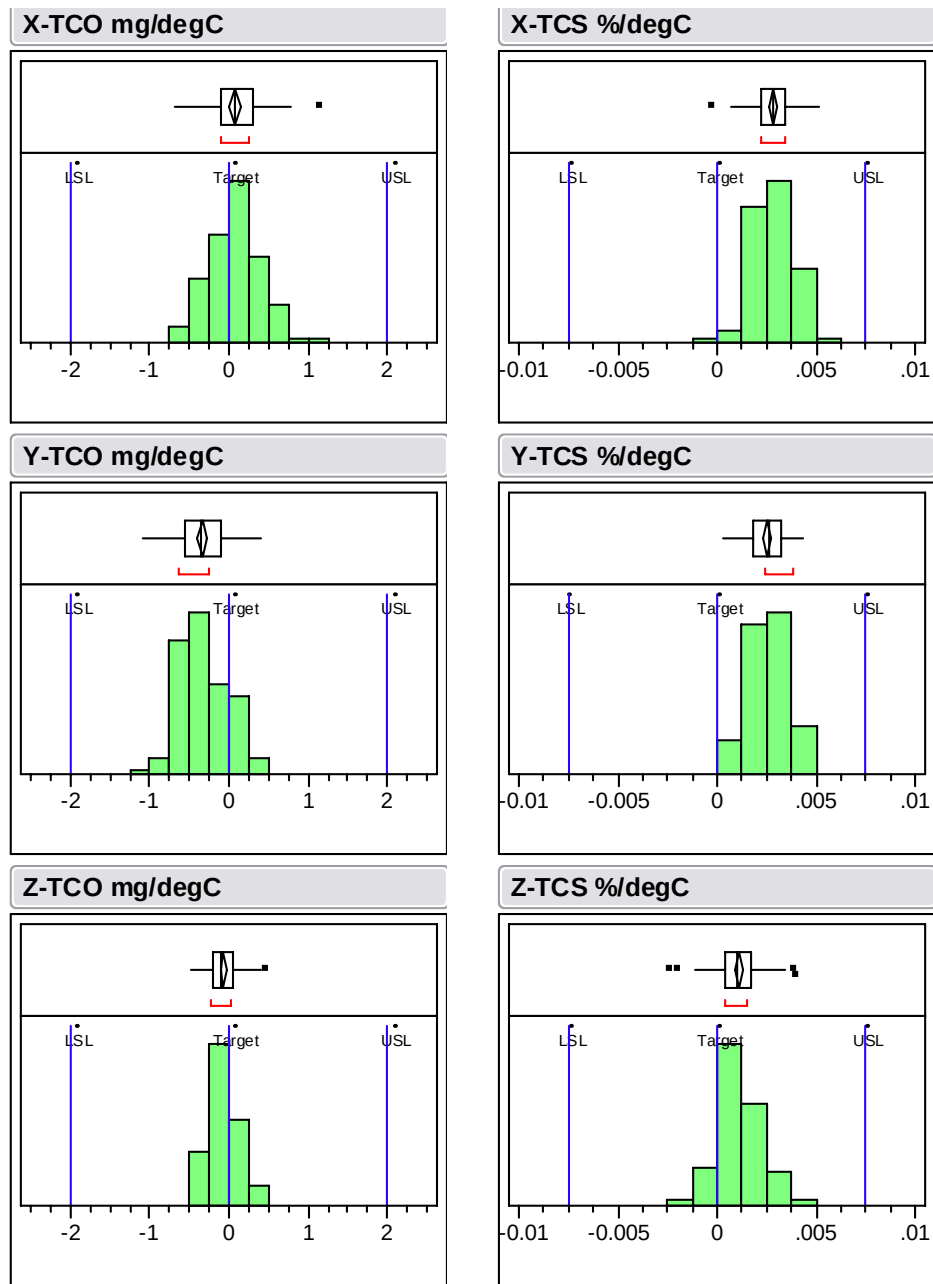


Figure 7. MMA7361L Temperature Coefficient of Offset (TCO) and Temperature Coefficient of Sensitivity (TCS) Distribution Charts

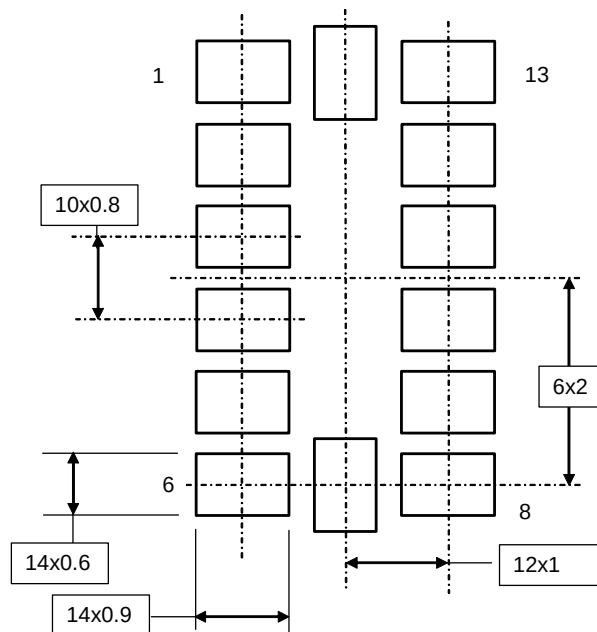
MINIMUM RECOMMENDED FOOTPRINT FOR SURFACE MOUNTED APPLICATIONS

PCB Mounting Recommendations

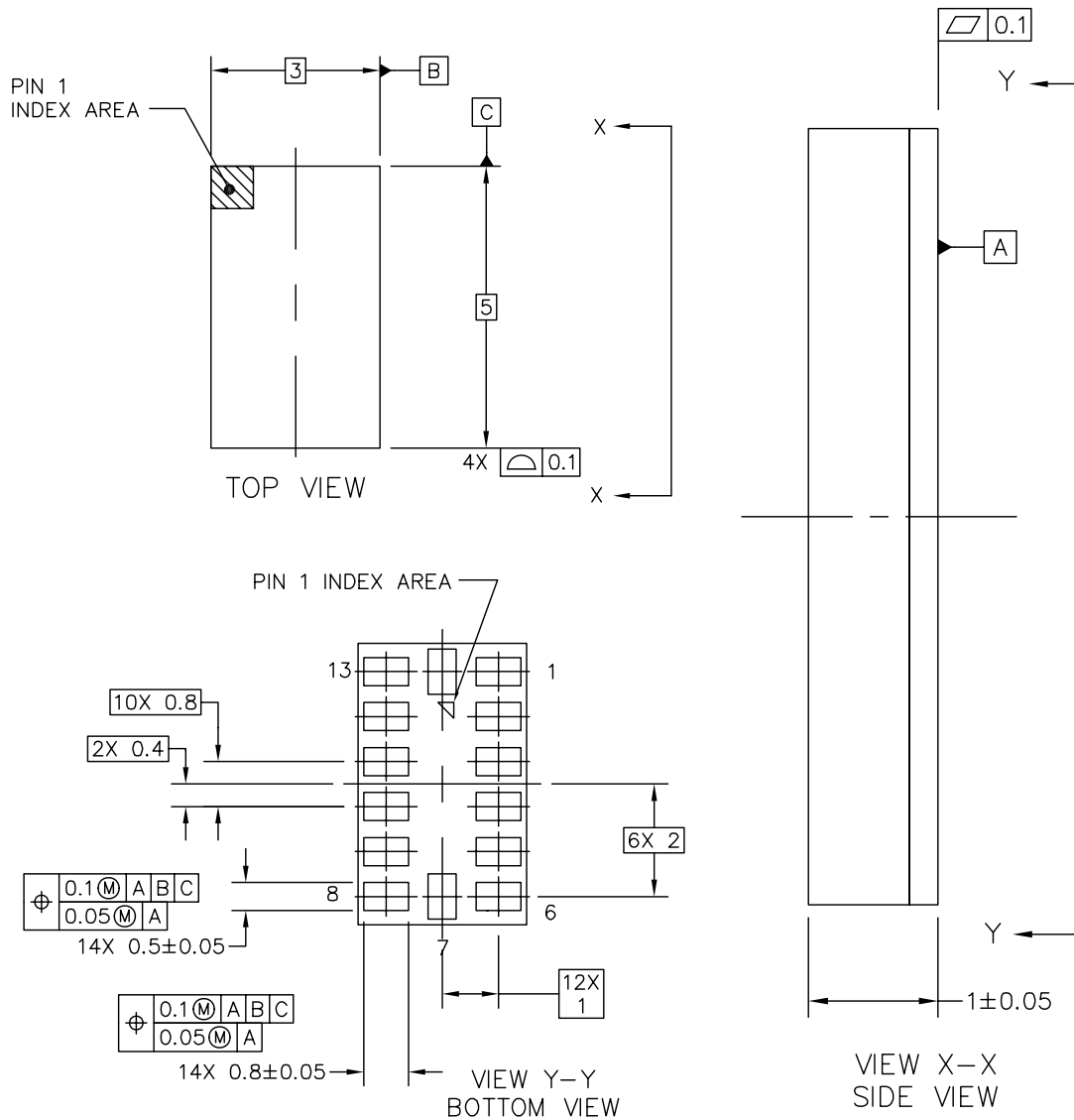
MEMS based sensors are sensitive to Printed Circuit Board (PCB) reflow processes. For optimal zero-g offset after PCB mounting, care must be taken to PCB layout and reflow conditions. Reference application note AN3484 for best practices to minimize the zero-g offset shift after PCB mounting.

Surface mount board layout is a critical portion of the total design. The footprint for the surface mount packages must be the correct size to ensure proper solder connection interface between the board and the package.

With the correct footprint, the packages will self-align when subjected to a solder reflow process. It is always recommended to design boards with a solder mask layer to avoid bridging and shorting between solder pads.



PACKAGE DIMENSIONS



© FREESCALE SEMICONDUCTOR, INC. ALL RIGHTS RESERVED.		MECHANICAL OUTLINE		PRINT VERSION NOT TO SCALE	
TITLE: LGA 14 I/O, 3 X 5 X 1 PITCH 0.8, SENSOR 1.0MM PKG		DOCUMENT NO: 98ASA10801D		REV: A	
		CASE NUMBER: 1977-01		09 JAN 2008	
		STANDARD: NON-JEDEC			

**CASE 1977-01
ISSUE A
14-LEAD LGA**

PACKAGE DIMENSIONS

NOTES:

1. ALL DIMENSIONS IN MILLIMETERS.
2. DIMENSIONING AND TOLERANCING PER ASME Y14.5M-1994.

© FREESCALE SEMICONDUCTOR, INC. ALL RIGHTS RESERVED.		MECHANICAL OUTLINE		PRINT VERSION NOT TO SCALE	
TITLE: LGA 14 I/O, 3 X 5 X 1 PITCH 0.8, SENSOR 1.0MM PKG		DOCUMENT NO: 98ASA10801D		REV: A	
		CASE NUMBER: 1977-01		09 JAN 2008	
		STANDARD: NON-JEDEC			

**CASE 1977-01
ISSUE A
14-LEAD LGA**

How to Reach Us:

Home Page:

www.freescale.com

Web Support:

<http://www.freescale.com/support>

USA/Europe or Locations Not Listed:

Freescale Semiconductor, Inc.
Technical Information Center, EL516
2100 East Elliot Road
Tempe, Arizona 85284
+1-800-521-6274 or +1-480-768-2130
www.freescale.com/support

Europe, Middle East, and Africa:

Freescale Halbleiter Deutschland GmbH
Technical Information Center
Schatzbogen 7
81829 Muenchen, Germany
+44 1296 380 456 (English)
+46 8 52200080 (English)
+49 89 92103 559 (German)
+33 1 69 35 48 48 (French)
www.freescale.com/support

Japan:

Freescale Semiconductor Japan Ltd.
Headquarters
ARCO Tower 15F
1-8-1, Shimo-Meguro, Meguro-ku,
Tokyo 153-0064
Japan
0120 191014 or +81 3 5437 9125
support.japan@freescale.com

Asia/Pacific:

Freescale Semiconductor Hong Kong Ltd.
Technical Information Center
2 Dai King Street
Tai Po Industrial Estate
Tai Po, N.T., Hong Kong
+800 2666 8080
support.asia@freescale.com

For Literature Requests Only:

Freescale Semiconductor Literature Distribution Center
P.O. Box 5405
Denver, Colorado 80217
1-800-441-2447 or 303-675-2140
Fax: 303-675-2150
LDCForFreescaleSemiconductor@hibbertgroup.com

Information in this document is provided solely to enable system and software implementers to use Freescale Semiconductor products. There are no express or implied copyright licenses granted hereunder to design or fabricate any integrated circuits or integrated circuits based on the information in this document.

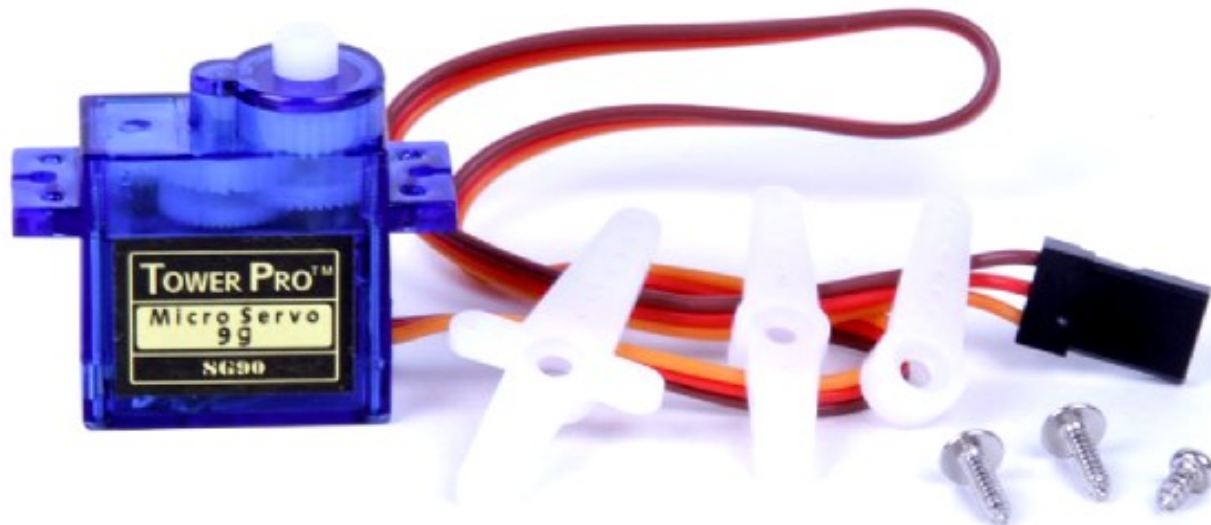
Freescale Semiconductor reserves the right to make changes without further notice to any products herein. Freescale Semiconductor makes no warranty, representation or guarantee regarding the suitability of its products for any particular purpose, nor does Freescale Semiconductor assume any liability arising out of the application or use of any product or circuit, and specifically disclaims any and all liability, including without limitation consequential or incidental damages. "Typical" parameters that may be provided in Freescale Semiconductor data sheets and/or specifications can and do vary in different applications and actual performance may vary over time. All operating parameters, including "Typicals", must be validated for each customer application by customer's technical experts. Freescale Semiconductor does not convey any license under its patent rights nor the rights of others. Freescale Semiconductor products are not designed, intended, or authorized for use as components in systems intended for surgical implant into the body, or other applications intended to support or sustain life, or for any other application in which the failure of the Freescale Semiconductor product could create a situation where personal injury or death may occur. Should Buyer purchase or use Freescale Semiconductor products for any such unintended or unauthorized application, Buyer shall indemnify and hold Freescale Semiconductor and its officers, employees, subsidiaries, affiliates, and distributors harmless against all claims, costs, damages, and expenses, and reasonable attorney fees arising out of, directly or indirectly, any claim of personal injury or death associated with such unintended or unauthorized use, even if such claim alleges that Freescale Semiconductor was negligent regarding the design or manufacture of the part.

Freescale™ and the Freescale logo are trademarks of Freescale Semiconductor, Inc. All other product or service names are the property of their respective owners.

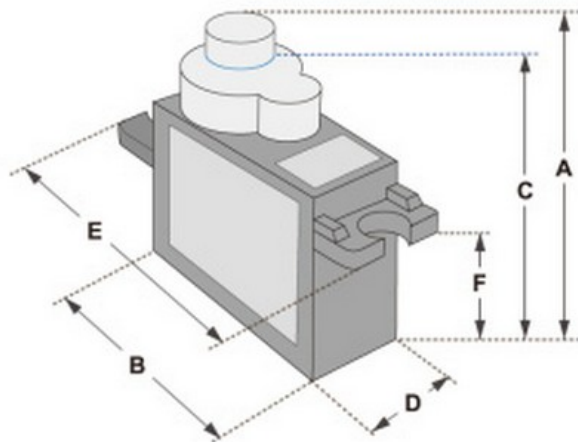
© Freescale Semiconductor, Inc. 2008. All rights reserved.

RoHS-compliant and/or Pb-free versions of Freescale products have the functionality and electrical characteristics of their non-RoHS-compliant and/or non-Pb-free counterparts. For further information, see <http://www.freescale.com> or contact your Freescale sales representative.

For information on Freescale's Environmental Products program, go to <http://www.freescale.com/epp>.



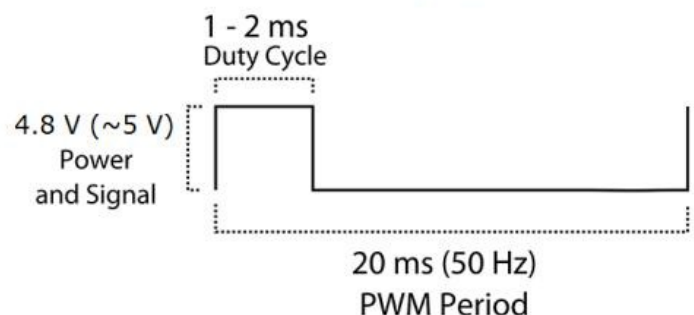
Tiny and lightweight with high output power. Servo can rotate approximately 180 degrees (90 in each direction), and works just like the standard kinds but smaller. You can use any servo code, hardware or library to control these servos. Good for beginners who want to make stuff move without building a motor controller with feedback & gear box, especially since it will fit in small places. It comes with a 3 horns (arms) and hardware.



Dimensions & Specifications	
A (mm) :	32
B (mm) :	23
C (mm) :	28.5
D (mm) :	12
E (mm) :	32
F (mm) :	19.5
Speed (sec) :	0.1
Torque (kg-cm) :	2.5
Weight (g) :	14.7
Voltage :	4.8 - 6

Position "0" (1.5 ms pulse) is middle, "90" (~2ms pulse) is middle, is all the way to the right, "-90" (~1ms pulse) is all the way to the left.

PWM=Orange (⏏)
Vcc=Red (+)
Ground=Brown (-)





Servomotor FUTABA

[PROD 013A- Servomotor Futaba -04/15]

Especificaciones Técnicas del Servomotor Futaba

Diseñado por:

N-MADA®

Características

- Engranajes de Nylon
- Voltaje de operación 4.8-6 Volts.
- Peso 38 gr.
- Velocidad de trabajo
 - o 0.23 seg/60 grados (4.8Volts)
 - o 0.19 seg/60 grados (6 Volts)
- Torque
 - o 3.2 kg/cm (4.8 volts)
 - o 4.1 kg/cm (6 Volts)
- Frecuencia de funcionamiento 50Hz
- Incluye accesorios

Dimensiones (en milímetros)

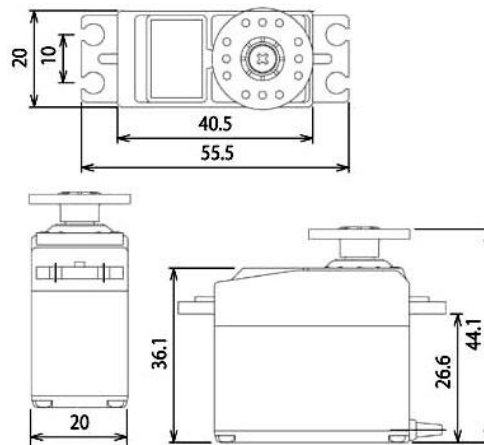


Figura 1. Dimensiones del Servomotor FUTABA [S3003].

Descripción de los pines

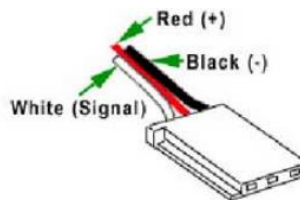


Figura 1. Conector del Servomotor FUTABA [S3003].

Pin	Función
Negro	Tierra
Rojo	Alimentación
Blanco	Control (PWM)

Tabla 1. Descripción del conector del Servomotor FUTABA [S3003].

Código de muestra con NoMADA®

El código de muestra y librerías para este módulo y muchos más se encuentran disponibles en el siguiente enlace en la sección de recursos NoMADA®:

<http://nomada-e.com/videotutoriales.html>

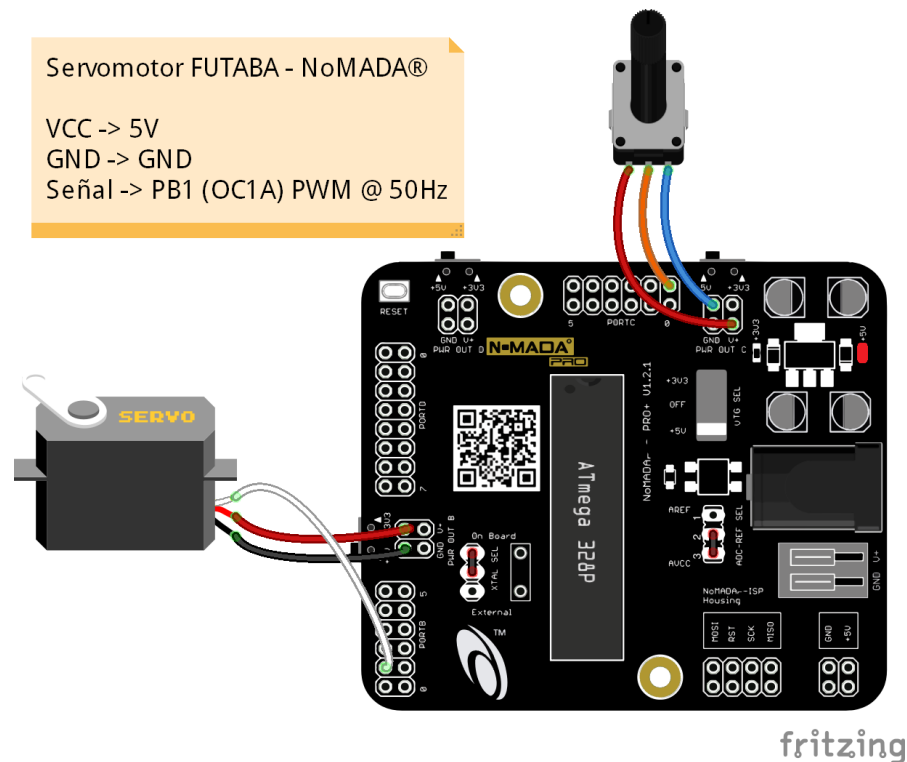


Figura 2. Conexión del Servomotor FUTABA [S3003] con NoMADA®.

Historial de Revisiones del Archivo.

PROD 013A Servomotor FUTABA [S3003] 04/15

1. Revisión Inicial



Diseño Embebido...

un Paso Adelante

*Av. Francisco García Salinas #329 Int. 9
Col. Lomas del Convento
01 492 491 36 82
Guadalupe, Zacatecas, México.*

©2015 NoMADA Store. Todos los derechos reservados | PROD 013A –Servomotor FUTABA [S3003] - 04/2015.





LM35 Precision Centigrade Temperature Sensors

1 Features

- Calibrated Directly in Celsius (Centigrade)
- Linear + 10-mV/°C Scale Factor
- 0.5°C Ensured Accuracy (at 25°C)
- Rated for Full –55°C to 150°C Range
- Suitable for Remote Applications
- Low-Cost Due to Wafer-Level Trimming
- Operates From 4 V to 30 V
- Less Than 60-μA Current Drain
- Low Self-Heating, 0.08°C in Still Air
- Non-Linearity Only ±¼°C Typical
- Low-Impedance Output, 0.1 Ω for 1-mA Load

2 Applications

- Power Supplies
- Battery Management
- HVAC
- Appliances

3 Description

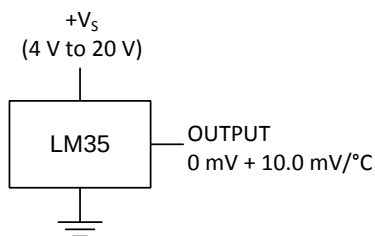
The LM35 series are precision integrated-circuit temperature devices with an output voltage linearly-proportional to the Centigrade temperature. The LM35 device has an advantage over linear temperature sensors calibrated in Kelvin, as the user is not required to subtract a large constant voltage from the output to obtain convenient Centigrade scaling. The LM35 device does not require any external calibration or trimming to provide typical accuracies of ±¼°C at room temperature and ±¾°C over a full –55°C to 150°C temperature range. Lower cost is assured by trimming and calibration at the wafer level. The low-output impedance, linear output, and precise inherent calibration of the LM35 device makes interfacing to readout or control circuitry especially easy. The device is used with single power supplies, or with plus and minus supplies. As the LM35 device draws only 60 μA from the supply, it has very low self-heating of less than 0.1°C in still air. The LM35 device is rated to operate over a –55°C to 150°C temperature range, while the LM35C device is rated for a –40°C to 110°C range (–10° with improved accuracy). The LM35-series devices are available packaged in hermetic TO transistor packages, while the LM35C, LM35CA, and LM35D devices are available in the plastic TO-92 transistor package. The LM35D device is available in an 8-lead surface-mount small-outline package and a plastic TO-220 package.

Device Information⁽¹⁾

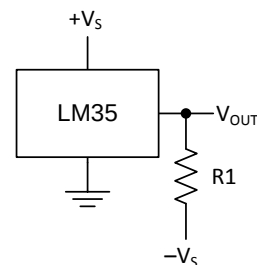
PART NUMBER	PACKAGE	BODY SIZE (NOM)
LM35	TO-CAN (3)	4.699 mm × 4.699 mm
	TO-92 (3)	4.30 mm × 4.30 mm
	SOIC (8)	4.90 mm × 3.91 mm
	TO-220 (3)	14.986 mm × 10.16 mm

(1) For all available packages, see the orderable addendum at the end of the datasheet.

Basic Centigrade Temperature Sensor (2°C to 150°C)



Full-Range Centigrade Temperature Sensor



Choose $R_1 = -V_S / 50 \mu\text{A}$
 $V_{\text{OUT}} = 1500 \text{ mV at } 150^\circ\text{C}$
 $V_{\text{OUT}} = 250 \text{ mV at } 25^\circ\text{C}$
 $V_{\text{OUT}} = -550 \text{ mV at } -55^\circ\text{C}$



Table of Contents

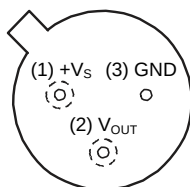
1 Features	1	7.3 Feature Description	13
2 Applications	1	7.4 Device Functional Modes	13
3 Description	1	8 Application and Implementation	14
4 Revision History	2	8.1 Application Information	14
5 Pin Configuration and Functions	3	8.2 Typical Application	15
6 Specifications	4	8.3 System Examples	16
6.1 Absolute Maximum Ratings	4	9 Power Supply Recommendations	19
6.2 ESD Ratings	4	10 Layout	19
6.3 Recommended Operating Conditions	4	10.1 Layout Guidelines	19
6.4 Thermal Information	4	10.2 Layout Example	20
6.5 Electrical Characteristics: LM35A, LM35CA Limits ...	5	11 Device and Documentation Support	21
6.6 Electrical Characteristics: LM35A, LM35CA	6	11.1 Receiving Notification of Documentation Updates	21
6.7 Electrical Characteristics: LM35, LM35C, LM35D Limits	8	11.2 Community Resources	21
6.8 Electrical Characteristics: LM35, LM35C, LM35D ...	9	11.3 Trademarks	21
6.9 Typical Characteristics	11	11.4 Electrostatic Discharge Caution	21
7 Detailed Description	13	11.5 Glossary	21
7.1 Overview	13	12 Mechanical, Packaging, and Orderable Information	21
7.2 Functional Block Diagram	13		

4 Revision History

Changes from Revision G (August 2016) to Revision H	Page
• Changed NDV Package (TO-CAN) pinout from bottom view back to top view; added textnote to pinout	3
• Added pin numbers to the TO-CAN (TO46) pinout	3
Changes from Revision F (January 2016) to Revision G	Page
• Equation 1 , changed From: 10 mV/°F To: 10mV/°C	13
• Power Supply Recommendations , changed From: "4-V to 5.5-V power supply" To: "4-V to 30-V power supply:	19
Changes from Revision E (January 2015) to Revision F	Page
• Changed NDV Package (TO-CAN) pinout from Top View to Bottom View	3
Changes from Revision D (October 2013) to Revision E	Page
• Added <i>Pin Configuration and Functions</i> section, <i>ESD Ratings</i> table, <i>Feature Description</i> section, <i>Device Functional Modes</i> , <i>Application and Implementation</i> section, <i>Power Supply Recommendations</i> section, <i>Layout</i> section, <i>Device and Documentation Support</i> section, and <i>Mechanical, Packaging, and Orderable Information</i> section	1
Changes from Revision C (July 2013) to Revision D	Page
• Changed <i>W</i> to Ω	1
• Changed <i>W</i> to Ω in <i>Abs Max</i> tablenote.	4

5 Pin Configuration and Functions

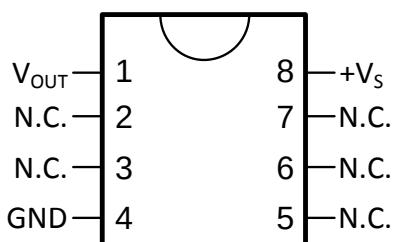
NDV Package
3-Pin TO-CAN
(Top View)



Case is connected to negative pin (GND)

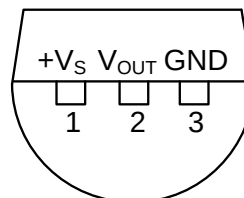
Refer the second NDV0003H page for reference

D Package
8-PIN SOIC
(Top View)

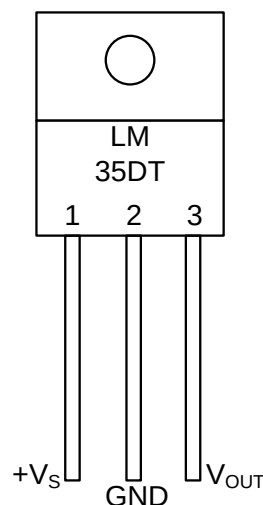


N.C. = No connection

LP Package
3-Pin TO-92
(Bottom View)



NEB Package
3-Pin TO-220
(Top View)



Tab is connected to the negative pin (GND).

NOTE: The LM35DT pinout is different than the discontinued LM35DP

Pin Functions

NAME	PIN				TYPE	DESCRIPTION
	TO46	TO92	TO220	SO8		
V _{OUT}	2	2	3	1	O	Temperature Sensor Analog Output
N.C.	—	—	—	2	—	No Connection
	—	—	—	3		
GND	3	3	2	4	GROUND	Device ground pin, connect to power supply negative terminal
N.C.	—	—	—	5	—	No Connection
	—	—	—	6		
	—	—	—	7		
+V _S	1	1	1	8	POWER	Positive power supply pin

6 Specifications

6.1 Absolute Maximum Ratings

over operating free-air temperature range (unless otherwise noted)⁽¹⁾⁽²⁾

		MIN	MAX	UNIT
Supply voltage		−0.2	35	V
Output voltage		−1	6	V
Output current			10	mA
Maximum Junction Temperature, T_{Jmax}			150	°C
Storage Temperature, T_{stg}	TO-CAN, TO-92 Package	−60	150	°C
	TO-220, SOIC Package	−65	150	

- (1) If Military/Aerospace specified devices are required, please contact the Texas Instruments Sales Office/ Distributors for availability and specifications.
- (2) Absolute Maximum Ratings indicate limits beyond which damage to the device may occur. DC and AC electrical specifications do not apply when operating the device beyond its rated operating conditions.

6.2 ESD Ratings

		VALUE	UNIT
$V_{(ESD)}$	Electrostatic discharge Human-body model (HBM), per ANSI/ESDA/JEDEC JS-001 ⁽¹⁾	±2500	V

- (1) JEDEC document JEP155 states that 500-V HBM allows safe manufacturing with a standard ESD control process.

6.3 Recommended Operating Conditions

over operating free-air temperature range (unless otherwise noted)

		MIN	MAX	UNIT
Specified operating temperature: T_{MIN} to T_{MAX}	LM35, LM35A	−55	150	°C
	LM35C, LM35CA	−40	110	
	LM35D	0	100	
Supply Voltage (+ V_S)		4	30	V

6.4 Thermal Information

THERMAL METRIC ⁽¹⁾⁽²⁾		LM35				UNIT
		NDV	LP	D	NEB	
		3 PINS		8 PINS	3 PINS	
$R_{\theta JA}$	Junction-to-ambient thermal resistance	400	180	220	90	°C/W
$R_{\theta JC(top)}$	Junction-to-case (top) thermal resistance	24	—	—	—	

- (1) For more information about traditional and new thermal metrics, see the *IC Package Thermal Metrics* application report, [SPRA953](#).
- (2) For additional thermal resistance information, see [Typical Application](#).

6.5 Electrical Characteristics: LM35A, LM35CA Limits

Unless otherwise noted, these specifications apply: $-55^{\circ}\text{C} \leq T_J \leq 150^{\circ}\text{C}$ for the LM35 and LM35A; $-40^{\circ}\text{C} \leq T_J \leq 110^{\circ}\text{C}$ for the LM35C and LM35CA; and $0^{\circ}\text{C} \leq T_J \leq 100^{\circ}\text{C}$ for the LM35D. $V_S = 5\text{ Vdc}$ and $I_{\text{LOAD}} = 50\text{ }\mu\text{A}$, in the circuit of [Full-Range Centigrade Temperature Sensor](#). These specifications also apply from 2°C to T_{MAX} in the circuit of [Figure 14](#).

PARAMETER	TEST CONDITIONS	LM35A			LM35CA			UNIT
		TYP	TESTED LIMIT ⁽¹⁾	DESIGN LIMIT ⁽²⁾	TYP	TESTED LIMIT ⁽¹⁾	DESIGN LIMIT ⁽²⁾	
Accuracy ⁽³⁾	$T_A = 25^{\circ}\text{C}$	± 0.2	± 0.5		± 0.2	± 0.5		$^{\circ}\text{C}$
	$T_A = -10^{\circ}\text{C}$	± 0.3			± 0.3		± 1	
	$T_A = T_{\text{MAX}}$	± 0.4	± 1		± 0.4	± 1		
	$T_A = T_{\text{MIN}}$	± 0.4	± 1		± 0.4		± 1.5	
Nonlinearity ⁽⁴⁾	$T_{\text{MIN}} \leq T_A \leq T_{\text{MAX}}$, $-40^{\circ}\text{C} \leq T_J \leq 125^{\circ}\text{C}$	± 0.18		± 0.35	± 0.15		± 0.3	$^{\circ}\text{C}$
Sensor gain (average slope)	$T_{\text{MIN}} \leq T_A \leq T_{\text{MAX}}$	10	9.9		10		9.9	$\text{mV}/^{\circ}\text{C}$
	$-40^{\circ}\text{C} \leq T_J \leq 125^{\circ}\text{C}$	10	10.1		10		10.1	
Load regulation ⁽⁵⁾ $0 \leq I_L \leq 1\text{ mA}$	$T_A = 25^{\circ}\text{C}$	± 0.4	± 1		± 0.4	± 1		mV/mA
	$T_{\text{MIN}} \leq T_A \leq T_{\text{MAX}}$, $-40^{\circ}\text{C} \leq T_J \leq 125^{\circ}\text{C}$	± 0.5		± 3	± 0.5		± 3	
Line regulation ⁽⁵⁾	$T_A = 25^{\circ}\text{C}$	± 0.01	± 0.05		± 0.01	± 0.05		mV/V
	$4\text{ V} \leq V_S \leq 30\text{ V}$, $-40^{\circ}\text{C} \leq T_J \leq 125^{\circ}\text{C}$	± 0.02		± 0.1	± 0.02		± 0.1	
Quiescent current ⁽⁶⁾	$V_S = 5\text{ V}$, 25°C	56	67		56	67		μA
	$V_S = 5\text{ V}$, $-40^{\circ}\text{C} \leq T_J \leq 125^{\circ}\text{C}$	105		131	91		114	
	$V_S = 30\text{ V}$, 25°C	56.2	68		56.2	68		
	$V_S = 30\text{ V}$, $-40^{\circ}\text{C} \leq T_J \leq 125^{\circ}\text{C}$	105.5		133	91.5		116	
Change of quiescent current ⁽⁵⁾	$4\text{ V} \leq V_S \leq 30\text{ V}$, 25°C	0.2	1		0.2	1		μA
	$4\text{ V} \leq V_S \leq 30\text{ V}$, $-40^{\circ}\text{C} \leq T_J \leq 125^{\circ}\text{C}$	0.5		2	0.5		2	
Temperature coefficient of quiescent current	$-40^{\circ}\text{C} \leq T_J \leq 125^{\circ}\text{C}$	0.39		0.5	0.39		0.5	$\mu\text{A}/^{\circ}\text{C}$
Minimum temperature for rate accuracy	In circuit of Figure 14 , $I_L = 0$	1.5		2	1.5		2	$^{\circ}\text{C}$
Long term stability	$T_J = T_{\text{MAX}}$, for 1000 hours	± 0.08			± 0.08			$^{\circ}\text{C}$

- (1) Tested Limits are ensured and 100% tested in production.
- (2) Design Limits are ensured (but not 100% production tested) over the indicated temperature and supply voltage ranges. These limits are not used to calculate outgoing quality levels.
- (3) Accuracy is defined as the error between the output voltage and $10\text{ mV}/^{\circ}\text{C}$ times the case temperature of the device, at specified conditions of voltage, current, and temperature (expressed in $^{\circ}\text{C}$).
- (4) Non-linearity is defined as the deviation of the output-voltage-versus-temperature curve from the best-fit straight line, over the rated temperature range of the device.
- (5) Regulation is measured at constant junction temperature, using pulse testing with a low duty cycle. Changes in output due to heating effects can be computed by multiplying the internal dissipation by the thermal resistance.
- (6) Quiescent current is defined in the circuit of [Figure 14](#).

6.6 Electrical Characteristics: LM35A, LM35CA

Unless otherwise noted, these specifications apply: $-55^{\circ}\text{C} \leq T_J \leq 150^{\circ}\text{C}$ for the LM35 and LM35A; $-40^{\circ}\text{C} \leq T_J \leq 110^{\circ}\text{C}$ for the LM35C and LM35CA; and $0^{\circ}\text{C} \leq T_J \leq 100^{\circ}\text{C}$ for the LM35D. $V_S = 5\text{ Vdc}$ and $I_{\text{LOAD}} = 50\text{ }\mu\text{A}$, in the circuit of [Full-Range Centigrade Temperature Sensor](#). These specifications also apply from 2°C to T_{MAX} in the circuit of [Figure 14](#).

PARAMETER	TEST CONDITIONS		LM35A			LM35CA			UNIT
			MIN	TYP	MAX	TYP	TYP	MAX	
Accuracy ⁽¹⁾	T _A = 25°C		±0.2			±0.2			°C
		Tested Limit ⁽²⁾	±0.5			±0.5			
		Design Limit ⁽³⁾							
	T _A = −10°C		±0.3			±0.3			
		Tested Limit ⁽²⁾							
		Design Limit ⁽³⁾				±1			
	T _A = T _{MAX}		±0.4			±0.4			
		Tested Limit ⁽²⁾	±1			±1			
		Design Limit ⁽³⁾							
	T _A = T _{MIN}		±0.4			±0.4			
		Tested Limit ⁽²⁾	±1						
		Design Limit ⁽³⁾				±1.5			
Nonlinearity ⁽⁴⁾	T _{MIN} ≤ T _A ≤ T _{MAX} , −40°C ≤ T _J ≤ 125°C		±0.18			±0.15			°C
		Tested Limit ⁽²⁾							
		Design Limit ⁽³⁾	±0.35			±0.3			
Sensor gain (average slope)	T _{MIN} ≤ T _A ≤ T _{MAX}		10			10			mV/°C
		Tested Limit ⁽²⁾	9.9						
		Design Limit ⁽³⁾				9.9			
	−40°C ≤ T _J ≤ 125°C		10			10			
		Tested Limit ⁽²⁾	10.1						
		Design Limit ⁽³⁾				10.1			
Load regulation ⁽⁵⁾ 0 ≤ I _L ≤ 1 mA	T _A = 25°C		±0.4			±0.4			mV/mA
		Tested Limit ⁽²⁾	±1			±1			
		Design Limit ⁽³⁾							
	T _{MIN} ≤ T _A ≤ T _{MAX} , −40°C ≤ T _J ≤ 125°C		±0.5			±0.5			
		Tested Limit ⁽²⁾							
		Design Limit ⁽³⁾	±3			±3			
Line regulation ⁽⁵⁾	T _A = 25°C		±0.01			±0.01			mV/V
		Tested Limit ⁽²⁾	±0.05			±0.05			
		Design Limit ⁽³⁾							
	4 V ≤ V _S ≤ 30 V, −40°C ≤ T _J ≤ 125°C		±0.02			±0.02			
		Tested Limit ⁽²⁾							
		Design Limit ⁽³⁾	±0.1			±0.1			

- (1) Accuracy is defined as the error between the output voltage and $10\text{ mV}/^{\circ}\text{C}$ times the case temperature of the device, at specified conditions of voltage, current, and temperature (expressed in $^{\circ}\text{C}$).
- (2) Tested Limits are ensured and 100% tested in production.
- (3) Design Limits are ensured (but not 100% production tested) over the indicated temperature and supply voltage ranges. These limits are not used to calculate outgoing quality levels.
- (4) Non-linearity is defined as the deviation of the output-voltage-versus-temperature curve from the best-fit straight line, over the rated temperature range of the device.
- (5) Regulation is measured at constant junction temperature, using pulse testing with a low duty cycle. Changes in output due to heating effects can be computed by multiplying the internal dissipation by the thermal resistance.

Electrical Characteristics: LM35A, LM35CA (continued)

Unless otherwise noted, these specifications apply: $-55^{\circ}\text{C} \leq T_J \leq 150^{\circ}\text{C}$ for the LM35 and LM35A; $-40^{\circ}\text{C} \leq T_J \leq 110^{\circ}\text{C}$ for the LM35C and LM35CA; and $0^{\circ}\text{C} \leq T_J \leq 100^{\circ}\text{C}$ for the LM35D. $V_S = 5\text{ Vdc}$ and $I_{\text{LOAD}} = 50\text{ }\mu\text{A}$, in the circuit of [Full-Range Centigrade Temperature Sensor](#). These specifications also apply from 2°C to T_{MAX} in the circuit of [Figure 14](#).

PARAMETER	TEST CONDITIONS		LM35A			LM35CA			UNIT
			MIN	TYP	MAX	TYP	TYP	MAX	
Quiescent current ⁽⁶⁾	$V_S = 5\text{ V}$, 25°C			56			56		μA
		Tested Limit ⁽²⁾			67			67	
		Design Limit ⁽³⁾							
	$V_S = 5\text{ V}$, $-40^{\circ}\text{C} \leq T_J \leq 125^{\circ}\text{C}$			105			91		
		Tested Limit ⁽²⁾							
		Design Limit ⁽³⁾			131			114	
	$V_S = 30\text{ V}$, 25°C			56.2			56.2		
		Tested Limit ⁽²⁾			68			68	
		Design Limit ⁽³⁾							
	$V_S = 30\text{ V}$, $-40^{\circ}\text{C} \leq T_J \leq 125^{\circ}\text{C}$			105.5			91.5		
		Tested Limit ⁽²⁾							
		Design Limit ⁽³⁾			133			116	
Change of quiescent current ⁽⁵⁾	$4\text{ V} \leq V_S \leq 30\text{ V}$, 25°C			0.2			0.2		μA
		Tested Limit ⁽²⁾			1			1	
		Design Limit ⁽³⁾							
	$4\text{ V} \leq V_S \leq 30\text{ V}$, $-40^{\circ}\text{C} \leq T_J \leq 125^{\circ}\text{C}$			0.5			0.5		
		Tested Limit ⁽²⁾							
		Design Limit ⁽³⁾			2			2	
Temperature coefficient of quiescent current	$-40^{\circ}\text{C} \leq T_J \leq 125^{\circ}\text{C}$			0.39			0.39		$\mu\text{A}/^{\circ}\text{C}$
		Tested Limit ⁽²⁾							
		Design Limit ⁽³⁾			0.5			0.5	
Minimum temperature for rate accuracy	In circuit of Figure 14 , $I_L = 0$			1.5			1.5		$^{\circ}\text{C}$
		Tested Limit ⁽²⁾							
		Design Limit ⁽³⁾			2			2	
Long term stability	$T_J = T_{\text{MAX}}$, for 1000 hours			± 0.08			± 0.08		$^{\circ}\text{C}$

(6) Quiescent current is defined in the circuit of [Figure 14](#).

LM35

SNIS159H – AUGUST 1999 – REVISED DECEMBER 2017

www.ti.com

6.7 Electrical Characteristics: LM35, LM35C, LM35D Limits

Unless otherwise noted, these specifications apply: $-55^{\circ}\text{C} \leq T_J \leq 150^{\circ}\text{C}$ for the LM35 and LM35A; $-40^{\circ}\text{C} \leq T_J \leq 110^{\circ}\text{C}$ for the LM35C and LM35CA; and $0^{\circ}\text{C} \leq T_J \leq 100^{\circ}\text{C}$ for the LM35D. $V_S = 5\text{ Vdc}$ and $I_{\text{LOAD}} = 50\text{ }\mu\text{A}$, in the circuit of [Full-Range Centigrade Temperature Sensor](#). These specifications also apply from 2°C to T_{MAX} in the circuit of [Figure 14](#).

PARAMETER	TEST CONDITIONS	LM35			LM35C, LM35D			UNIT
		TYP	TESTED LIMIT ⁽¹⁾	DESIGN LIMIT ⁽²⁾	TYP	TESTED LIMIT ⁽¹⁾	DESIGN LIMIT ⁽²⁾	
Accuracy, LM35, LM35C ⁽³⁾	$T_A = 25^{\circ}\text{C}$	± 0.4	± 1		± 0.4	± 1		$^{\circ}\text{C}$
	$T_A = -10^{\circ}\text{C}$	± 0.5			± 0.5		± 1.5	
	$T_A = T_{\text{MAX}}$	± 0.8	± 1.5		± 0.8		± 1.5	
	$T_A = T_{\text{MIN}}$	± 0.8		± 1.5	± 0.8		± 2	
Accuracy, LM35D ⁽³⁾	$T_A = 25^{\circ}\text{C}$				± 0.6	± 1.5		$^{\circ}\text{C}$
	$T_A = T_{\text{MAX}}$				± 0.9		± 2	
	$T_A = T_{\text{MIN}}$				± 0.9		± 2	
Nonlinearity ⁽⁴⁾	$T_{\text{MIN}} \leq T_A \leq T_{\text{MAX}}$, $-40^{\circ}\text{C} \leq T_J \leq 125^{\circ}\text{C}$	± 0.3		± 0.5	± 0.2		± 0.5	$^{\circ}\text{C}$
Sensor gain (average slope)	$T_{\text{MIN}} \leq T_A \leq T_{\text{MAX}}$, $-40^{\circ}\text{C} \leq T_J \leq 125^{\circ}\text{C}$	10	9.8		10		9.8	$\text{mV}/^{\circ}\text{C}$
		10	10.2		10		10.2	
Load regulation ⁽⁵⁾ $0 \leq I_L \leq 1\text{ mA}$	$T_A = 25^{\circ}\text{C}$	± 0.4	± 2		± 0.4	± 2		mV/mA
	$T_{\text{MIN}} \leq T_A \leq T_{\text{MAX}}$, $-40^{\circ}\text{C} \leq T_J \leq 125^{\circ}\text{C}$	± 0.5		± 5	± 0.5		± 5	
Line regulation ⁽⁵⁾	$T_A = 25^{\circ}\text{C}$	± 0.01	± 0.1		± 0.01	± 0.1		mV/V
	$4\text{ V} \leq V_S \leq 30\text{ V}$, $-40^{\circ}\text{C} \leq T_J \leq 125^{\circ}\text{C}$	± 0.02		± 0.2	± 0.02		± 0.2	
Quiescent current ⁽⁶⁾	$V_S = 5\text{ V}$, 25°C	56	80		56	80		μA
	$V_S = 5\text{ V}$, $-40^{\circ}\text{C} \leq T_J \leq 125^{\circ}\text{C}$	105		158	91		138	
	$V_S = 30\text{ V}$, 25°C	56.2	82		56.2	82		
	$V_S = 30\text{ V}$, $-40^{\circ}\text{C} \leq T_J \leq 125^{\circ}\text{C}$	105.5		161	91.5		141	
Change of quiescent current ⁽⁵⁾	$4\text{ V} \leq V_S \leq 30\text{ V}$, 25°C	0.2	2		0.2	2		μA
	$4\text{ V} \leq V_S \leq 30\text{ V}$, $-40^{\circ}\text{C} \leq T_J \leq 125^{\circ}\text{C}$	0.5		3	0.5		3	
Temperature coefficient of quiescent current	$-40^{\circ}\text{C} \leq T_J \leq 125^{\circ}\text{C}$	0.39		0.7	0.39		0.7	$\mu\text{A}/^{\circ}\text{C}$
Minimum temperature for rate accuracy	In circuit of Figure 14 , $I_L = 0$	1.5		2	1.5		2	$^{\circ}\text{C}$
Long term stability	$T_J = T_{\text{MAX}}$, for 1000 hours	± 0.08			± 0.08			$^{\circ}\text{C}$

- (1) Tested Limits are ensured and 100% tested in production.
- (2) Design Limits are ensured (but not 100% production tested) over the indicated temperature and supply voltage ranges. These limits are not used to calculate outgoing quality levels.
- (3) Accuracy is defined as the error between the output voltage and $10\text{ mV}/^{\circ}\text{C}$ times the case temperature of the device, at specified conditions of voltage, current, and temperature (expressed in $^{\circ}\text{C}$).
- (4) Non-linearity is defined as the deviation of the output-voltage-versus-temperature curve from the best-fit straight line, over the rated temperature range of the device.
- (5) Regulation is measured at constant junction temperature, using pulse testing with a low duty cycle. Changes in output due to heating effects can be computed by multiplying the internal dissipation by the thermal resistance.
- (6) Quiescent current is defined in the circuit of [Figure 14](#).

6.8 Electrical Characteristics: LM35, LM35C, LM35D

Unless otherwise noted, these specifications apply: $-55^{\circ}\text{C} \leq T_J \leq 150^{\circ}\text{C}$ for the LM35 and LM35A; $-40^{\circ}\text{C} \leq T_J \leq 110^{\circ}\text{C}$ for the LM35C and LM35CA; and $0^{\circ}\text{C} \leq T_J \leq 100^{\circ}\text{C}$ for the LM35D. $V_S = 5\text{ Vdc}$ and $I_{\text{LOAD}} = 50\text{ }\mu\text{A}$, in the circuit of [Full-Range Centigrade Temperature Sensor](#). These specifications also apply from 2°C to T_{MAX} in the circuit of [Figure 14](#).

PARAMETER	TEST CONDITIONS		LM35			LM35C, LM35D			UNIT
			MIN	TYP	MAX	MIN	TYP	MAX	
Accuracy, LM35, LM35C ⁽¹⁾	T _A = 25°C		±0.4			±0.4			°C
		Tested Limit ⁽²⁾	±1			±1			
		Design Limit ⁽³⁾							
	T _A = −10°C		±0.5			±0.5			
		Tested Limit ⁽²⁾							
		Design Limit ⁽³⁾				±1.5			
	T _A = T _{MAX}		±0.8			±0.8			
		Tested Limit ⁽²⁾	±1.5						
		Design Limit ⁽³⁾				±1.5			
	T _A = T _{MIN}		±0.8			±0.8			
		Tested Limit ⁽²⁾							
		Design Limit ⁽³⁾	±1.5			±2			
Accuracy, LM35D ⁽¹⁾	T _A = 25°C					±0.6			°C
		Tested Limit ⁽²⁾				±1.5			
		Design Limit ⁽³⁾							
	T _A = T _{MAX}					±0.9			
		Tested Limit ⁽²⁾							
		Design Limit ⁽³⁾				±2			
	T _A = T _{MIN}					±0.9			
		Tested Limit ⁽²⁾							
		Design Limit ⁽³⁾				±2			
Nonlinearity ⁽⁴⁾	T _{MIN} ≤ T _A ≤ T _{MAX} , −40°C ≤ T _J ≤ 125°C		±0.3			±0.2			°C
		Tested Limit ⁽²⁾							
		Design Limit ⁽³⁾	±0.5			±0.5			
Sensor gain (average slope)	T _{MIN} ≤ T _A ≤ T _{MAX} , −40°C ≤ T _J ≤ 125°C		10			10			mV/°C
		Tested Limit ⁽²⁾	9.8						
		Design Limit ⁽³⁾				9.8			
			10			10			
		Tested Limit ⁽²⁾	10.2						
		Design Limit ⁽³⁾				10.2			
Load regulation ⁽⁵⁾ 0 ≤ I _L ≤ 1 mA	T _A = 25°C		±0.4			±0.4			mV/mA
		Tested Limit ⁽²⁾	±2						
		Design Limit ⁽³⁾							
	T _{MIN} ≤ T _A ≤ T _{MAX} , −40°C ≤ T _J ≤ 125°C		±0.5			±0.5			
		Tested Limit ⁽²⁾							
		Design Limit ⁽³⁾	±5			±5			

- (1) Accuracy is defined as the error between the output voltage and $10\text{ mV}/^{\circ}\text{C}$ times the case temperature of the device, at specified conditions of voltage, current, and temperature (expressed in $^{\circ}\text{C}$).
- (2) Tested Limits are ensured and 100% tested in production.
- (3) Design Limits are ensured (but not 100% production tested) over the indicated temperature and supply voltage ranges. These limits are not used to calculate outgoing quality levels.
- (4) Non-linearity is defined as the deviation of the output-voltage-versus-temperature curve from the best-fit straight line, over the rated temperature range of the device.
- (5) Regulation is measured at constant junction temperature, using pulse testing with a low duty cycle. Changes in output due to heating effects can be computed by multiplying the internal dissipation by the thermal resistance.

Electrical Characteristics: LM35, LM35C, LM35D (continued)

Unless otherwise noted, these specifications apply: $-55^{\circ}\text{C} \leq T_J \leq 150^{\circ}\text{C}$ for the LM35 and LM35A; $-40^{\circ}\text{C} \leq T_J \leq 110^{\circ}\text{C}$ for the LM35C and LM35CA; and $0^{\circ}\text{C} \leq T_J \leq 100^{\circ}\text{C}$ for the LM35D. $V_S = 5\text{ Vdc}$ and $I_{\text{LOAD}} = 50\text{ }\mu\text{A}$, in the circuit of [Full-Range Centigrade Temperature Sensor](#). These specifications also apply from 2°C to T_{MAX} in the circuit of [Figure 14](#).

PARAMETER	TEST CONDITIONS		LM35			LM35C, LM35D			UNIT
			MIN	TYP	MAX	MIN	TYP	MAX	
Line regulation ⁽⁵⁾	T _A = 25°C		±0.01			±0.01			mV/V
		Tested Limit ⁽²⁾	±0.1						
		Design Limit ⁽³⁾				±0.1			
	4 V ≤ V _S ≤ 30 V, −40°C ≤ T _J ≤ 125°C		±0.02			±0.02			
		Tested Limit ⁽²⁾							
		Design Limit ⁽³⁾	±0.2			±0.2			
Quiescent current ⁽⁶⁾	V _S = 5 V, 25°C		56			56			μA
		Tested Limit ⁽²⁾	80			80			
		Design Limit ⁽³⁾							
	V _S = 5 V, −40°C ≤ T _J ≤ 125°C		105			91			
		Tested Limit ⁽²⁾							
		Design Limit ⁽³⁾	158			138			
	V _S = 30 V, 25°C		56.2			56.2			
		Tested Limit ⁽²⁾	82			82			
		Design Limit ⁽³⁾							
	V _S = 30 V, −40°C ≤ T _J ≤ 125°C		105.5			91.5			
		Tested Limit ⁽²⁾							
		Design Limit ⁽³⁾	161			141			
Change of quiescent current ⁽⁵⁾	4 V ≤ V _S ≤ 30 V, 25°C		0.2			0.2			μA
		Tested Limit ⁽²⁾				2			
		Design Limit ⁽³⁾	2						
	4 V ≤ V _S ≤ 30 V, −40°C ≤ T _J ≤ 125°C		0.5			0.5			
		Tested Limit ⁽²⁾							
		Design Limit ⁽³⁾	3			3			
Temperature coefficient of quiescent current	−40°C ≤ T _J ≤ 125°C		0.39			0.39			μA/°C
		Tested Limit ⁽²⁾							
		Design Limit ⁽³⁾	0.7			0.7			
Minimum temperature for rate accuracy	In circuit of Figure 14 , I _L = 0		1.5			1.5			°C
		Tested Limit ⁽²⁾							
		Design Limit ⁽³⁾	2			2			
Long term stability	T _J = T _{MAX} , for 1000 hours		±0.08			±0.08			°C

(6) Quiescent current is defined in the circuit of [Figure 14](#).

6.9 Typical Characteristics

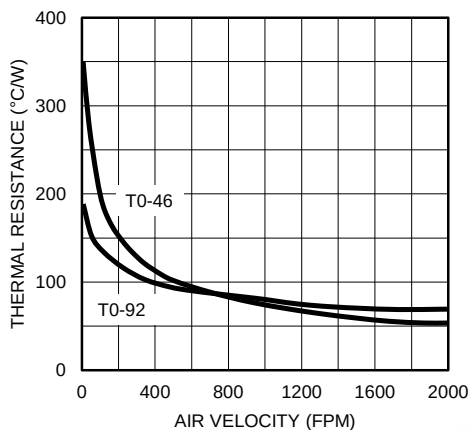


Figure 1. Thermal Resistance Junction To Air

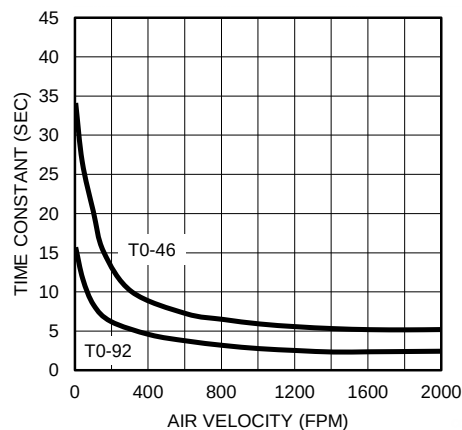


Figure 2. Thermal Time Constant

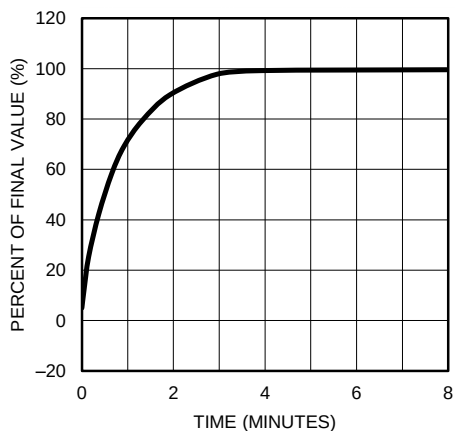


Figure 3. Thermal Response In Still Air

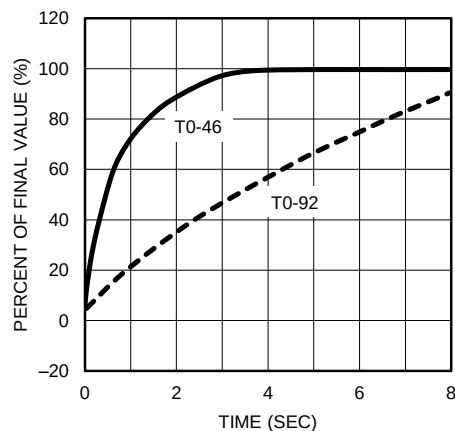


Figure 4. Thermal Response In Stirred Oil Bath

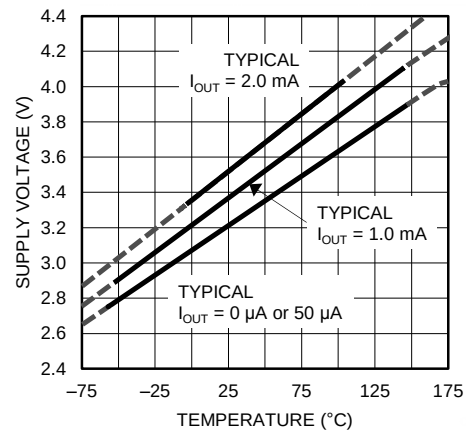


Figure 5. Minimum Supply Voltage vs Temperature

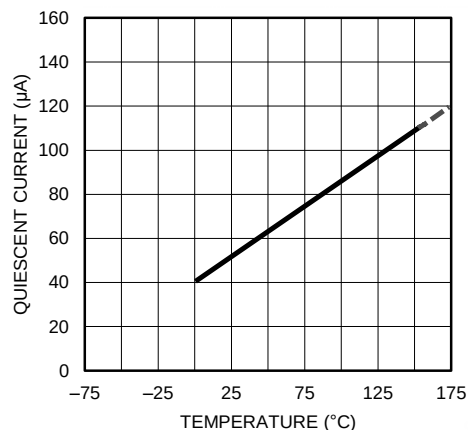


Figure 6. Quiescent Current vs Temperature (in Circuit of Figure 14)

Typical Characteristics (continued)

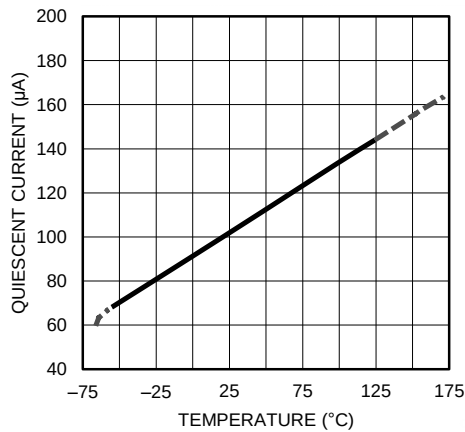


Figure 7. Quiescent Current vs Temperature (in Circuit of Full-Range Centigrade Temperature Sensor)

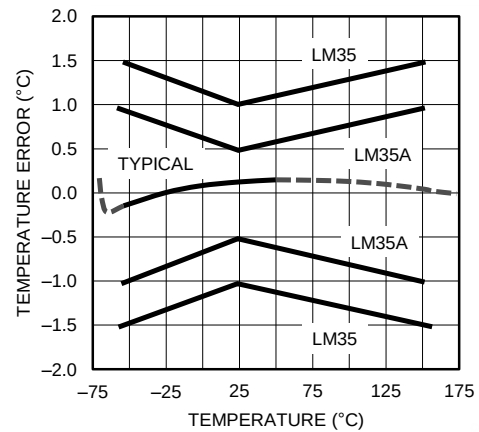


Figure 8. Accuracy vs Temperature (Ensured)

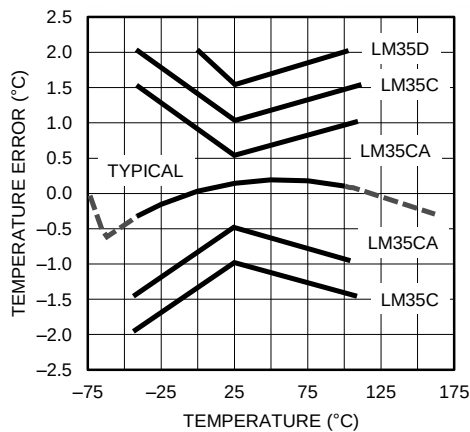


Figure 9. Accuracy vs Temperature (Ensured)

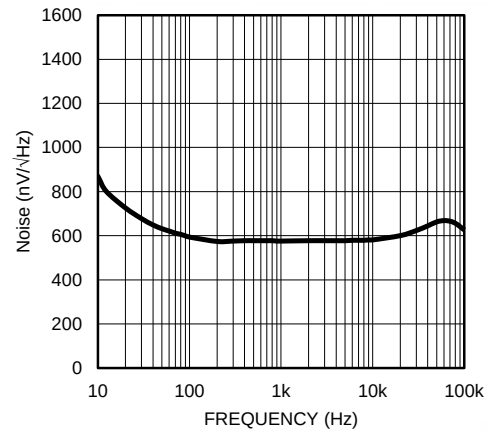


Figure 10. Noise Voltage

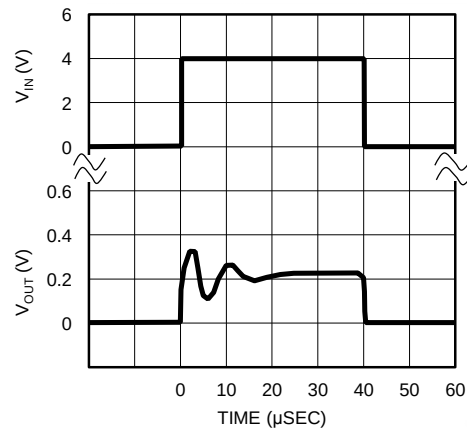


Figure 11. Start-Up Response

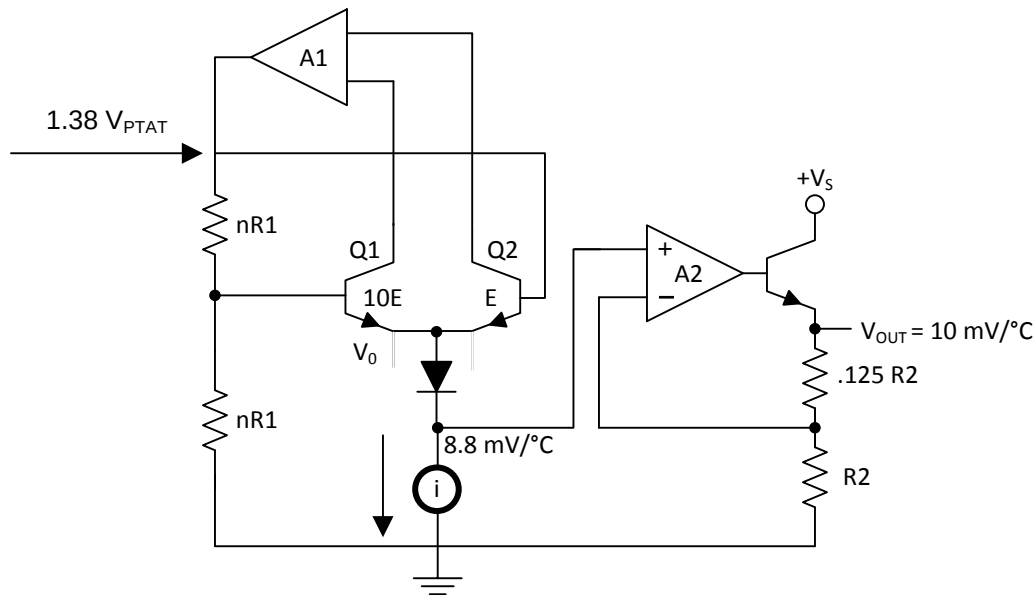
7 Detailed Description

7.1 Overview

The LM35-series devices are precision integrated-circuit temperature sensors, with an output voltage linearly proportional to the Centigrade temperature. The LM35 device has an advantage over linear temperature sensors calibrated in Kelvin, as the user is not required to subtract a large constant voltage from the output to obtain convenient Centigrade scaling. The LM35 device does not require any external calibration or trimming to provide typical accuracies of $\pm \frac{1}{4}^{\circ}\text{C}$ at room temperature and $\pm \frac{3}{4}^{\circ}\text{C}$ over a full -55°C to 150°C temperature range. Lower cost is assured by trimming and calibration at the wafer level. The low output impedance, linear output, and precise inherent calibration of the LM35 device makes interfacing to readout or control circuitry especially easy. The device is used with single power supplies, or with plus and minus supplies. As the LM35 device draws only $60\text{ }\mu\text{A}$ from the supply, it has very low self-heating of less than 0.1°C in still air. The LM35 device is rated to operate over a -55°C to 150°C temperature range, while the LM35C device is rated for a -40°C to 110°C range (-10° with improved accuracy). The temperature-sensing element is comprised of a delta-V BE architecture.

The temperature-sensing element is then buffered by an amplifier and provided to the VOUT pin. The amplifier has a simple class A output stage with typical $0.5\text{-}\Omega$ output impedance as shown in the [Functional Block Diagram](#). Therefore the LM35 can only source current and its sinking capability is limited to $1\text{ }\mu\text{A}$.

7.2 Functional Block Diagram



7.3 Feature Description

7.3.1 LM35 Transfer Function

The accuracy specifications of the LM35 are given with respect to a simple linear transfer function:

$$V_{\text{OUT}} = 10\text{ mV}/^{\circ}\text{C} \times T$$

where

- V_{OUT} is the LM35 output voltage
- T is the temperature in $^{\circ}\text{C}$

(1)

7.4 Device Functional Modes

The only functional mode of the LM35 is that it has an analog output directly proportional to temperature.

8 Application and Implementation

NOTE

Information in the following applications sections is not part of the TI component specification, and TI does not warrant its accuracy or completeness. TI's customers are responsible for determining suitability of components for their purposes. Customers should validate and test their design implementation to confirm system functionality.

8.1 Application Information

The features of the LM35 make it suitable for many general temperature sensing applications. Multiple package options expand on its flexibility.

8.1.1 Capacitive Drive Capability

Like most micropower circuits, the LM35 device has a limited ability to drive heavy capacitive loads. Alone, the LM35 device is able to drive 50 pF without special precautions. If heavier loads are anticipated, isolating or decoupling the load with a resistor is easy (see Figure 12). The tolerance of capacitance can be improved with a series R-C damper from output to ground (see Figure 13).

When the LM35 device is applied with a 200-Ω load resistor as shown in Figure 16, Figure 17, or Figure 19, the device is relatively immune to wiring capacitance because the capacitance forms a bypass from ground to input and not on the output. However, as with any linear circuit connected to wires in a hostile environment, performance is affected adversely by intense electromagnetic sources (such as relays, radio transmitters, motors with arcing brushes, and SCR transients), because the wiring acts as a receiving antenna and the internal junctions act as rectifiers. For best results in such cases, a bypass capacitor from V_{IN} to ground and a series R-C damper, such as 75 Ω in series with 0.2 or 1 μF from output to ground, are often useful. Examples are shown in Figure 13, Figure 24, and Figure 25.

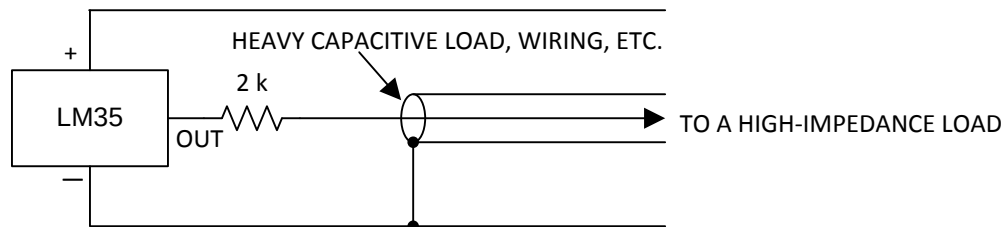


Figure 12. LM35 with Decoupling from Capacitive Load

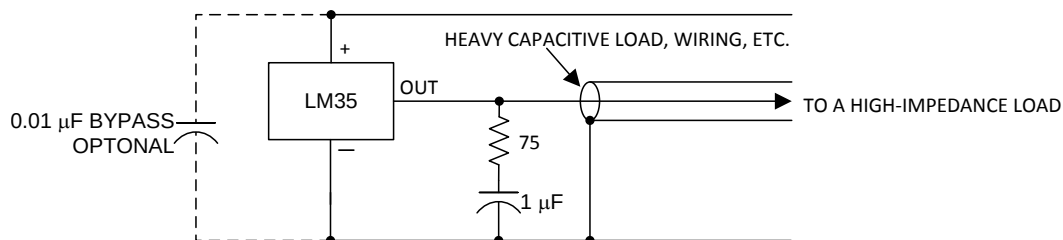


Figure 13. LM35 with R-C Damper

8.2 Typical Application

8.2.1 Basic Centigrade Temperature Sensor

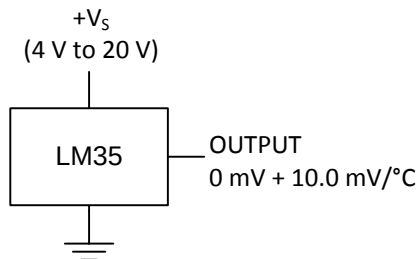


Figure 14. Basic Centigrade Temperature Sensor (2 °C to 150 °C)

8.2.1.1 Design Requirements

Table 1. Design Parameters

PARAMETER	VALUE
Accuracy at 25°C	±0.5°C
Accuracy from –55 °C to 150°C	±1°C
Temperature Slope	10 mV/°C

8.2.1.2 Detailed Design Procedure

Because the LM35 device is a simple temperature sensor that provides an analog output, design requirements related to layout are more important than electrical requirements. For a detailed description, refer to the [Layout](#).

8.2.1.3 Application Curve

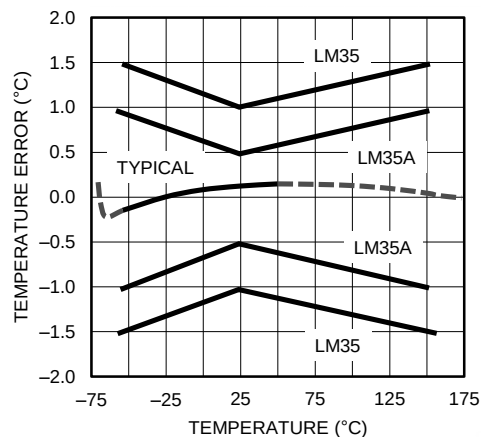


Figure 15. Accuracy vs Temperature (Ensured)

8.3 System Examples

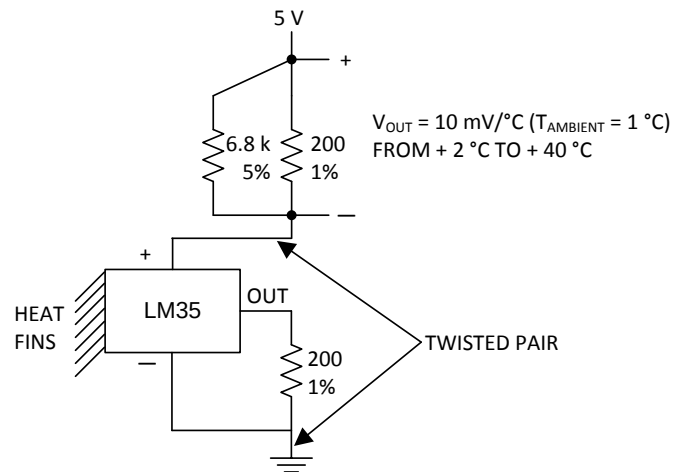


Figure 16. Two-Wire Remote Temperature Sensor (Grounded Sensor)

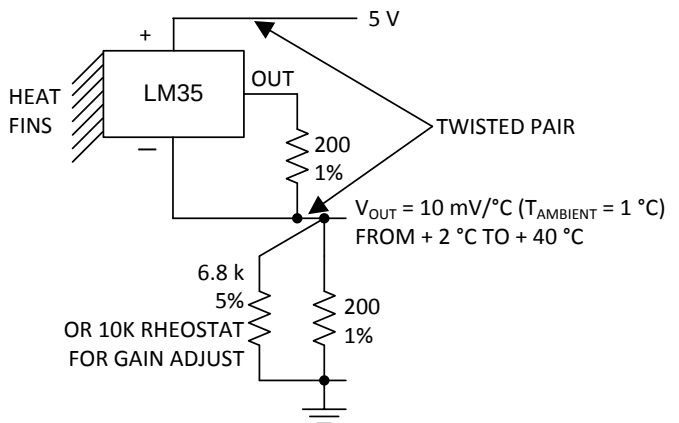


Figure 17. Two-Wire Remote Temperature Sensor (Output Referred to Ground)

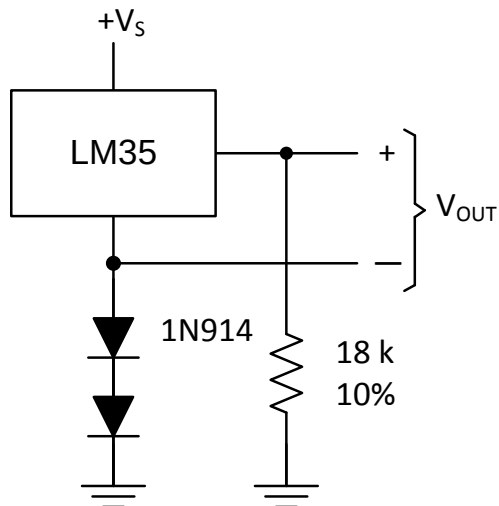


Figure 18. Temperature Sensor, Single Supply (-55° to +150°C)

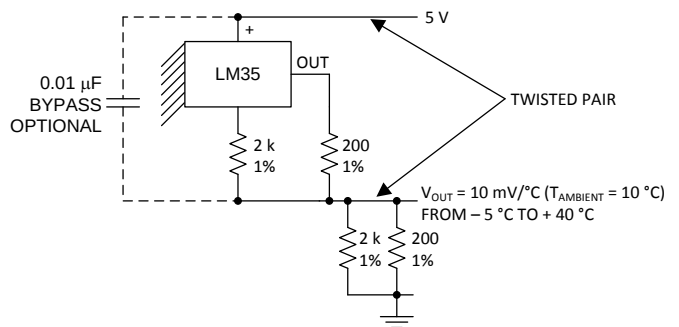
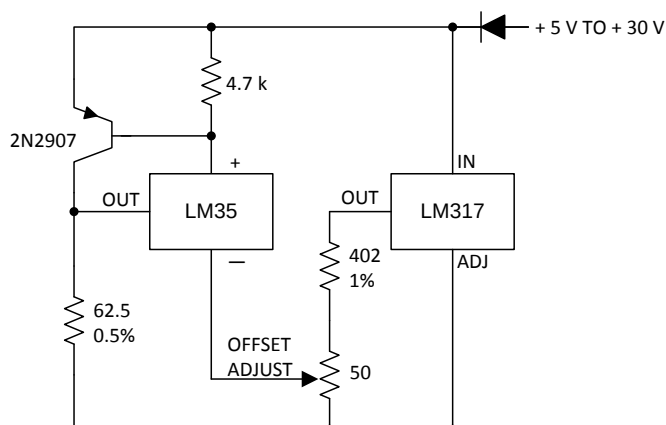
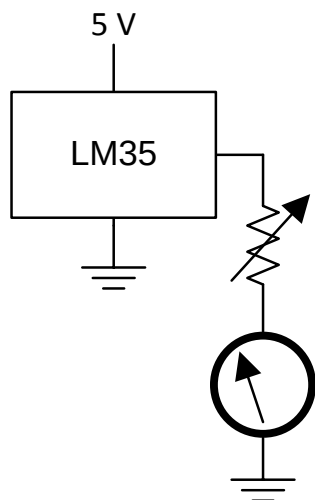


Figure 19. Two-Wire Remote Temperature Sensor (Output Referred to Ground)

System Examples (continued)



**Figure 20. 4-To-20 mA Current Source
(0°C to 100°C)**



**Figure 22. Centigrade Thermometer
(Analog Meter)**

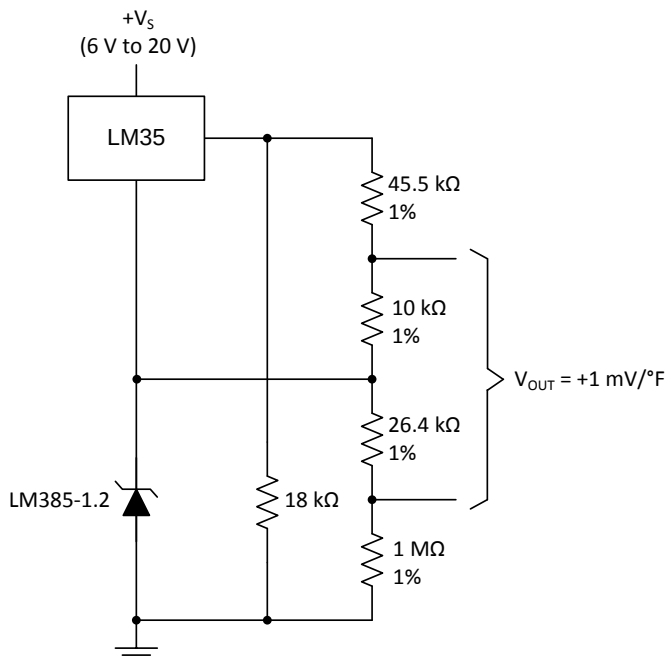
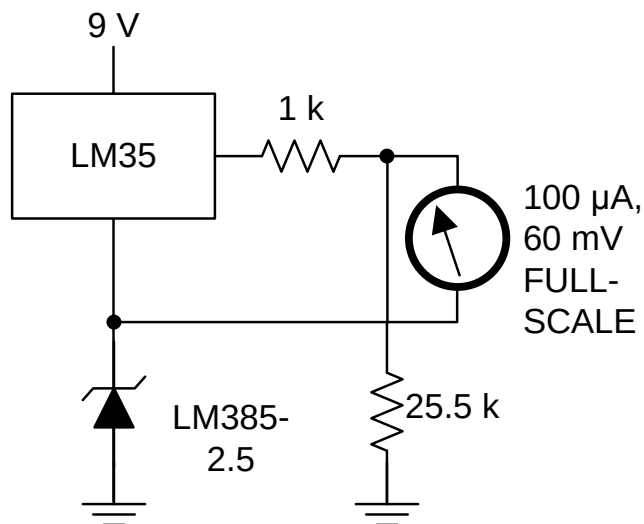


Figure 21. Fahrenheit Thermometer



**Figure 23. Fahrenheit Thermometer, Expanded
Scale Thermometer
(50°F to 80°F, for Example Shown)**

System Examples (continued)

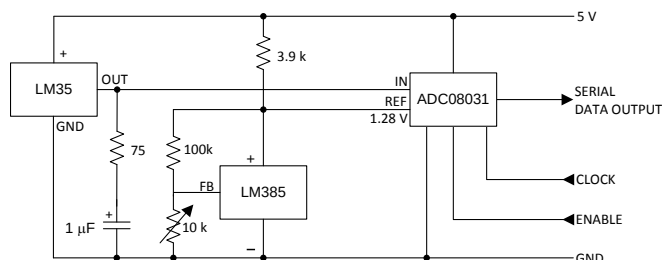


Figure 24. Temperature to Digital Converter (Serial Output) (128°C Full Scale)

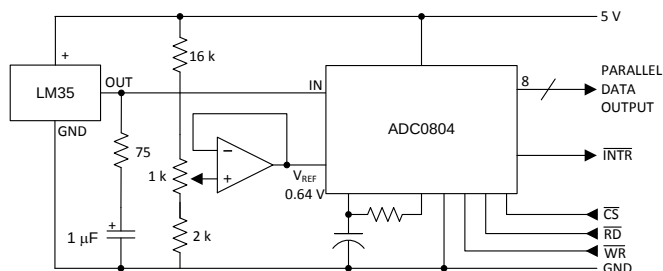
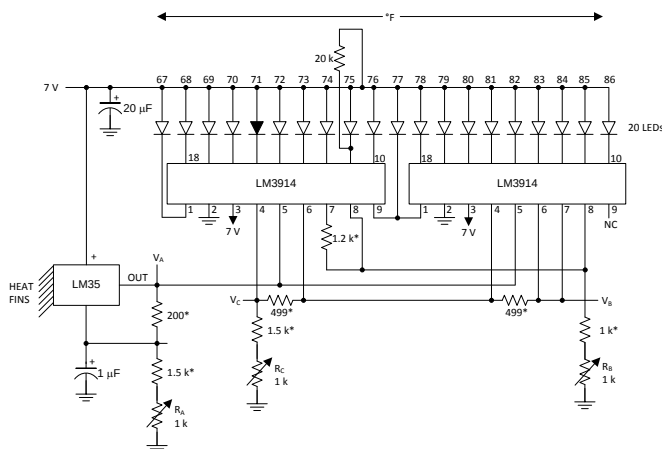


Figure 25. Temperature to Digital Converter (Parallel TRI-STATE Outputs for Standard Data Bus to μ P Interface) (128°C Full Scale)



*=1% or 2% film resistor

Trim R_B for $V_B = 3.075$ V

Trim R_C for $V_C = 1.955$ V

Trim R_A for $V_A = 0.075$ V + 100 mV/°C $\times T_{\text{ambient}}$

Example, $V_A = 2.275$ V at 22°C

Figure 26. Bar-Graph Temperature Display (Dot Mode)

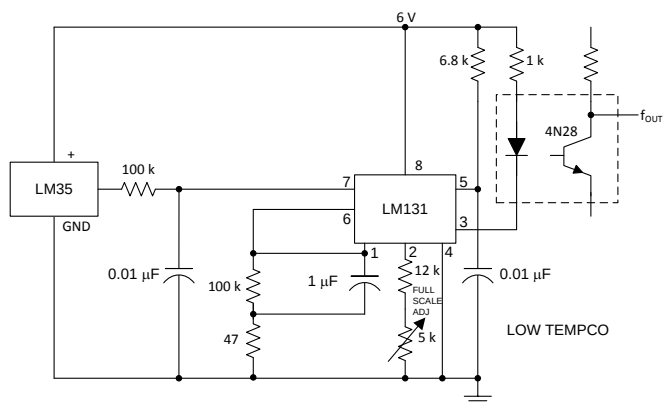


Figure 27. LM35 With Voltage-To-Frequency Converter and Isolated Output (2°C to 150°C; 20 to 1500 Hz)

9 Power Supply Recommendations

The LM35 device has a very wide 4-V to 30-V power supply voltage range, which makes it ideal for many applications. In noisy environments, TI recommends adding a 0.1 μF from V+ to GND to bypass the power supply voltage. Larger capacitances may be required and are dependent on the power-supply noise.

10 Layout

10.1 Layout Guidelines

The LM35 is easily applied in the same way as other integrated-circuit temperature sensors. Glue or cement the device to a surface and the temperature should be within about 0.01°C of the surface temperature.

The 0.01°C proximity presumes that the ambient air temperature is almost the same as the surface temperature. If the air temperature were much higher or lower than the surface temperature, the actual temperature of the LM35 die would be at an intermediate temperature between the surface temperature and the air temperature; this is especially true for the TO-92 plastic package. The copper leads in the TO-92 package are the principal thermal path to carry heat into the device, so its temperature might be closer to the air temperature than to the surface temperature.

Ensure that the wiring leaving the LM35 device is held at the same temperature as the surface of interest to minimize the temperature problem. The easiest fix is to cover up these wires with a bead of epoxy. The epoxy bead will ensure that the leads and wires are all at the same temperature as the surface, and that the temperature of the LM35 die is not affected by the air temperature.

The TO-46 metal package can also be soldered to a metal surface or pipe without damage. Of course, in that case the V– terminal of the circuit will be grounded to that metal. Alternatively, mount the LM35 inside a sealed-end metal tube, and then dip into a bath or screw into a threaded hole in a tank. As with any IC, the LM35 device and accompanying wiring and circuits must be kept insulated and dry, to avoid leakage and corrosion. This is especially true if the circuit may operate at cold temperatures where condensation can occur. Printed-circuit coatings and varnishes such as a conformal coating and epoxy paints or dips are often used to insure that moisture cannot corrode the LM35 device or its connections.

These devices are sometimes soldered to a small light-weight heat fin to decrease the thermal time constant and speed up the response in slowly-moving air. On the other hand, a small thermal mass may be added to the sensor, to give the steadiest reading despite small deviations in the air temperature.

Table 2. Temperature Rise of LM35 Due To Self-heating (Thermal Resistance, $R_{\theta JA}$)

	TO, no heat sink	TO ⁽¹⁾ , small heat fin	TO-92, no heat sink	TO-92 ⁽²⁾ , small heat fin	SOIC-8, no heat sink	SOIC-8 ⁽²⁾ , small heat fin	TO-220, no heat sink
Still air	400°C/W	100°C/W	180°C/W	140°C/W	220°C/W	110°C/W	90°C/W
Moving air	100°C/W	40°C/W	90°C/W	70°C/W	105°C/W	90°C/W	26°C/W
Still oil	100°C/W	40°C/W	90°C/W	70°C/W	—	—	—
Stirred oil	50°C/W	30°C/W	45°C/W	40°C/W	—	—	—
(Clamped to metal, Infinite heat sink)	(24°C/W)		—	—	(55°C/W)		—

(1) Wakefield type 201, or 1-in disc of 0.02-in sheet brass, soldered to case, or similar.

(2) TO-92 and SOIC-8 packages glued and leads soldered to 1-in square of 1/16-in printed circuit board with 2-oz foil or similar.

10.2 Layout Example

○ VIA to ground plane

○ VIA to power plane

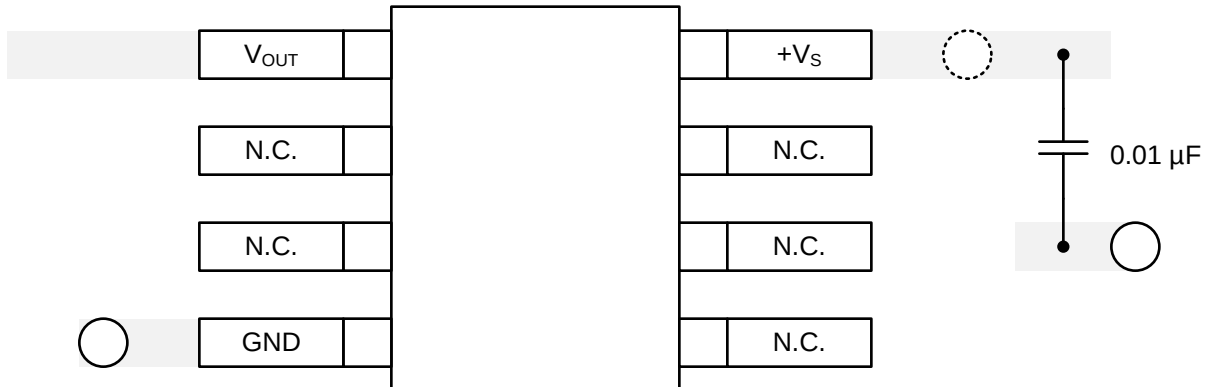


Figure 28. Layout Example

11 Device and Documentation Support

11.1 Receiving Notification of Documentation Updates

To receive notification of documentation updates, navigate to the device product folder on ti.com. In the upper right corner, click on *Alert me* to register and receive a weekly digest of any product information that has changed. For change details, review the revision history included in any revised document

11.2 Community Resources

The following links connect to TI community resources. Linked contents are provided "AS IS" by the respective contributors. They do not constitute TI specifications and do not necessarily reflect TI's views; see TI's [Terms of Use](#).

TI E2E™ Online Community *TI's Engineer-to-Engineer (E2E) Community*. Created to foster collaboration among engineers. At e2e.ti.com, you can ask questions, share knowledge, explore ideas and help solve problems with fellow engineers.

Design Support *TI's Design Support* Quickly find helpful E2E forums along with design support tools and contact information for technical support.

11.3 Trademarks

E2E is a trademark of Texas Instruments.
All other trademarks are the property of their respective owners.

11.4 Electrostatic Discharge Caution



These devices have limited built-in ESD protection. The leads should be shorted together or the device placed in conductive foam during storage or handling to prevent electrostatic damage to the MOS gates.

11.5 Glossary

[SLYZ022](#) — *TI Glossary*.

This glossary lists and explains terms, acronyms, and definitions.

12 Mechanical, Packaging, and Orderable Information

The following pages include mechanical, packaging, and orderable information. This information is the most current data available for the designated devices. This data is subject to change without notice and revision of this document. For browser-based versions of this data sheet, refer to the left-hand navigation.

PACKAGING INFORMATION

Orderable Device	Status (1)	Package Type	Package Drawing	Pins	Package Qty	Eco Plan (2)	Lead/Ball Finish (6)	MSL Peak Temp (3)	Op Temp (°C)	Device Marking (4/5)	Samples
LM35AH	ACTIVE	TO	NDV	3	500	TBD	Call TI	Call TI	-55 to 150	(LM35AH, LM35AH)	Samples
LM35AH/NOPB	ACTIVE	TO	NDV	3	500	Green (RoHS & no Sb/Br)	Call TI	Level-1-NA-UNLIM	-55 to 150	(LM35AH, LM35AH)	Samples
LM35CAH	ACTIVE	TO	NDV	3	500	TBD	Call TI	Call TI	-40 to 110	(LM35CAH, LM35CAH)	Samples
LM35CAH/NOPB	ACTIVE	TO	NDV	3	500	Green (RoHS & no Sb/Br)	Call TI	Level-1-NA-UNLIM	-40 to 110	(LM35CAH, LM35CAH)	Samples
LM35CAZ/LFT4	ACTIVE	TO-92	LP	3	2000	Green (RoHS & no Sb/Br)	SN	N / A for Pkg Type		LM35 CAZ	Samples
LM35CAZ/NOPB	ACTIVE	TO-92	LP	3	1800	Green (RoHS & no Sb/Br)	SN	N / A for Pkg Type	-40 to 110	LM35 CAZ	Samples
LM35CH	ACTIVE	TO	NDV	3	500	TBD	Call TI	Call TI	-40 to 110	(LM35CH, LM35CH)	Samples
LM35CH/NOPB	ACTIVE	TO	NDV	3	500	Green (RoHS & no Sb/Br)	Call TI	Level-1-NA-UNLIM	-40 to 110	(LM35CH, LM35CH)	Samples
LM35CZ/LFT1	ACTIVE	TO-92	LP	3	2000	Green (RoHS & no Sb/Br)	SN	N / A for Pkg Type		LM35 CZ	Samples
LM35CZ/NOPB	ACTIVE	TO-92	LP	3	1800	Green (RoHS & no Sb/Br)	SN	N / A for Pkg Type	-40 to 110	LM35 CZ	Samples
LM35DH	ACTIVE	TO	NDV	3	1000	TBD	Call TI	Call TI	0 to 70	(LM35DH, LM35DH)	Samples
LM35DH/NOPB	ACTIVE	TO	NDV	3	1000	Green (RoHS & no Sb/Br)	Call TI	Level-1-NA-UNLIM	0 to 70	(LM35DH, LM35DH)	Samples
LM35DM	NRND	SOIC	D	8	95	TBD	Call TI	Call TI	0 to 100	LM35D M	
LM35DM/NOPB	ACTIVE	SOIC	D	8	95	Green (RoHS & no Sb/Br)	SN	Level-1-260C-UNLIM	0 to 100	LM35D M	Samples
LM35DMX	NRND	SOIC	D	8	2500	TBD	Call TI	Call TI	0 to 100	LM35D M	
LM35DMX/NOPB	ACTIVE	SOIC	D	8	2500	Green (RoHS & no Sb/Br)	SN	Level-1-260C-UNLIM	0 to 100	LM35D M	Samples
LM35DT	NRND	TO-220	NEB	3	45	TBD	Call TI	Call TI	0 to 100	LM35DT	
LM35DT/NOPB	ACTIVE	TO-220	NEB	3	45	Green (RoHS & no Sb/Br)	SN	Level-1-NA-UNLIM	0 to 100	LM35DT	Samples

Orderable Device	Status (1)	Package Type	Package Drawing	Pins	Package Qty	Eco Plan (2)	Lead/Ball Finish (6)	MSL Peak Temp (3)	Op Temp (°C)	Device Marking (4/5)	Samples
LM35DZ/LFT1	ACTIVE	TO-92	LP	3	2000	Green (RoHS & no Sb/Br)	SN	N / A for Pkg Type		LM35 DZ	Samples
LM35DZ/LFT4	ACTIVE	TO-92	LP	3	2000	Green (RoHS & no Sb/Br)	SN	N / A for Pkg Type		LM35 DZ	Samples
LM35DZ/NOPB	ACTIVE	TO-92	LP	3	1800	Green (RoHS & no Sb/Br)	SN	N / A for Pkg Type	0 to 100	LM35 DZ	Samples
LM35H	ACTIVE	TO	NDV	3	500	TBD	Call TI	Call TI	-55 to 150	(LM35H, LM35H)	Samples
LM35H/NOPB	ACTIVE	TO	NDV	3	500	Green (RoHS & no Sb/Br)	Call TI	Level-1-NA-UNLIM	-55 to 150	(LM35H, LM35H)	Samples

(1) The marketing status values are defined as follows:

ACTIVE: Product device recommended for new designs.

LIFEBUY: TI has announced that the device will be discontinued, and a lifetime-buy period is in effect.

NRND: Not recommended for new designs. Device is in production to support existing customers, but TI does not recommend using this part in a new design.

PREVIEW: Device has been announced but is not in production. Samples may or may not be available.

OBsolete: TI has discontinued the production of the device.

(2) **RoHS:** TI defines "RoHS" to mean semiconductor products that are compliant with the current EU RoHS requirements for all 10 RoHS substances, including the requirement that RoHS substance do not exceed 0.1% by weight in homogeneous materials. Where designed to be soldered at high temperatures, "RoHS" products are suitable for use in specified lead-free processes. TI may reference these types of products as "Pb-Free".

RoHS Exempt: TI defines "RoHS Exempt" to mean products that contain lead but are compliant with EU RoHS pursuant to a specific EU RoHS exemption.

Green: TI defines "Green" to mean the content of Chlorine (Cl) and Bromine (Br) based flame retardants meet JS709B low halogen requirements of <=1000ppm threshold. Antimony trioxide based flame retardants must also meet the <=1000ppm threshold requirement.

(3) MSL, Peak Temp. - The Moisture Sensitivity Level rating according to the JEDEC industry standard classifications, and peak solder temperature.

(4) There may be additional marking, which relates to the logo, the lot trace code information, or the environmental category on the device.

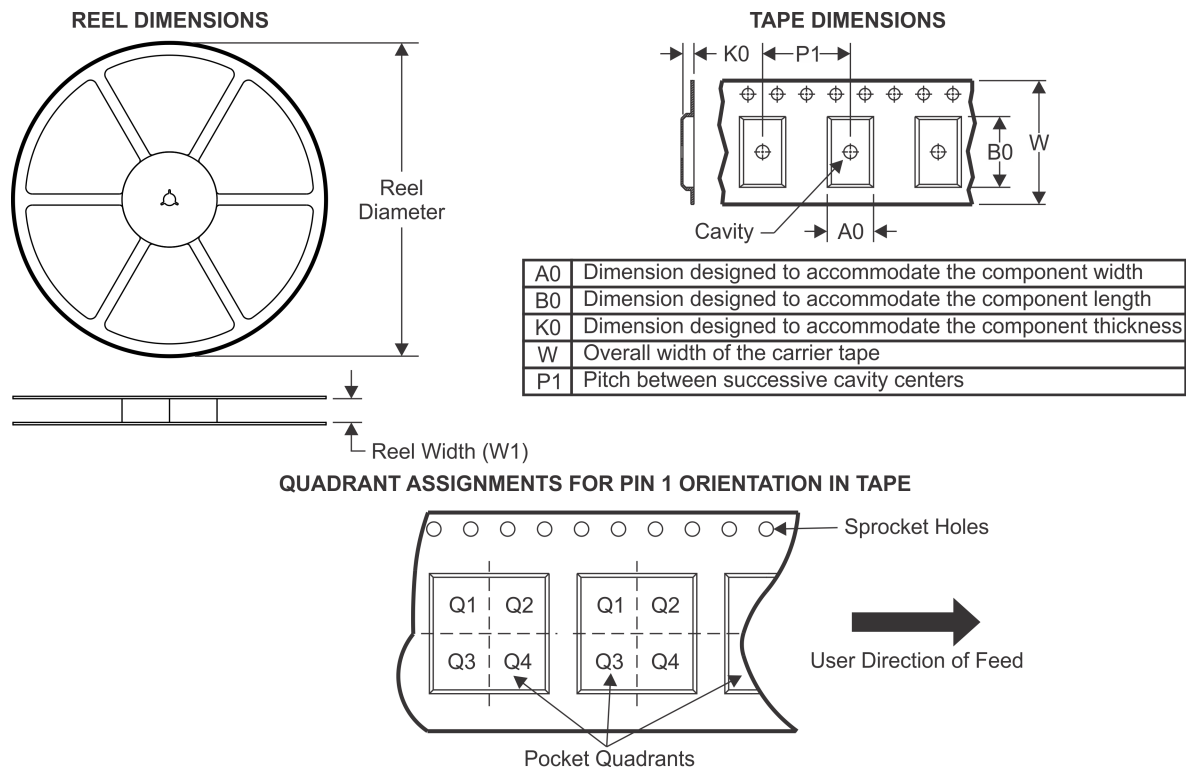
(5) Multiple Device Markings will be inside parentheses. Only one Device Marking contained in parentheses and separated by a "~" will appear on a device. If a line is indented then it is a continuation of the previous line and the two combined represent the entire Device Marking for that device.

(6) Lead/Ball Finish - Orderable Devices may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead/Ball Finish values may wrap to two lines if the finish value exceeds the maximum column width.

Important Information and Disclaimer: The information provided on this page represents TI's knowledge and belief as of the date that it is provided. TI bases its knowledge and belief on information provided by third parties, and makes no representation or warranty as to the accuracy of such information. Efforts are underway to better integrate information from third parties. TI has taken and

continues to take reasonable steps to provide representative and accurate information but may not have conducted destructive testing or chemical analysis on incoming materials and chemicals. TI and TI suppliers consider certain information to be proprietary, and thus CAS numbers and other limited information may not be available for release.

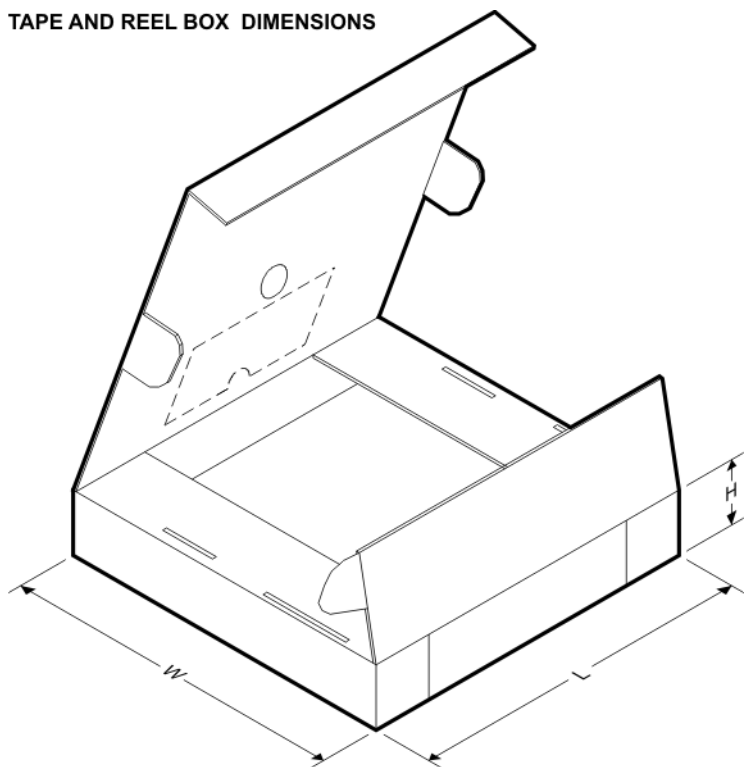
In no event shall TI's liability arising out of such information exceed the total purchase price of the TI part(s) at issue in this document sold by TI to Customer on an annual basis.

TAPE AND REEL INFORMATION


*All dimensions are nominal

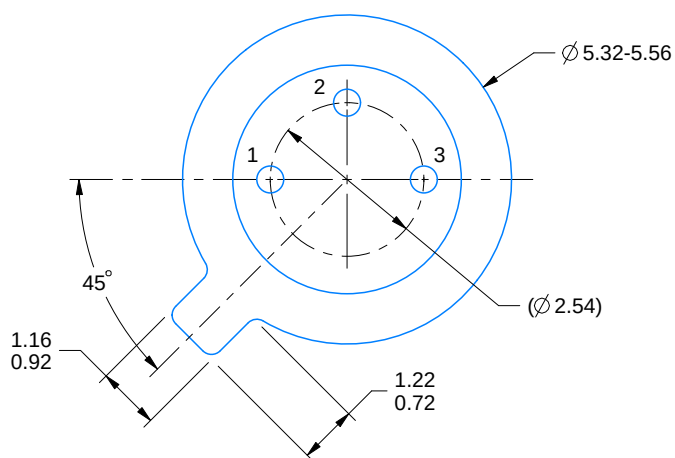
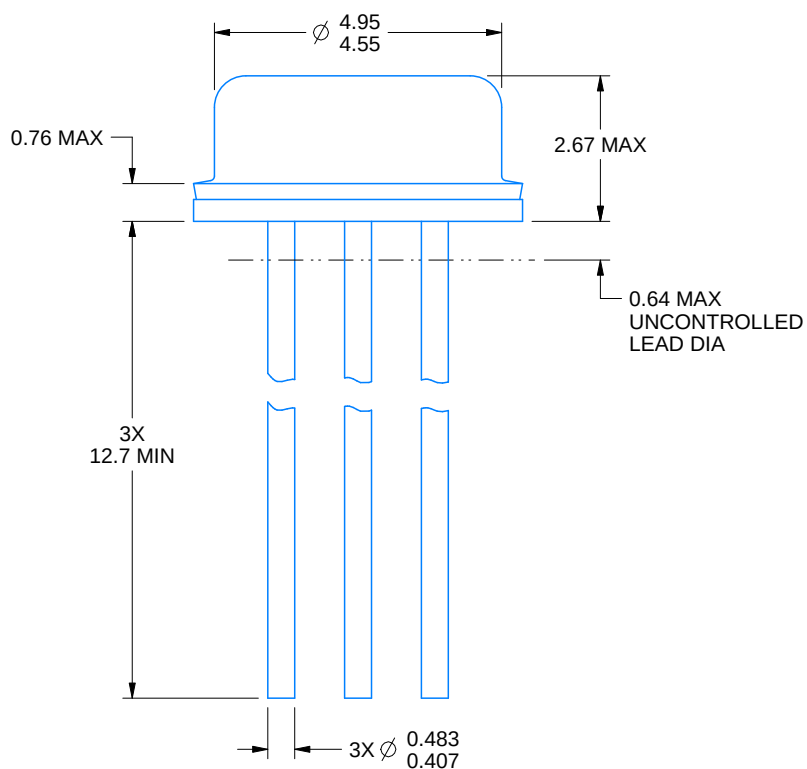
Device	Package Type	Package Drawing	Pins	SPQ	Reel Diameter (mm)	Reel Width W1 (mm)	A0 (mm)	B0 (mm)	K0 (mm)	P1 (mm)	W (mm)	Pin1 Quadrant
LM35DMX	SOIC	D	8	2500	330.0	12.4	6.5	5.4	2.0	8.0	12.0	Q1
LM35DMX/NOPB	SOIC	D	8	2500	330.0	12.4	6.5	5.4	2.0	8.0	12.0	Q1

TAPE AND REEL BOX DIMENSIONS



*All dimensions are nominal

Device	Package Type	Package Drawing	Pins	SPQ	Length (mm)	Width (mm)	Height (mm)
LM35DMX	SOIC	D	8	2500	367.0	367.0	35.0
LM35DMX/NOPB	SOIC	D	8	2500	367.0	367.0	35.0



4219876/A 01/2017

NOTES:

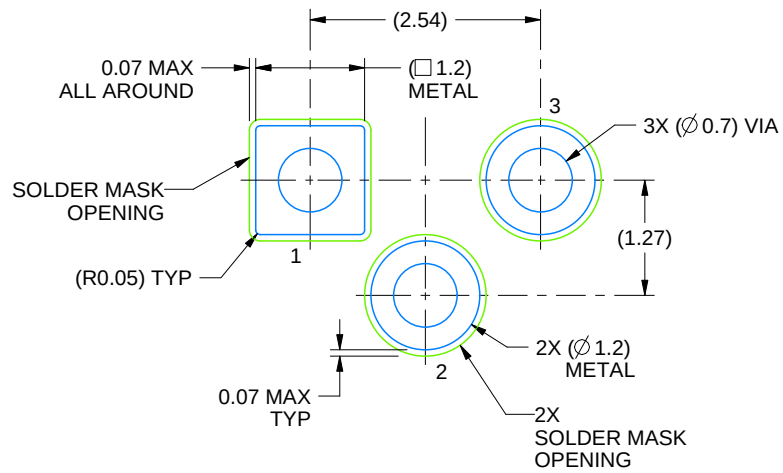
1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. Reference JEDEC registration TO-46.

EXAMPLE BOARD LAYOUT

NDV0003H

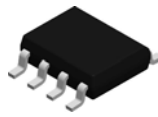
TO-CAN - 2.67 mm max height

TO-46



LAND PATTERN EXAMPLE
NON-SOLDER MASK DEFINED
SCALE:12X

4219876/A 01/2017

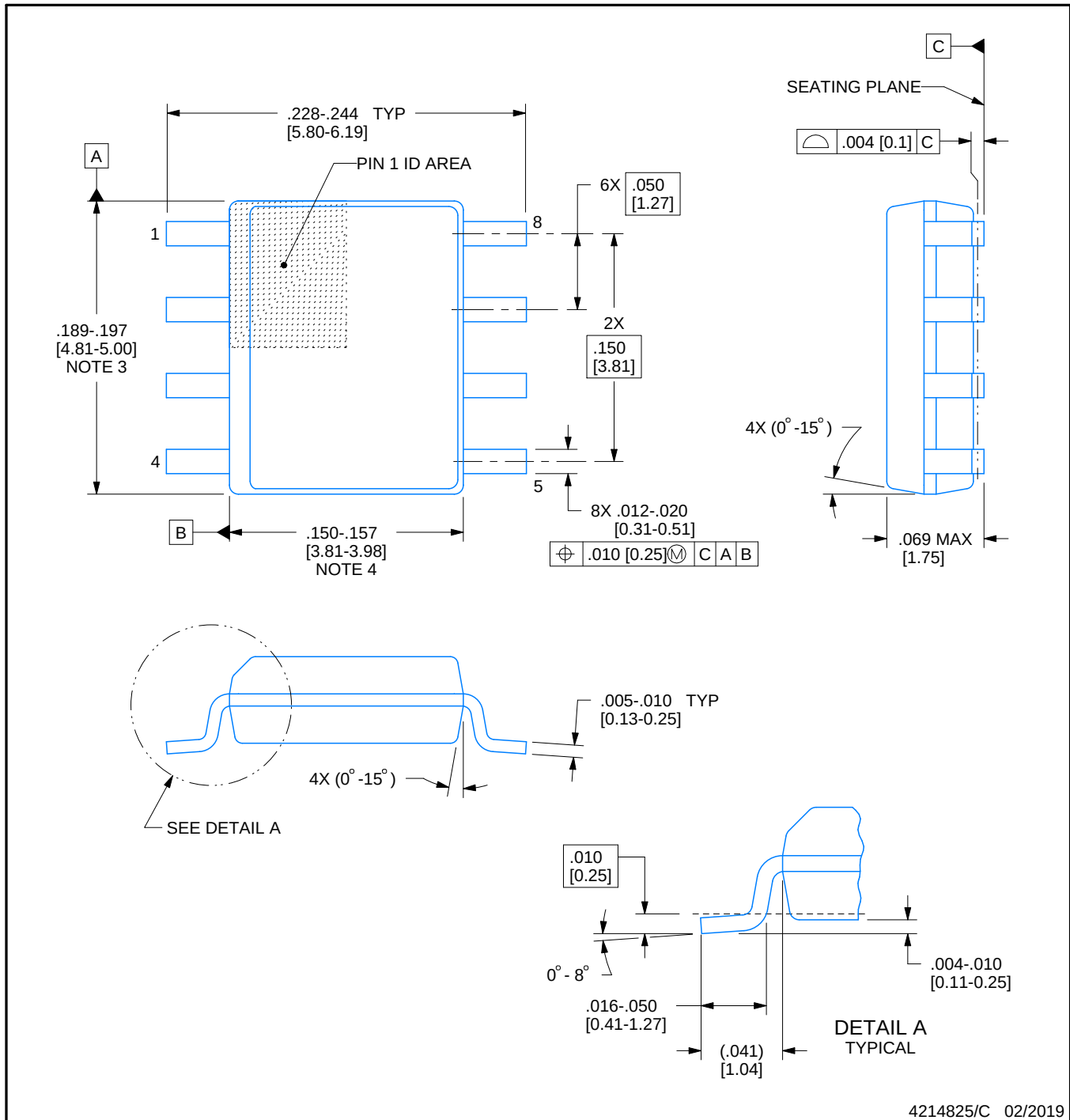


D0008A

PACKAGE OUTLINE

SOIC - 1.75 mm max height

SMALL OUTLINE INTEGRATED CIRCUIT



4214825/C 02/2019

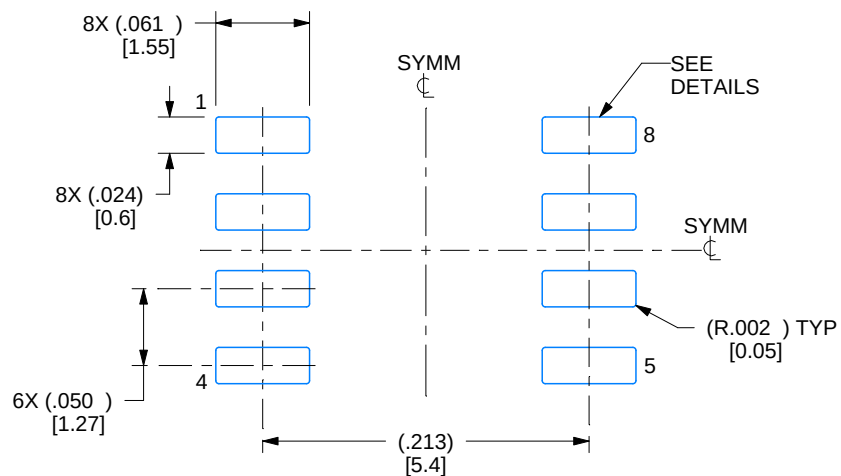
NOTES:

1. Linear dimensions are in inches [millimeters]. Dimensions in parenthesis are for reference only. Controlling dimensions are in inches. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This dimension does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed .006 [0.15] per side.
4. This dimension does not include interlead flash.
5. Reference JEDEC registration MS-012, variation AA.

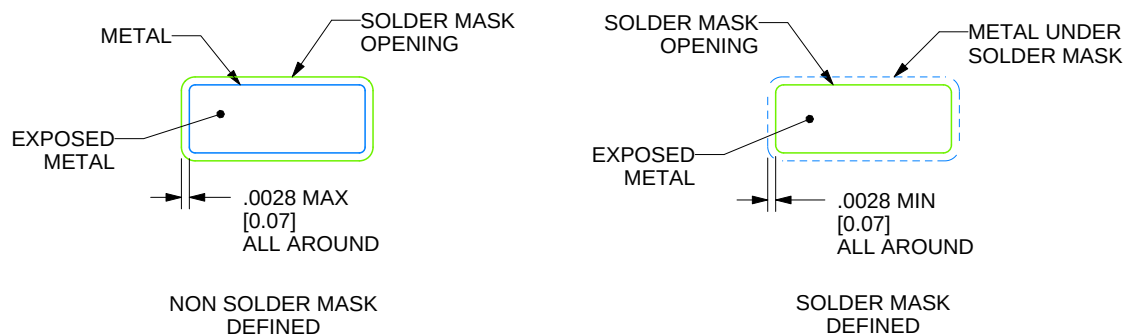
D0008A

SOIC - 1.75 mm max height

SMALL OUTLINE INTEGRATED CIRCUIT



LAND PATTERN EXAMPLE
EXPOSED METAL SHOWN
SCALE:8X



SOLDER MASK DETAILS

4214825/C 02/2019

NOTES: (continued)

6. Publication IPC-7351 may have alternate designs.

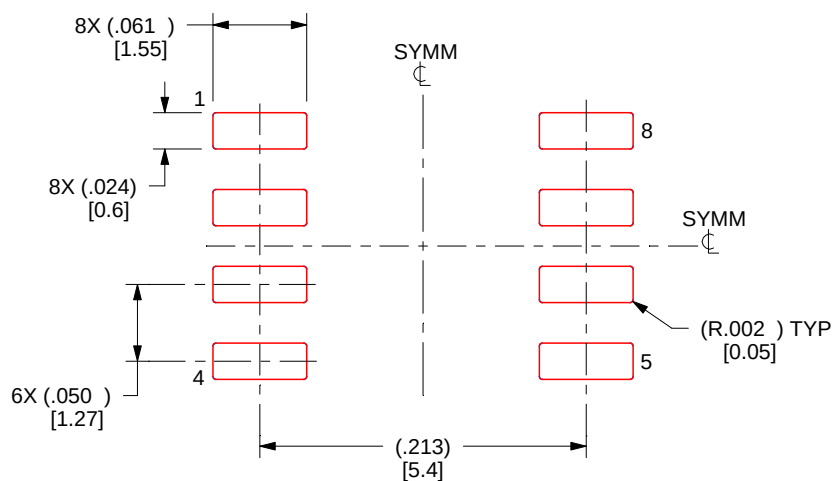
7. Solder mask tolerances between and around signal pads can vary based on board fabrication site.

EXAMPLE STENCIL DESIGN

D0008A

SOIC - 1.75 mm max height

SMALL OUTLINE INTEGRATED CIRCUIT

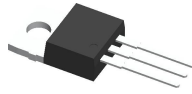


SOLDER PASTE EXAMPLE
BASED ON .005 INCH [0.125 MM] THICK STENCIL
SCALE:8X

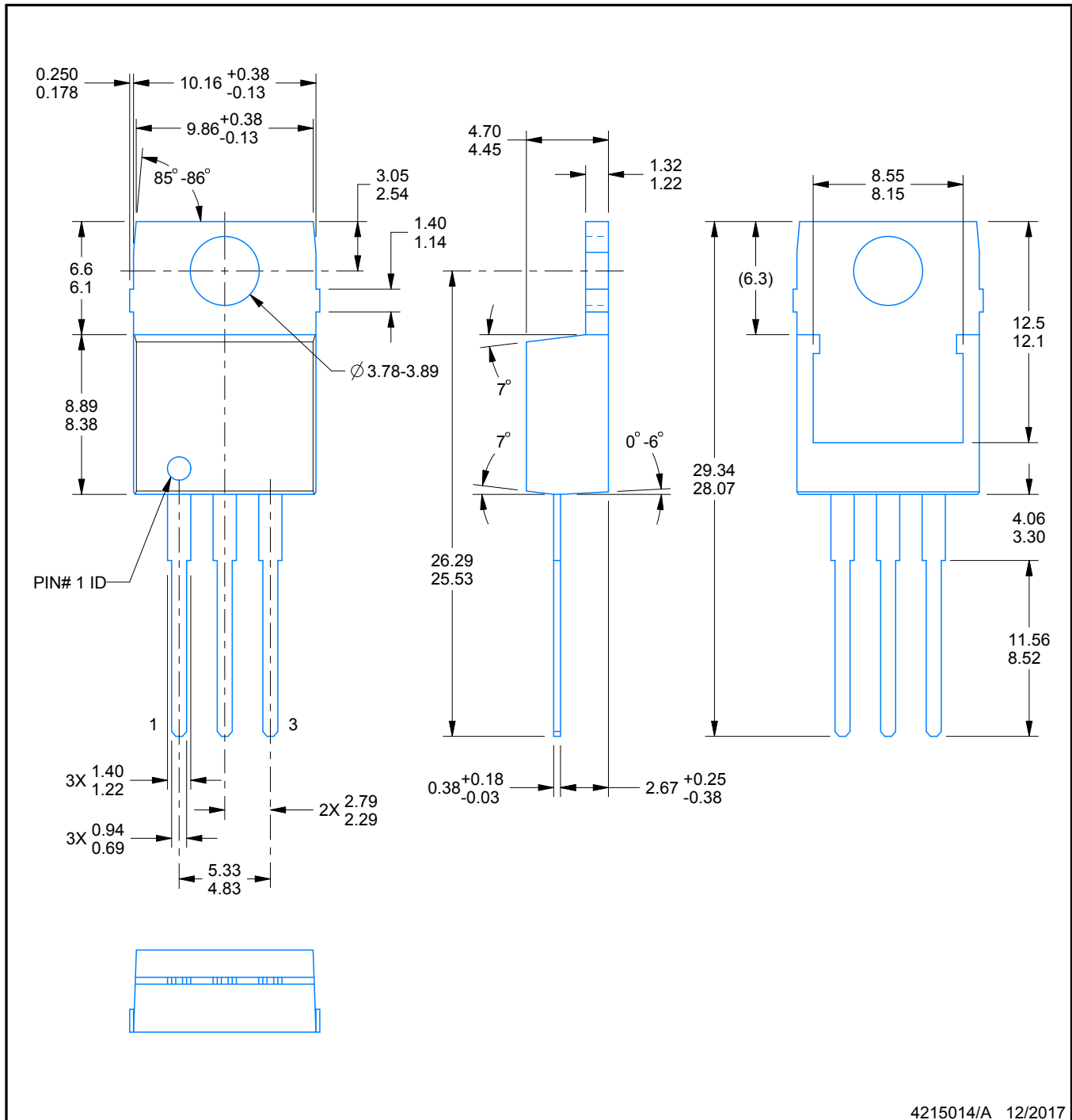
4214825/C 02/2019

NOTES: (continued)

8. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
9. Board assembly site may have different recommendations for stencil design.

NEB0003F**PACKAGE OUTLINE****TO-220 - 19.65 mm max height**

TRANSISTOR OUTLINE



4215014/A 12/2017

NOTES:

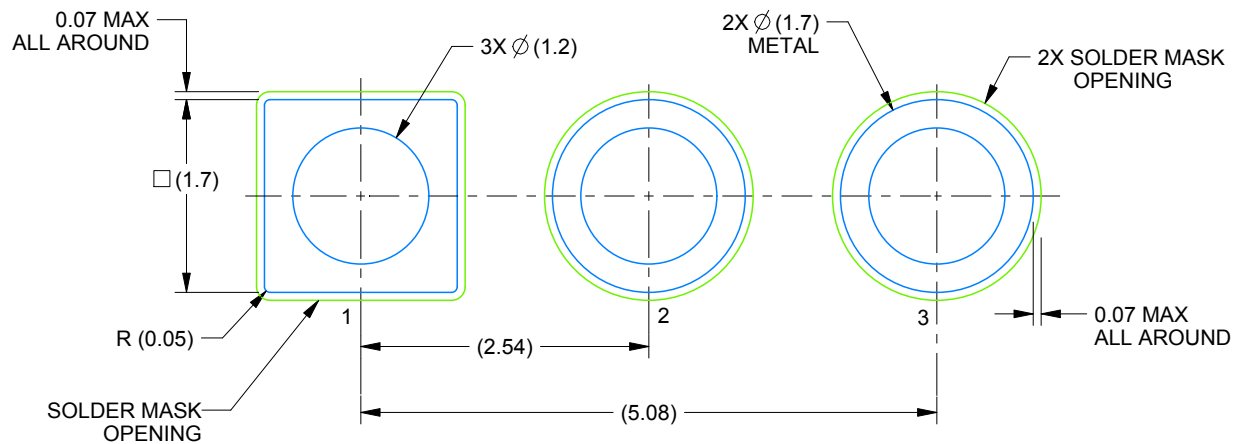
1. All controlling linear dimensions are in inches. Dimensions in brackets are in millimeters. Any dimension in brackets or parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. Reference JEDEC registration TO-220.

EXAMPLE BOARD LAYOUT

NEB0003F

TO-220 - 19.65 mm max height

TRANSISTOR OUTLINE

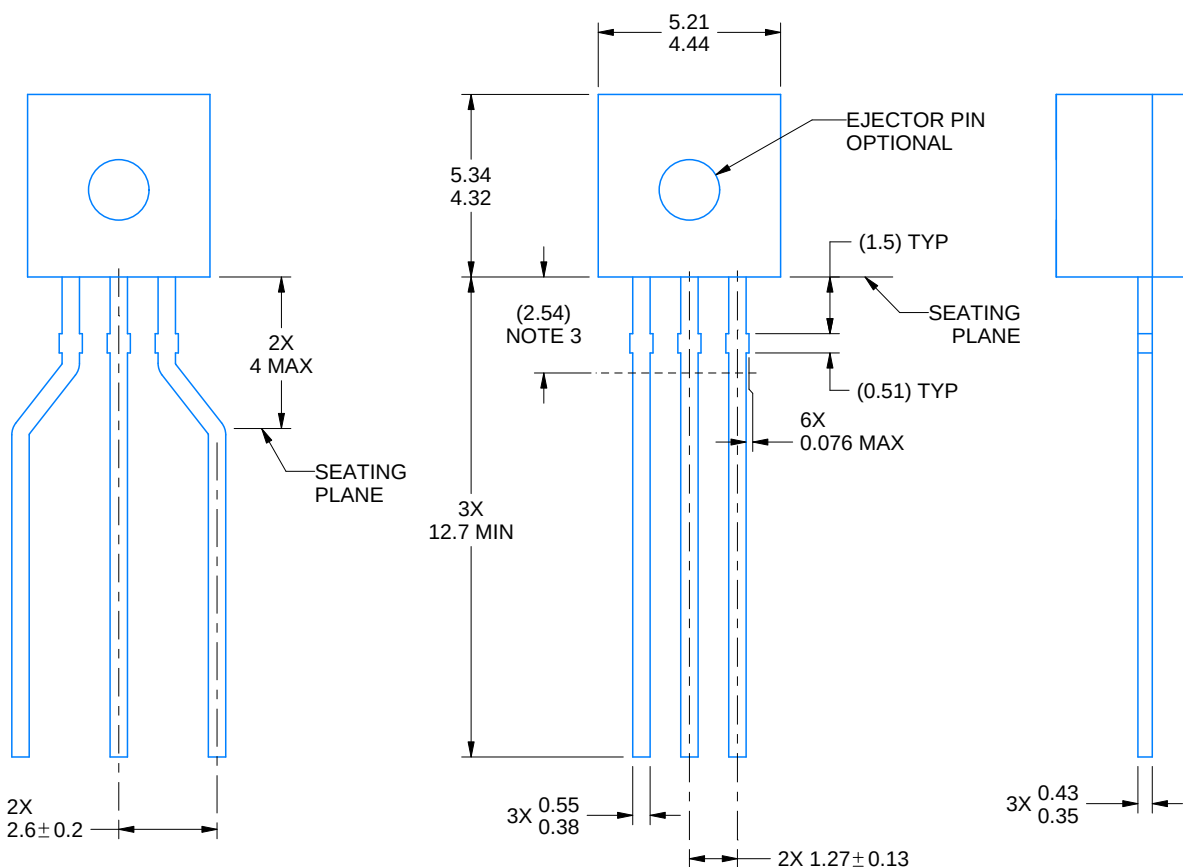


LAND PATTERN EXAMPLE
NON-SOLDER MASK DEFINED
SCALE:15X

4215014/A 12/2017



Images above are just a representation of the package family, actual package may vary.
Refer to the product data sheet for package details.



4215214/B 04/2017

NOTES:

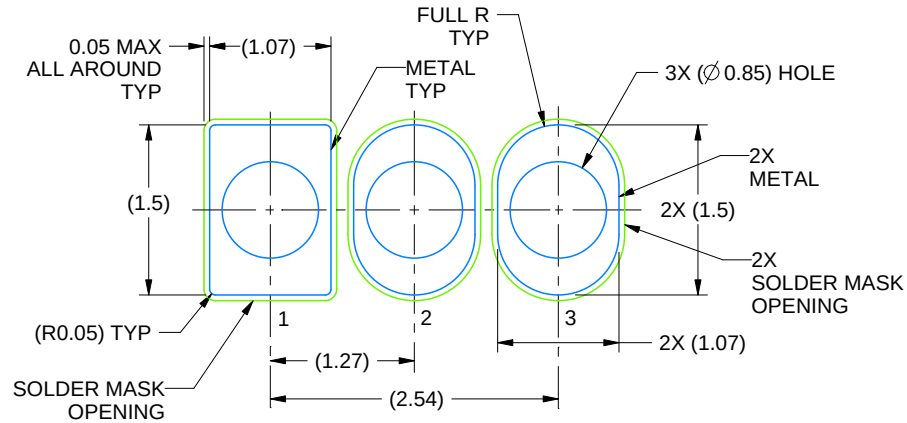
1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. Lead dimensions are not controlled within this area.
4. Reference JEDEC TO-226, variation AA.
5. Shipping method:
 - a. Straight lead option available in bulk pack only.
 - b. Formed lead option available in tape and reel or ammo pack.
 - c. Specific products can be offered in limited combinations of shipping medium and lead options.
 - d. Consult product folder for more information on available options.

EXAMPLE BOARD LAYOUT

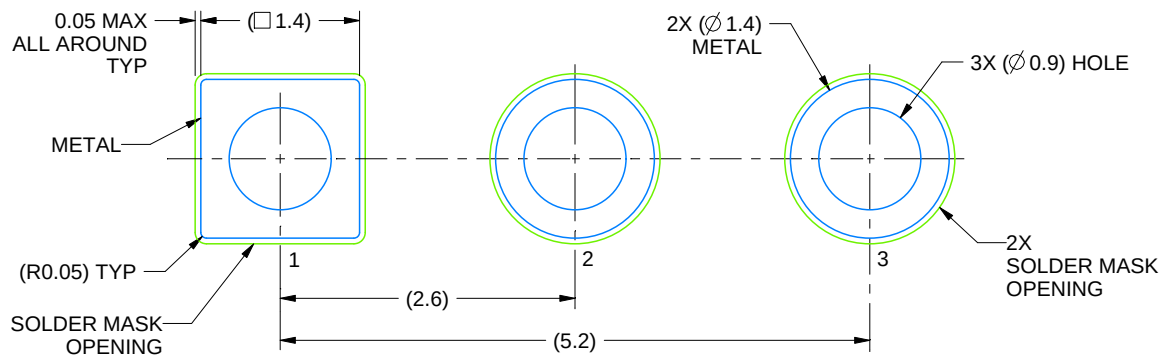
LP0003A

TO-92 - 5.34 mm max height

TO-92



LAND PATTERN EXAMPLE
STRAIGHT LEAD OPTION
NON-SOLDER MASK DEFINED
SCALE:15X



LAND PATTERN EXAMPLE
FORMED LEAD OPTION
NON-SOLDER MASK DEFINED
SCALE:15X

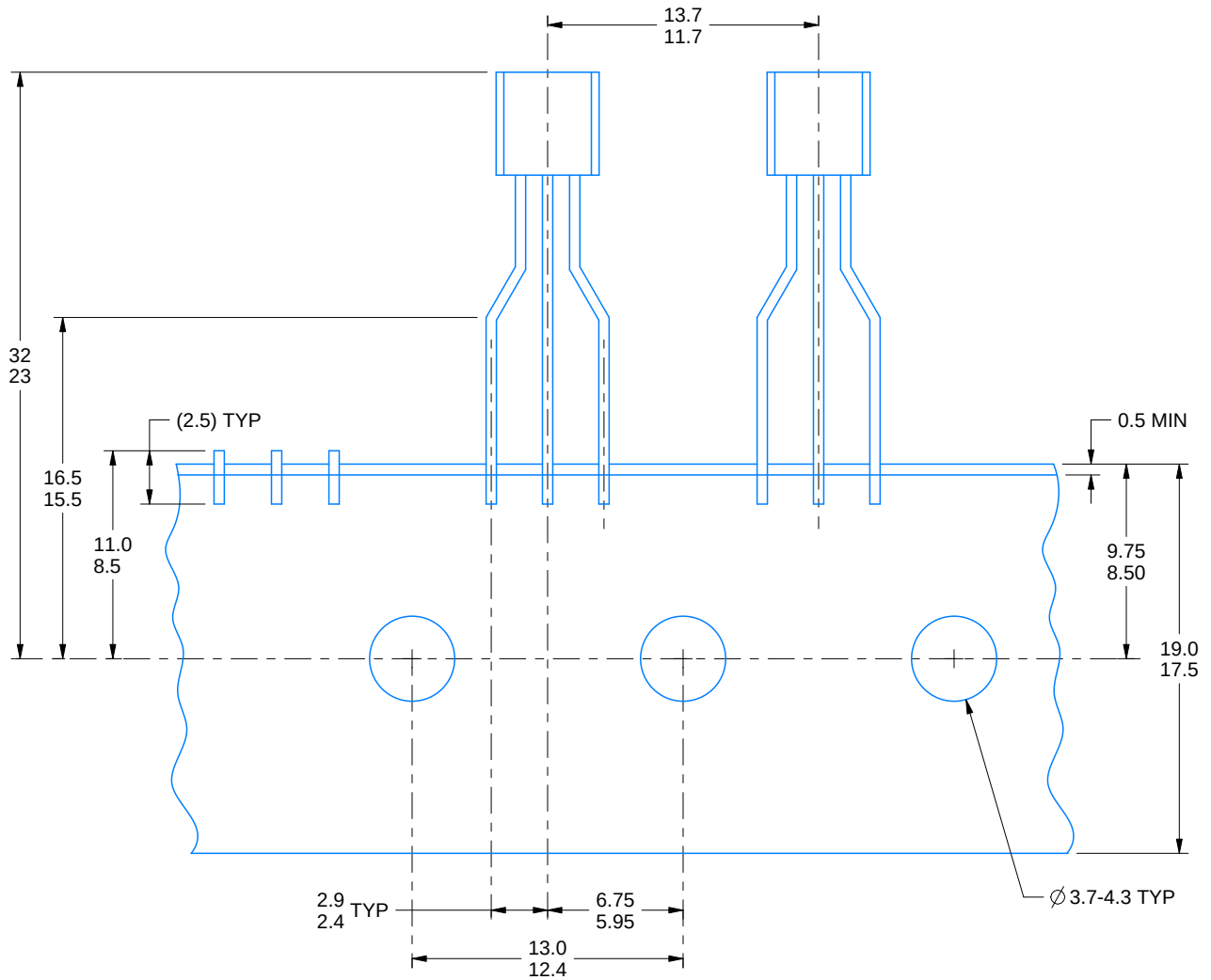
4215214/B 04/2017

TAPE SPECIFICATIONS

LP0003A

TO-92 - 5.34 mm max height

TO-92



FOR FORMED LEAD OPTION PACKAGE

4215214/B 04/2017

IMPORTANT NOTICE AND DISCLAIMER

TI PROVIDES TECHNICAL AND RELIABILITY DATA (INCLUDING DATASHEETS), DESIGN RESOURCES (INCLUDING REFERENCE DESIGNS), APPLICATION OR OTHER DESIGN ADVICE, WEB TOOLS, SAFETY INFORMATION, AND OTHER RESOURCES "AS IS" AND WITH ALL FAULTS, AND DISCLAIMS ALL WARRANTIES, EXPRESS AND IMPLIED, INCLUDING WITHOUT LIMITATION ANY IMPLIED WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE OR NON-INFRINGEMENT OF THIRD PARTY INTELLECTUAL PROPERTY RIGHTS.

These resources are intended for skilled developers designing with TI products. You are solely responsible for (1) selecting the appropriate TI products for your application, (2) designing, validating and testing your application, and (3) ensuring your application meets applicable standards, and any other safety, security, or other requirements. These resources are subject to change without notice. TI grants you permission to use these resources only for development of an application that uses the TI products described in the resource. Other reproduction and display of these resources is prohibited. No license is granted to any other TI intellectual property right or to any third party intellectual property right. TI disclaims responsibility for, and you will fully indemnify TI and its representatives against, any claims, damages, costs, losses, and liabilities arising out of your use of these resources.

TI's products are provided subject to TI's Terms of Sale (www.ti.com/legal/termsofsale.html) or other applicable terms available either on ti.com or provided in conjunction with such TI products. TI's provision of these resources does not expand or otherwise alter TI's applicable warranties or warranty disclaimers for TI products.

Mailing Address: Texas Instruments, Post Office Box 655303, Dallas, Texas 75265
Copyright © 2020, Texas Instruments Incorporated

Guangzhou HC Information Technology Co., Ltd.

Product Data Sheet

Module Data Sheet

Rev 1

1. 0	2.0	2.1	2.2				
2006/6/18	2006/9/6	2010/4/22	2011/4/6				

DRAWN BY :	Ling Xin		MODEL : HC-06	
CHECKED BY :	Eric Huang		Description:: BC04 has external 8M Flash and EDR module HC-06 is industrial, and compatible with civil HC-04	
APPD. BY:	Simon Mok		REV: 2.0	Page :
Former version introduction	HC-06 is the higher version of LV_BC_2.0. Linvor is the former of wavesen.			

Contents

1. Product's picture
2. Feature
3. Pins description
4. The parameters and mode of product
5. Block diagram
6. Debugging device
7. Characteristic of test
8. Test diagram
9. AT command set

1. Product's picture

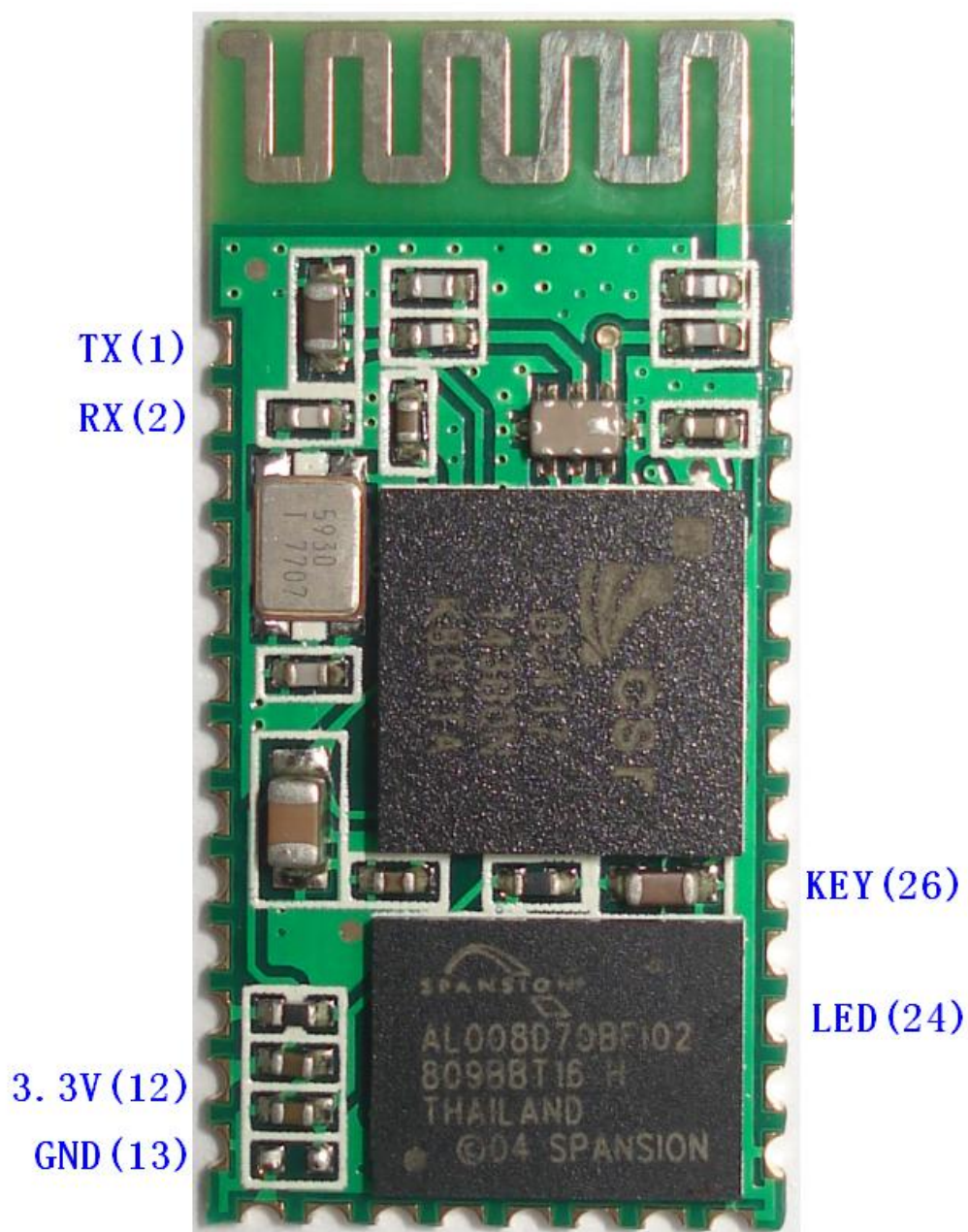


Figure 1 A Bluetooth module

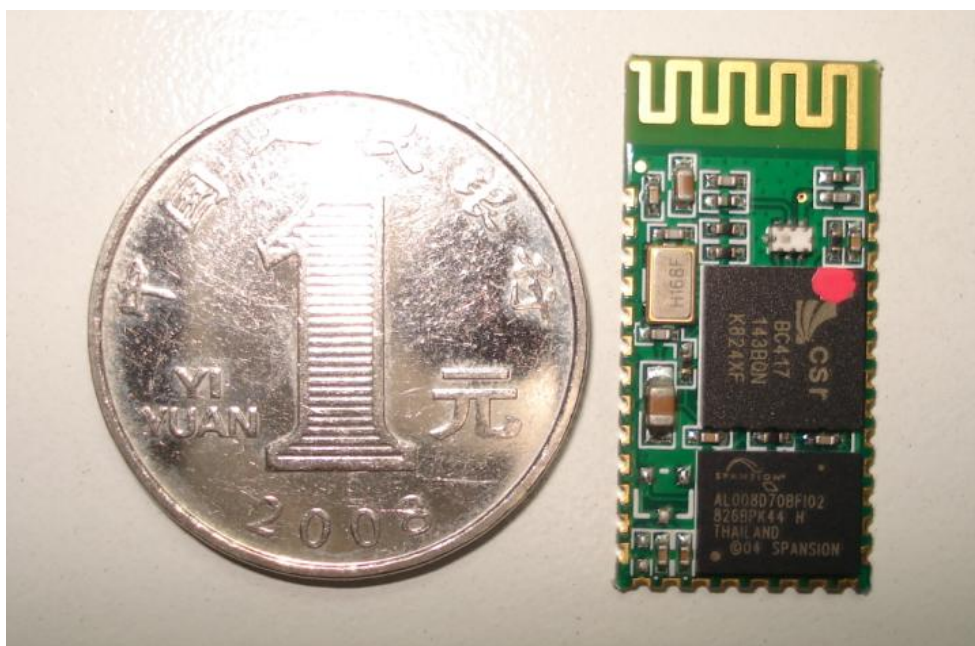


Figure 2. A Bluetooth module size



Figure 3 50 pieces chips in an anti-static blister package.

2. Feature

- Wireless transceiver
 - Sensitivity (Bit error rate) can reach -80dBm.
 - The change range of output's power: -4 - +6dBm.
- Function description (perfect Bluetooth solution)
 - Has an EDR module; and the change range of modulation depth: 2Mbps - 3Mbps.
 - Has a build-in 2.4GHz antenna; user needn't test antenna.
 - Has the external 8Mbit FLASH
 - Can work at the low voltage (3.1V~4.2V). The current in pairing is in the range of 30~40mA. The current in communication is 8mA.
 - Standard HCI Port (UART or USB)
 - USB Protocol: Full Speed USB1.1, Compliant With 2.0
 - This module can be used in the SMD.
 - It's made through RoHS process.
 - The board PIN is half hole size.
 - Has a 2.4GHz digital wireless transceiver.
 - Bases at CSR BC04 Bluetooth technology.
 - Has the function of adaptive frequency hopping.
 - Small (27mm×13mm×2mm)
 - Peripherals circuit is simple.
 - It's at the Bluetooth class 2 power level.
 - Storage temperature range: -40 °C - 85°C, work temperature range: -25 °C - +75°C
 - Any wave inter Interference: 2.4MHz, the power of emitting: 3 dBm.
 - Bit error rate: 0. Only the signal decays at the transmission link, bit error may be produced. For example, when RS232 or TTL is being processed, some signals may decay.
- Low power consumption
- Has high-performance wireless transceiver system
- Low Cost

- Application fields:
 - Bluetooth Car Handsfree Device
 - Bluetooth GPS
 - Bluetooth PCMCIA , USB Dongle
 - Bluetooth Data Transfer
- Software
 - CSR

3. PINs description

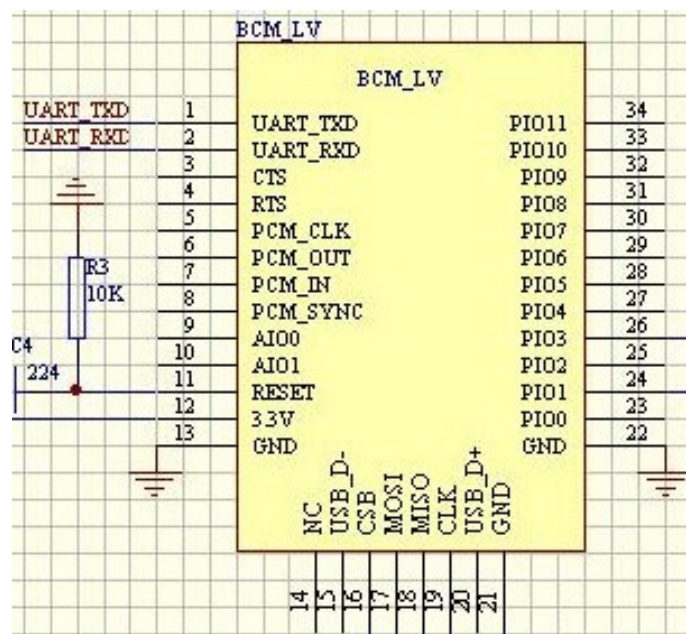


Figure 3 PIN configuration

The PINs at this block diagram is as same as the physical one.

PIN Name	PIN #	Pad type	Description	Note
GND	13 21 22	VSS	Ground pot	
1V8	14	VDD	Integrated 1.8V (+) supply with On-chip linear regulator output within 1.7-1.9V	
VCC	12	3.3V		
AIO0	9	Bi-Directional	Programmable input/output line	
AIO1	10	Bi-Directional	Programmable input/output line	

www.wavesen.com Phone: 020-84083341 Fax: 020-84332079 QQ:1043073574

Address: Room 527, No.13, Jiangong Road, Tianhe software park, Tianhe district, Guangzhou Post: 510660

Technology consultant: support@wavesen.com

Business consultant: sales@wavesen.com

Complaint and suggestion: sunbirdit@hotmail.com

PIO0	23	Bi-Directional RX EN	Programmable input/output line, control output for LNA(if fitted)	
PIO1	24	Bi-Directional TX EN	Programmable input/output line, control output for PA(if fitted)	
PIO2	25	Bi-Directional	Programmable input/output line	
PIO3	26	Bi-Directional	Programmable input/output line	
PIO4	27	Bi-Directional	Programmable input/output line	
PIO5	28	Bi-Directional	Programmable input/output line	
PIO6	29	Bi-Directional	Programmable input/output line	CLK_REQ
PIO7	30	Bi-Directional	Programmable input/output line	CLK_OUT
PIO8	31	Bi-Directional	Programmable input/output line	
PIO9	32	Bi-Directional	Programmable input/output line	
PIO10	33	Bi-Directional	Programmable input/output line	
PIO11	34	Bi-Directional	Programmable input/output line	
RESETB	11	CMOS Input with weak internal pull-down		
UART_RTS	4	CMOS output, tri-stable with weak internal pull-up	UART request to send, active low	
UART_CTS	3	CMOS input with weak internal pull-down	UART clear to send, active low	
UART_RX	2	CMOS input with weak internal pull-down	UART Data input	
UART_TX	1	CMOS output, Tri-stable with weak internal pull-up	UART Data output	
SPI_MOSI	17	CMOS input with weak internal pull-down	Serial peripheral interface data input	
SPI_CSB	16	CMOS input with weak internal	Chip select for serial peripheral interface, active low	

		pull-up		
SPI_CLK	19	CMOS input with weak internal pull-down	Serial peripheral interface clock	
SPI_MISO	18	CMOS input with weak internal pull-down	Serial peripheral interface data Output	
USB_-	15	Bi-Directional		
USB_+	20	Bi-Directional		
1.8V	14		1.8V external power supply input	Default : 1.8V internal power supply.
PCM_CLK	5	Bi-Directional		
PCM_OUT	6	CMOS output		
PCM_IN	7	CMOS Input		
PCM_SYNC	8	Bi-Directional		

4. The parameters and mode of product

LINVOR BLUE T

www.linvor.com

Bluetooth Module
 Bluetooth

CSR,BC417143B

V 2.0

2006/09/6

蓝牙 RF 模块

1. 采用 CSR BC4 +8M FLASH 方案
2. 具有 PIO0-PIO11、AIO0、AIO1、
USB、PCM、UART 及 SPI 接口，
模块内置 8MFLASH，功能强大，
用户可定制软件,适用于各种蓝牙
设备，内置 RF 天线,便于调试。

蓝牙协议版本	Bluetooth Specification V2.0 With EDR
USB 协议	Full Speed USB V1.1
USB Protocol	Compliant With USB V2.0
频率	2.4Ghz ISM band
调制方式	GFSK(Gaussian Frequency Shift Keying)
发射功率	-4 ->4 dBm, Class 2
灵敏度	≤ -80dBm at 0.1% BER
通讯速率	Asynchronous:2Mbps(Max)
供电电源	3.3V
工作温度	-20~+55 Centigrade
封装尺寸	27mmX13mmX2mm

Page 1 of 2

If you want more information, please visit www.wavesen.com.

www.wavesen.com Phone: 020-84083341 Fax: 020-84332079 QQ:1043073574

Address: Room 527, No.13, Jiangong Road, Tianhe software park, Tianhe district, Guangzhou Post: 510660

Technology consultant: support@wavesen.com

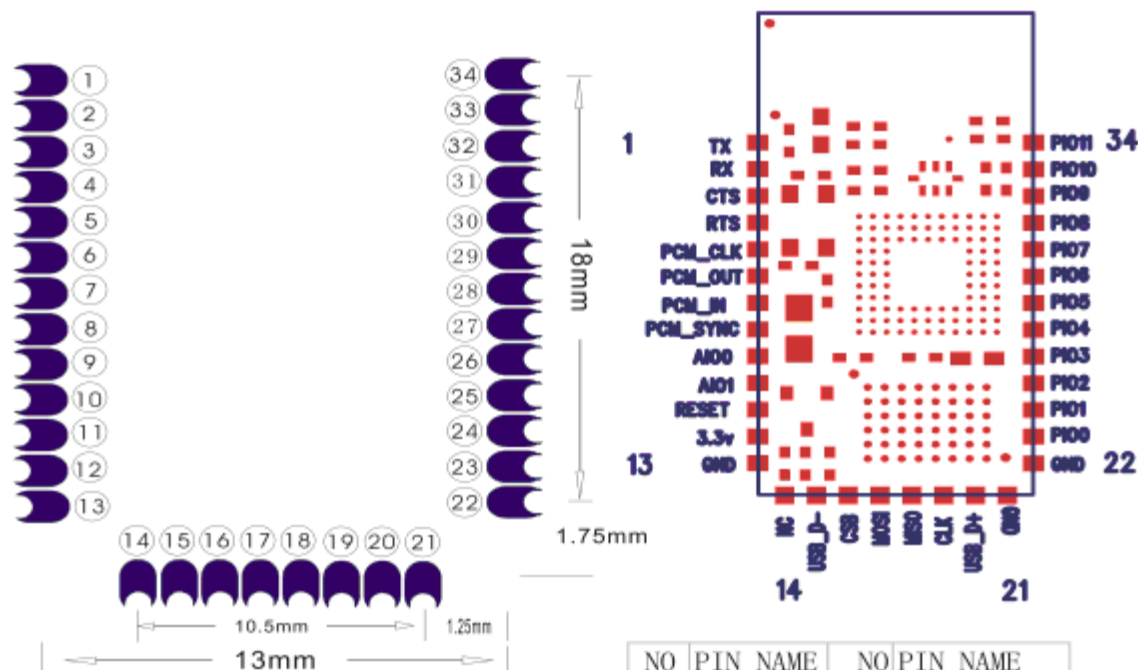
Business consultant: sales@wavesen.com

Complaint and suggestion: sunbirdit@hotmail.com

LINVOR BLUE T
www.linvor.com

LV-BC-2.0

单位: mm



PCB Layout 请参考实物

NO	PIN NAME	NO	PIN NAME
1	TX	20	USB D+
2	RX	21	GND
3	CTS	22	GND
4	RTS	23	PI00
5	PCM CLK	24	PI01
6	PCM OUT	25	PI02
7	PCM IN	26	PI03
8	PCM SYNC	27	PI04
9	AI00	28	PI05
10	AI01	29	PI06
11	RESET	30	PI07
12	3.3V	31	PI08
13	GND	32	PI09
14	NC	33	PI010
15	USB D-	34	PI011
16	CSB		
17	MOSI		
18	MISO		
19	CLK		

5. Block diagram

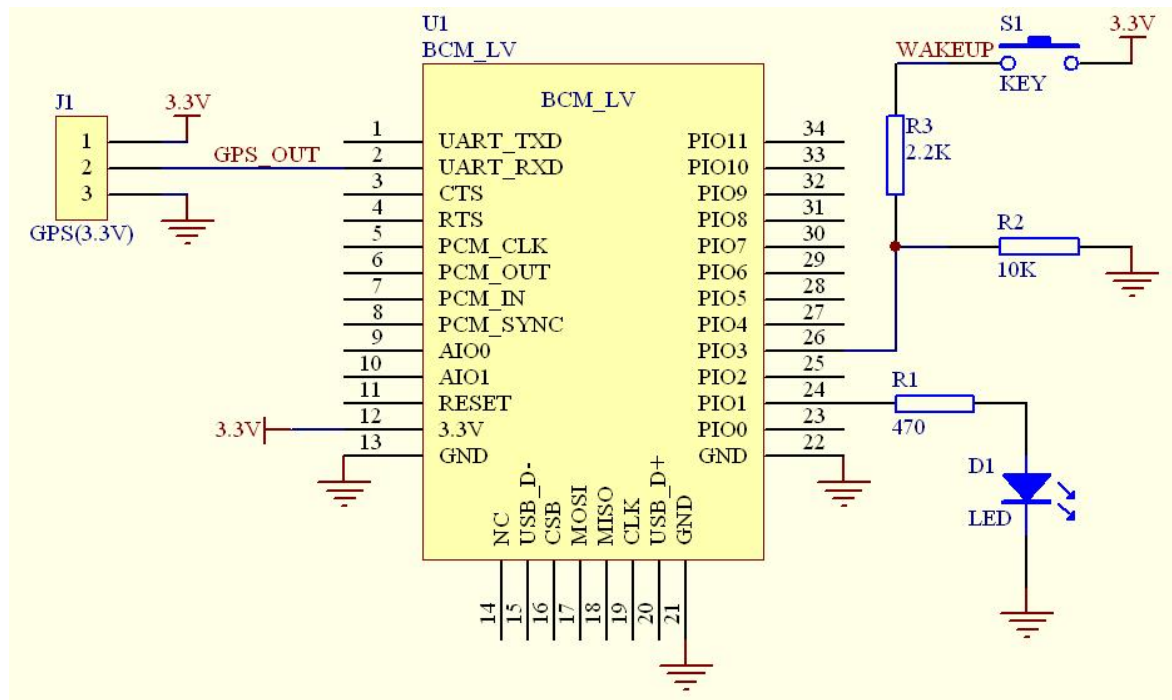


Figure 5 Block diagram 1

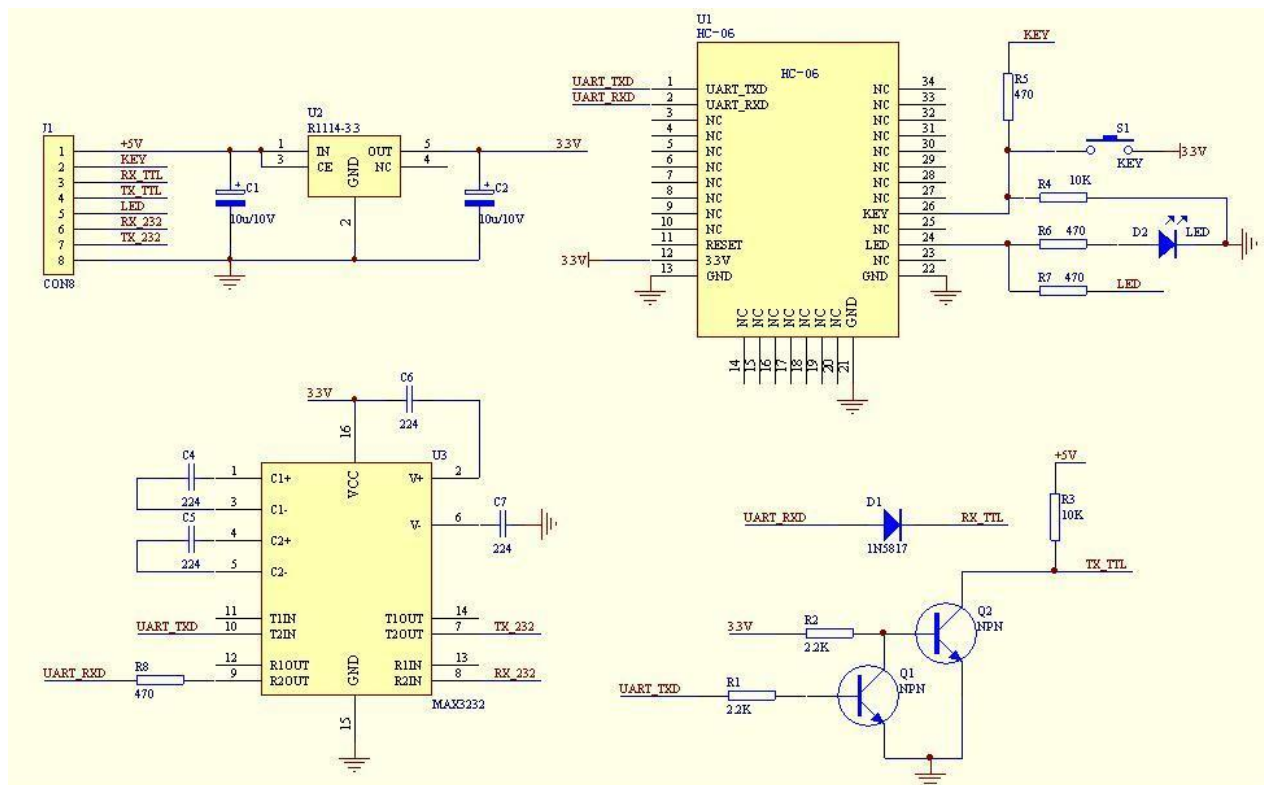


Figure 5 Block diagram 2

HC-04/06 master device has a function of remembering the last paired slave device. As a master device, it will search the last paired slave device until the connection is built. But if the WAKEUP button is pressed, HC-04/06 will lose the memory and research the new slave device.

6. Debugging device

6.1 Device

PC, hardware, 3G, 3G Frequency Counter (SP3386), 3.15V DC power supply, Shielding, Bluetooth Test box.

6.2 Software

7. Characteristic of test

Test Condition 25°C RH 65%				
	Min	Typ	Max	Unit
1. Carrier Freq. (<i>ISM Band</i>)	2.4		2.4835	MHz
2. RF O/P Power	-6	2	4	dBm
3. Step size of Power control	2		8	dB
4. Freq. Offset (<i>Typical Carrier freq.</i>)	-75		75	KHz
5. Carrier Freq. drift (<i>Hopping on, drift rate/50uS</i>)	-20		20	KHz
1 slot packet	-25		25	KHz
3 slot packet	-40		-40	KHz
6. Average Freq. Deviations (<i>Hopping off, modulation</i>)	140		175	KHz
Freq. Deviation	115			KHz
Ratio of Freq. Deviation	0.8			
7. Receive Sensitivity @< 0.1% BER(<i>Bit error rate</i>)	-83			dBm

8. Test diagram

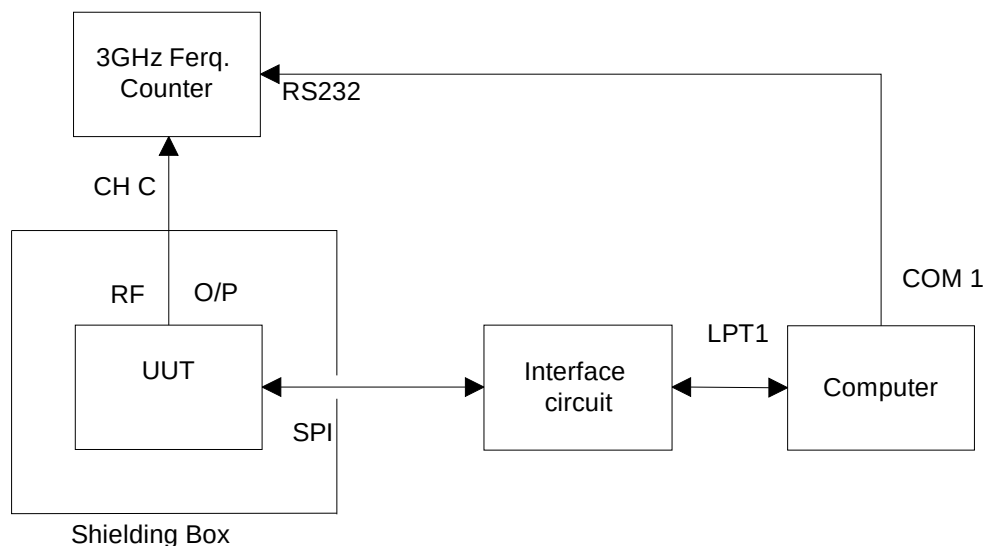


Fig 1. Programming and Freq. Alignment

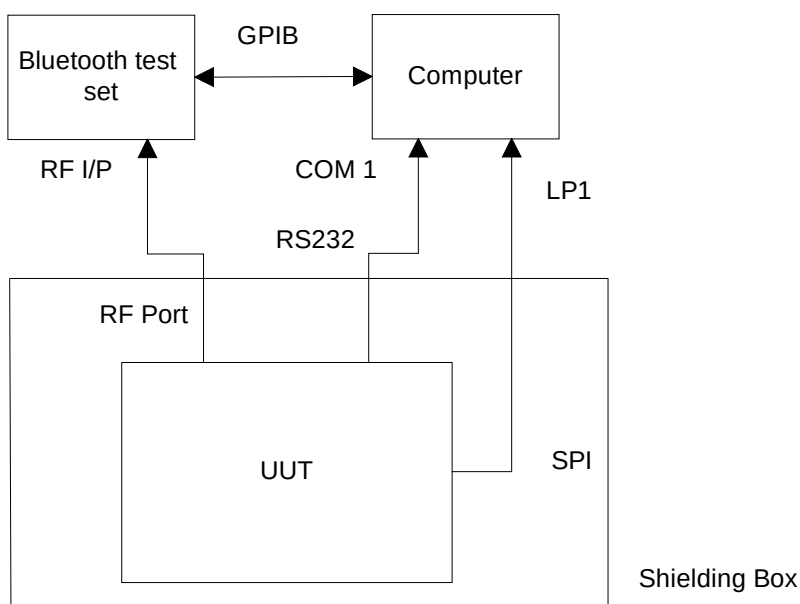


Fig 2 RF parameter Test Procedure

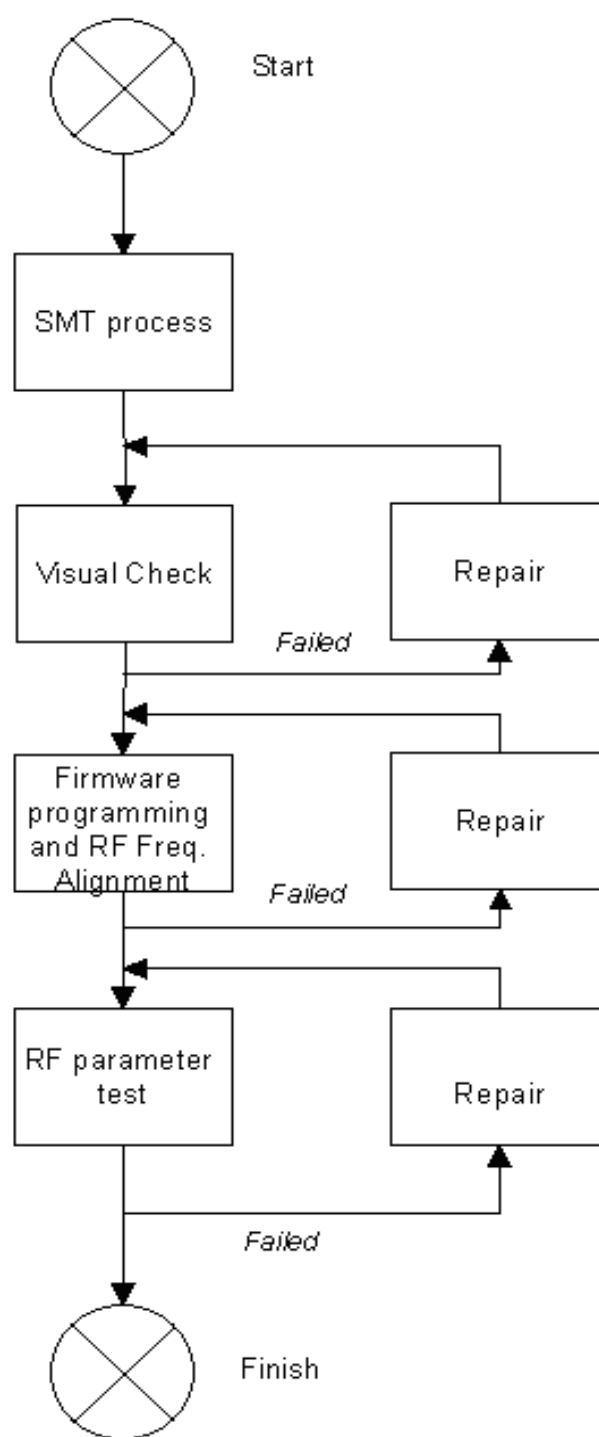


Fig 3 Assemble/Alignment/Testing Flow Chart

9. AT command set

The way to the AT command mode: supply power to the module, it will enter to the AT mode if it needn't pair. The interval of command is about 1 second.

Default parameter: Baud rate:9600N81, ID: linvor, Password:1234

1. Test communication

Send: AT (please send it every second)

Back: OK

2. Reset the Bluetooth serial baud rate

Send: AT+BAUD1

Back: OK1200

Send: AT+BAUD2

Back: OK2400

.....

1-----1200

2-----2400

3-----4800

4-----9600 (Default)

5-----19200

6-----38400

7-----57600

8-----115200

9-----230400

A-----460800

B-----921600

C-----1382400

PC can't support the baud rate lager than 115200. The solution is: make the MCU have higher baud rate (lager than 115200) through programming, and reset the baud rate to low level through the AT command.

The baud rate reset by the AT command can be kept for the next time even though the power is cut off.

3. Reset the Bluetooth name

Send: AT+NAMEname

Back: OKname

www.wavesen.com Phone: 020-84083341 Fax: 020-84332079 QQ:1043073574

Address: Room 527, No.13, Jiangong Road, Tianhe software park, Tianhe district, Guangzhou Post: 510660

Technology consultant: support@wavesen.com

Business consultant: sales@wavesen.com

Complaint and suggestion: sunbirdit@hotmail.com

Parameter name: Name needed to be set (20 characters limited)

Example:

Send: AT+NAMEbill_gates

Back: OKname

Now, the Bluetooth name is reset to be "bill_gates"

The parameter can be kept even though the power is cut off. User can see the new Bluetooth name in PDA refresh service. (Note: The name is limited in 20 characters.)

4. change the Bluetooth pair password

Send: AT+PINxxxx

Back:OKsetpin

Parameter xxxx: The pair password needed to be set, is a 4-bits number. This command can be used in the master and slave module. At some occasions, the master module may be asked to enter the password when the master module tries to connect the slave module (adapter or cell-phone). Only if the password is entered, the successful connection can be built. At the other occasions, the pair can be finish automatically if the master module can search the proper slave module and the password is correct.

Besides the paired slave module, the master can connect the other devices who have slave module, such as Bluetooth digital camera, Bluetooth GPS, Bluetooth serial printer etc.

Example:

Send: AT+PIN8888

Back: OKsetpin

Then the password is changed to be 8888, while the default is 1234.

This parameter can be kept even though the power is cut off.

5. No parity check (The version, higher than V1.5, can use this command)

Send: AT+PN (This is the default value)

Back: OK NONE

6. Set odd parity check (The version, higher than V1.5, can use this command)

Send: AT+PO

Back: OK ODD

7. Set even parity check(The version, higher than V1.5, can use this command)

Send: AT+PE

Back: OK EVEN

8. Get the AT version

Send: AT+VERSION

Back: LinvorV1.n